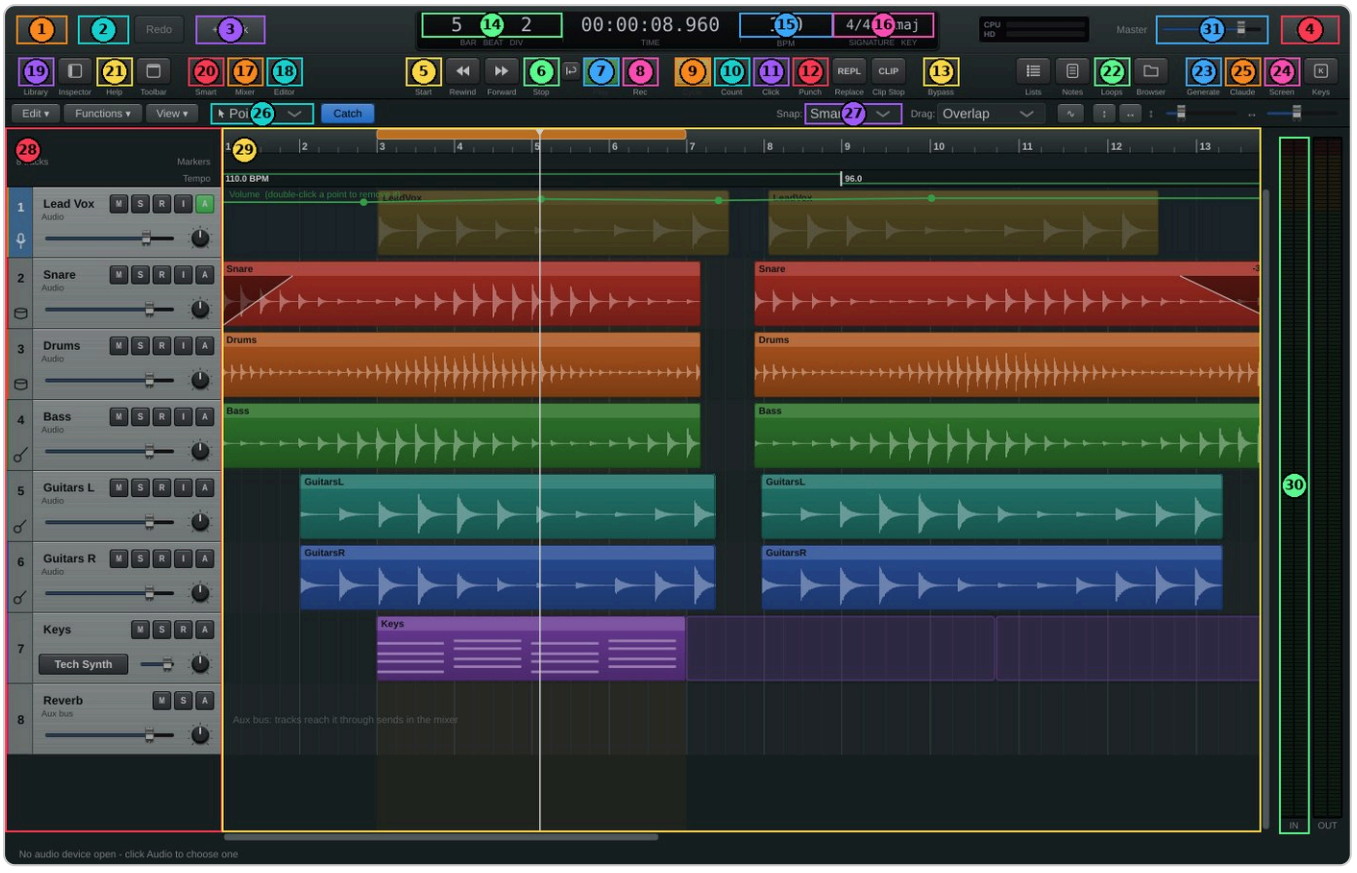
Audio Tech Pro

The Manual - the DAW and its plugins, and how to use them



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Start

Welcome to Audio Tech Pro

Audio Tech Pro is a complete studio: record audio and MIDI, edit, play instruments, mix with a full console and almost a hundred effects, and export your song. This manual explains every part of the program and every plugin that comes with it, with pictures.

> Press **F1** at any time to open this manual. Use the tabs at the top to move between its parts, the list on the left to pick a chapter, and the search box to find any word. **Help > Manual (PDF)** opens the same manual as a PDF you can print or read on your phone.

How this manual is organised

- **Start** - installing, a first song in ten steps, the main window, projects and files.
- **Record & Edit** - recording, tracks and regions, the piano roll, audio editing, automation.
- **Mix** - the mixer, the channel strip, using effects and plugins, mixing tools, exporting.
- **Instruments** - the built-in instruments, Nemix, session players, Live Loops, and the lists of sounds.
- **Plugins** - every built-in effect, with its picture, what it does, how to use it and all its controls.
- **More** - looks and themes, settings, the Claude assistant, screen recording, every key command and troubleshooting.

Pictures with **numbered dots** have a key under them: each number explains the part with that number.

Installing

Mac (Apple Silicon or Intel): open the `.dmg` and drag Audio Tech Pro to Applications, or run the `.pkg` installer. Because the app is not from the App Store, macOS may say it "can't be opened" or "is damaged". Open **Terminal** (Applications > Utilities) and paste this one line, then press Return and type your password:

```
APP="/Applications/Audio Tech Pro.app"; sudo xattr -cr "$APP"; sudo codesign --force --deep --sign - "$APP"; open "$APP"
```

You only need to do this once after each new version.

Windows: run `Setup.exe`. If Windows SmartScreen appears, click **More info > Run anyway**.

The recorded instruments for Nemix (the **Nemix Library**) are installed with the app; nothing else to download.

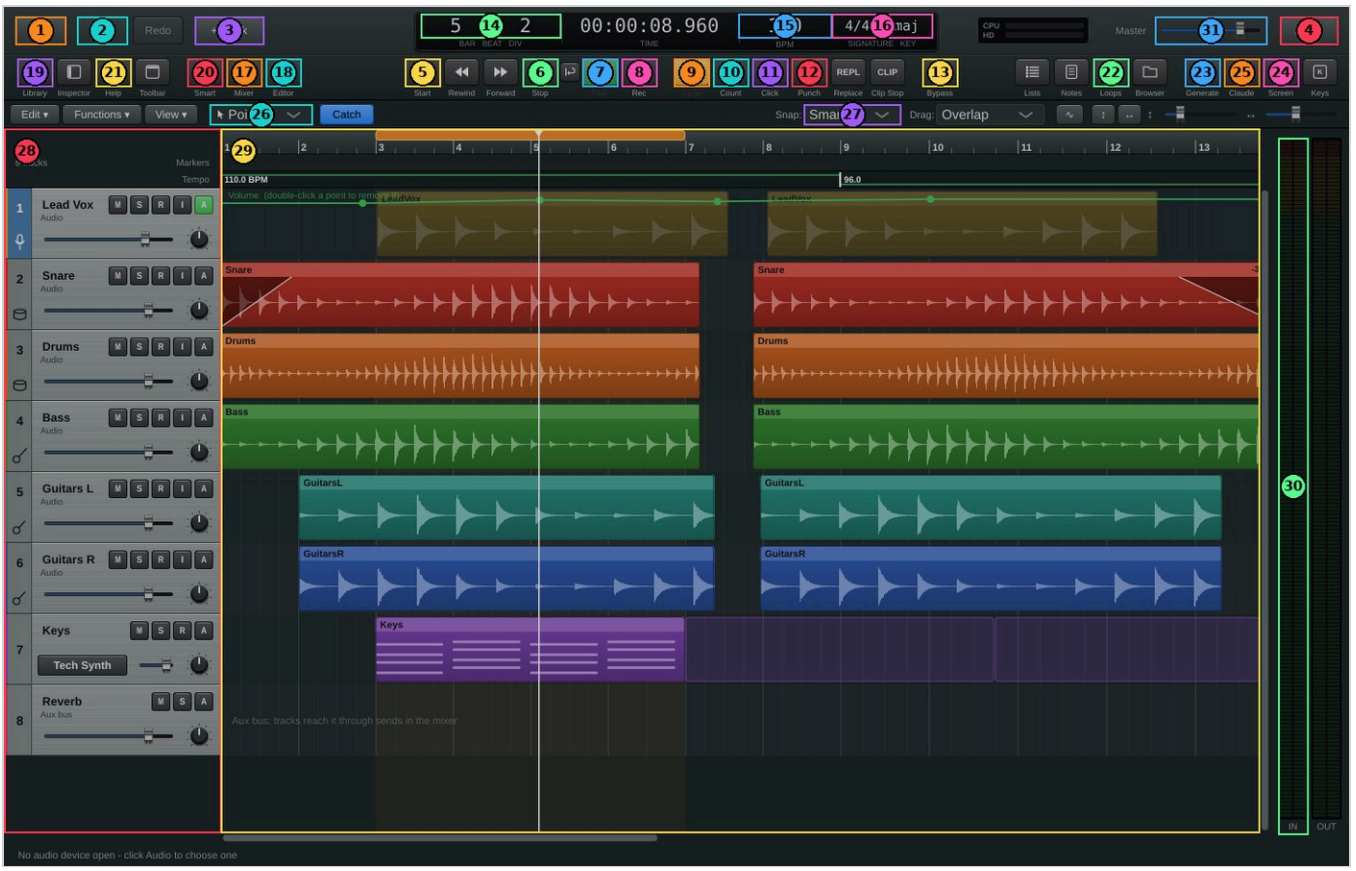
Your first song in ten steps

1. **Choose your audio interface.** Click **Audio** at the top right (or press `Cmd/Ctrl+,`). Pick your interface, a sample rate of 48 kHz and a buffer of 128 samples. Smaller buffers feel tighter when you play; larger ones let you use more plugins.
2. **Set the tempo and key.** Double-click the tempo in the display at the top (BPM) and type a number. Click the key (for example *Cmaj*) to choose the song's key.
3. **Add a drum track.** Choose **Track > New Drummer Track**. A drummer plays along with the song. In its editor below the tracks, drag the dot in the square to make it busier or louder.
4. **Add an instrument.** Press `Cmd/Ctrl+Option/Alt+S` for a new instrument track. Click the instrument key on its track header (it shows *Tech Synth*), or the input slot on its mixer strip, and choose **Nemix (wavetable synth)**, then pick a sound in Nemix's **BROWSE** page - "Grand Piano", "Finger Bass" or "Supersaw Lead", for example.
5. **Record MIDI.** Press `K` to turn the click on, then `R` to record. Play your MIDI keyboard. Press `Space` to stop.
6. **Fix the timing.** Select the new region and press `Q` to quantize it, or double-click it to open the **piano roll** and move notes by hand.
7. **Record a voice or guitar.** Press `Cmd/Ctrl+Option/Alt+A` for an audio track, choose its input on the track (or in the Inspector), arm it with the **R** key on the track, and press `R` to record.
8. **Mix.** Press `X` for the mixer. Set the faders, add effects by clicking an empty insert slot, and send tracks to a reverb with the sends. Or try **Mix > Automix...** for a balanced start.
9. **Master.** On the master strip add **Mastering Assistant** (Metering > Mastering Assistant), play the loudest part of the song and let it set the EQ and loudness.
10. **Export.** Choose **File > Bounce Song to WAV...** (`Cmd/Ctrl+B`). Save the project with `Cmd/Ctrl+S`.

> Not sure what a button does? Turn on **Quick Help** (the **?** key in the control bar) and point at it.

The Main Window

Everything starts in the main window: the control bar along the top, the tracks in the middle, and panels you can show around them (the Inspector on the left, the mixer or an editor below, the Library, Smart Controls and more).



1. **File menu** - new, open, save, import audio, bounce the song.
2. **Undo and Redo** - `Cmd/Ctrl+Z` and `Cmd/Ctrl+Shift+Z`. Hover over them to see what will be undone.
3. **Add tracks** - a new audio track, instrument track or aux bus.
4. **Audio settings** - your audio interface, sample rate and buffer size, latency calibration.
5. **Go to start** - `Return`.
6. **Stop** - press again to go back to the beginning.
7. **Play** - `Space` plays and stops.
8. **Record** - `R` records on the armed tracks (if none is armed, the selected track is armed for you).
9. **Cycle** - `C` turns the loop on and off. Drag in the top half of the ruler to set the range. Recording with Cycle on makes a new take on every pass.
10. **Count-in** - one bar of click before recording (`Shift+K`).
11. **Metronome** - the click (`K`).
12. **Punch in/out** - Autopunch: record only between the punch locators (the red bar under the cycle in the ruler).

13. **Bypass all** - hear the mix with all the processing switched off, to compare. Click again to bring it all back.
14. **Position** - bar, beat and division, and the time in hours, minutes and seconds.
15. **Tempo** - double-click to type a new tempo; right-click to detect it from a region, or to tap it.
16. **Time signature and key** - click to change them.
17. **Mixer** - show or hide the mixer under the tracks (`X`).
18. **Editor** - the piano roll for the selected MIDI region, or the audio editor for an audio region (`P`).
19. **Library** - patches and sounds for the selected track (`Y`).
20. **Smart Controls** - the main knobs of the selected track (`B`).
21. **Quick Help** - a box that explains whatever is under the mouse.
22. **Loops** - the Loop Browser: browse loops and samples, click to hear, drag onto a track (`O`).
23. **Generate** - write drums, bass, chords and a melody in a genre's style.
24. **Screen** - record the app with the mix's sound to a video.
25. **Claude** - the Claude assistant: ask questions or tell it what to do.
26. **Tool menu** - the tool the mouse uses in the tracks area (`T` opens a menu of all tools).
27. **Snap** - where regions land when you move them. *Smart* follows the zoom. Hold `Cmd` (Ctrl on Windows) while dragging to move freely.
28. **Track headers** - each track's number, name, icon and buttons: **M** mute, **S** solo, **R** record arm, **I** input monitoring, **A** automation, plus a volume fader and pan knob.
29. **Tracks area** - the regions (audio and MIDI) on a timeline, with the ruler, markers and tempo lane above.
30. **Input and output meters** - click to clear the peaks; right-click for peak hold and fall speed.
31. **Master volume** - the level of everything you hear.

The control bar keys

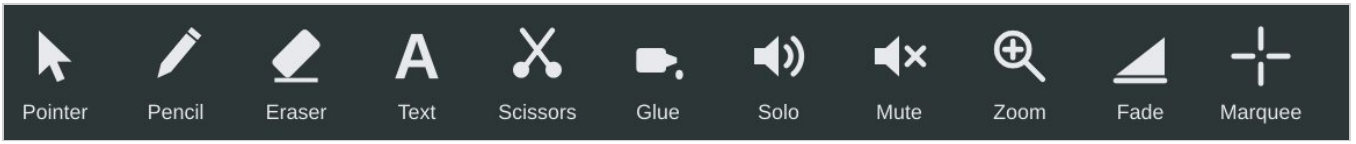
The second row of the control bar has keys for the panels and functions you use most. From the left: **Library**, **Inspector**, **Help** (Quick Help), **Toolbar**, **Smart** (Smart Controls), **Mixer**, **Editor**; the transport (**Start**, **Rewind**, **Forward**, **Stop**, **Return to start**, **Play**, **Rec**); **Cycle**, **Count**, **Click**, **Punch**, **Replace**, **Clip Stop** and **Bypass**; then **Lists** (Event List), **Notes** (Notepad), **Loops**, **Browser**, **Generate**, **Claude**, **Screen** and **Keys** (Key Commands).

- **Return to start** - when lit, stopping jumps back to where playback started (as some programs do). Off, the playhead stays where you stopped.
- **Replace** - a new recording replaces the parts of regions it covers.
- **Clip Stop** - recording stops by itself if an input clips (hits 0 dB).

The bar above the tracks

- **Edit, Functions, View** - the tracks area's own menus.
- **Tool** - Pointer, Pencil, Eraser, Text, Scissors, Glue, Solo, Mute, Zoom, Fade and Marquee. See *Tracks and Regions* for each tool.

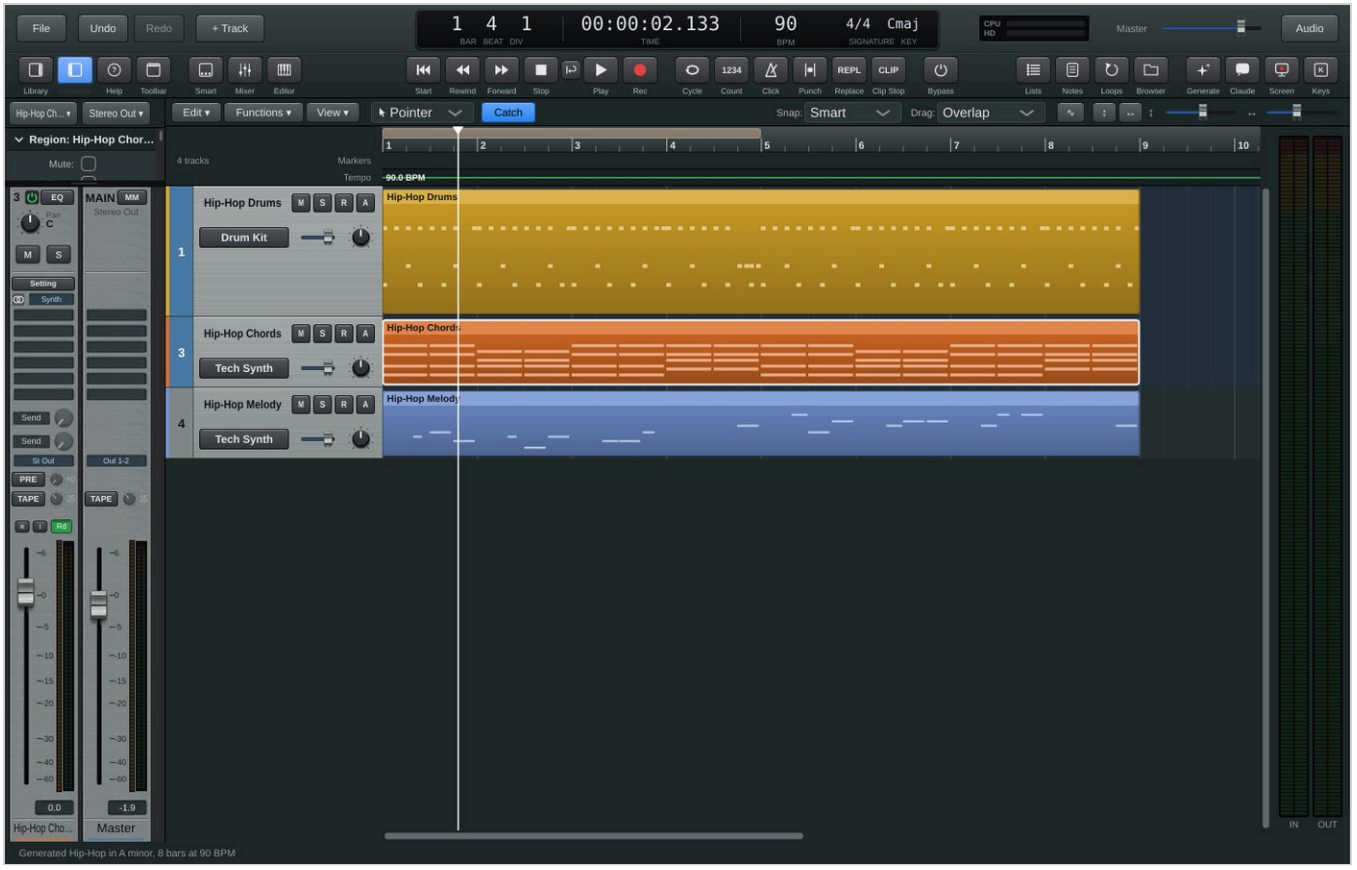
- **Catch** - the tracks area follows the playhead while it plays. Off, the view stays where you put it.
- **Snap** - Smart, Bar, Beat, Division or Off.
- **Drag** - what happens when regions meet: *Overlap* lets them lie on top of each other, *No Overlap* shortens what they cover, *X-Fade* crossfades where they overlap, *Shuffle R/L* closes the gaps.
- **Waveform Zoom**, **fit height**, **fit width**, and the **track height** and **horizontal zoom** sliders.



Panels you can show

- **Inspector** (`I`) - on the left: the selected region's settings, the selected track's settings, and its channel strip beside the master's.
- **Mixer** (`X`) - below the tracks, or in its own window (`Cmd/Ctrl+2`).
- **Editors** (`P`) - the piano roll, step editor and score for MIDI; the audio editor for audio. In their own window with `Cmd/Ctrl+4`.
- **Smart Controls** (`B`) - eight knobs for the selected track's sound.
- **Library** (`Y`) and **Loop Browser** (`O`) - on the right.
- **Browser** (`F`) - audio files on your computer.
- **Toolbar** - a row of keys for common edits (split, join, repeat, quantize, bounce, freeze...).
- **Status bar** - hints along the bottom.

View > Show All Panels and **View > Hide All Panels** switch them all at once. Hide All leaves just the tracks, for the biggest view.



Zooming and moving around

- `Cmd/Ctrl+Right` and `Cmd/Ctrl+Left` zoom in and out in time; `Cmd/Ctrl+Down` and `Cmd/Ctrl+Up` make the tracks taller or shorter.
- `Z` fits the whole song in the window.
- Scroll with the mouse wheel or trackpad; hold `Shift` with a mouse wheel to scroll sideways.
- `,` and `.` move one bar back and forward; `Return` goes to the start.
- Click in the ruler to move the playhead there.

Projects and Files

A **project** is your song: its tracks, regions, mixer, plugins and settings, saved in one file, with the audio it records in an **Audio Files** folder beside it.

New, open and save

- **File > New** (`Cmd/Ctrl+N`) - an empty project. **File > New from Template** starts from one of your templates.
- **File > Open...** (`Cmd/Ctrl+O`) - open a saved project. Recent projects are listed in the File menu.
- **File > Save** (`Cmd/Ctrl+S`) and **Save As...** (`Cmd/Ctrl+Shift+S`).
- **File > Close Project** - asks to save changes first.

Keeping your work safe

- **Autosave** - in **Settings** (`Cmd/Ctrl+Option/Alt+,`) choose how often the project saves itself: every minute up to every 10 minutes.
- **Backups** - each save keeps a backup (choose how many, 3 to 50, and whether more are made every few minutes). **File > Revert to Backup...** opens one.
- **Alternatives** - **File > New Alternative...** saves a copy beside the project, to try ideas. Switch between them under **File > Alternatives**.
- **Templates** - **File > Save as Template...** saves the current set-up (tracks, plugins, routing) to start new songs from.
- **Undo History** (`Cmd/Ctrl+Option/Alt+Z`) - every edit you can undo; click one to go back or forward to it.

Audio files

- **File > Import Audio File...** (`Cmd/Ctrl+Shift+I`) - puts audio files on the selected track at the playhead. You can also drag files from your computer, the **Browser** (`F`) or the **Loop Browser** (`O`) onto the tracks.
- **File > Consolidate Project...** - copies every audio file the project uses into its own folder, so it's all in one place.
- **File > Save a Copy As...** - a copy of the project with all its audio in one folder (and a .zip if you like), to open on any computer or send to someone.
- **File > Find Missing Audio Files...** - when a project can't find its audio (it was moved), look for it in a folder you choose.

Bringing in work from other programs

File > Open Project from Another Program... opens projects from Ableton Live, FL Studio, Studio One, Pro Tools, Logic, Cubase, DAWproject files and standard MIDI files. Tracks, regions,

tempo and as much of the mix as the file holds come in.

From **Logic Pro** (a `.logicx` project), the tracks with their names, the audio regions, the MIDI regions with their notes, the tempo changes, the time signatures and the markers come in. Logic keeps its plugins, instruments and mix settings in a form Apple hasn't published, so instrument tracks start on the built-in synth (choose Nemix or your plugin for each) and the faders at 0 dB. In Logic you can also use **File > Export > All Tracks as Audio Files** and open the result here, to bring each track's sound exactly as you mixed it.

File > Import Tracks from Project... brings chosen tracks - their regions, plugins, sends and automation - from another of your projects into this one.

Exporting

- **File > Bounce Song to WAV...** (`Cmd/Ctrl+B` or `Cmd/Ctrl+E`) - the whole song (plus two seconds for reverb tails) to a WAV file. With **Cycle** on, just the cycle range is exported.
- **File > Export All Tracks as Audio Files...** (`Cmd/Ctrl+Shift+E`) - each track and bus to its own WAV in a folder, with its plugins, fader and pan, ignoring solo. Use this to send stems to a mix engineer or to another program.
- **Track > Bounce in Place** - renders the selected track to a new audio track and mutes the original.
- **Edit > AudioSuite...** - processes selected audio regions with an effect into new files.

The Notepad

View > Notepad (`Cmd/Ctrl+Option/Alt+P`) keeps lyrics, ideas and to-do notes with the project.

Record & Edit

Recording

Set up your audio interface

Click **Audio** at the top right, or choose **Audio Settings...** (`Cmd/Ctrl+,`):

- **Device** - your interface. On Mac it's Core Audio, on Windows choose the ASIO driver for the lowest delay.
- **Sample rate** - 44.1 or 48 kHz for most music.
- **Buffer size** - 64 or 128 samples while recording (you hear yourself with little delay); 512 or 1024 while mixing with many plugins.

Calibrate Recording Latency... plays a click through your interface and measures how long it takes to come back, so recordings line up exactly with the music. Connect an output to an input with a cable (or hold a mic to a speaker) and click Measure.

Combine Audio Interfaces... uses several interfaces at once, as one (on Mac).

Mix > I/O Labels... gives the interface's inputs your own names ("Vocal mic", "Guitar DI"); they show in the input menus.

Record audio

1. Make an audio track: **Track > New Audio Track** (`Cmd/Ctrl+Option/Alt+A`).
2. Choose its **input** on the mixer strip or in the Inspector (In 1, In 2, or a stereo pair), and mono or stereo with the round button beside it.
3. Arm it: click **R** on the track header. Click **I** (input monitoring) to hear yourself through the track's plugins.
4. Check the level: sing or play your loudest; the meter should stay out of the red. Turn the gain on your interface, not the fader.
5. Press `R` to record, `Space` to stop.

The format (16-bit, 24-bit or 32-bit float WAV) is set in **Settings > Recording format**. 24-bit is the usual choice.

Record MIDI

Make an instrument track (`Cmd/Ctrl+Option/Alt+S`), choose its instrument, connect your MIDI keyboard, and press `R`. MIDI keyboards are found automatically.

- **Merge MIDI Recordings** (Record menu) - MIDI recorded over a MIDI region is added to it instead of making a new region.
- **Quantize** afterwards with `Q`, or in the piano roll.

Helpers

- **Metronome** (`K`) and **Count-in** (`Shift+K`, one bar of click before recording).
- **Autopunch** (`Cmd/Ctrl+Option/Alt+I`) - records only between the punch locators: the red bar under the cycle in the ruler. Drag its ends to set where recording starts and stops. Play from before it; recording drops in and out by itself.
- **Replace** - a new recording replaces the parts of regions it covers. Off, new recordings go on top (and become takes).
- **Stop Recording on Clip** - if an input clips (hits 0 dB), recording stops by itself, keeping what came before.

Takes and comping

Record over the same place several times - or turn on **Cycle** (`C`) and keep playing: each pass becomes a new **take**, and they're kept together in a **take folder**.

- Click the take folder's **name bar** for its menu: choose which take plays, **Open Take Folder**, **Flatten Comp**, **Clear Comp** or **Keep only this take**.
- With the folder open, each take is a lane. **Swipe** across the best parts of each take with the mouse - the parts you swipe are lit and play together as one performance (a *comp*). This is called *Quick Swipe comping*.
- When you're happy, **Flatten Comp** turns it into ordinary regions.

Markers

- `Option/Alt+'` makes a marker at the playhead; double-click the marker track to make one anywhere. Double-click a marker to rename it.
- `Option/Alt+.` and `Option/Alt+,` jump to the next and previous marker.
- Use them for Verse, Chorus, Bridge - then jump between parts while you work.

Tracks and Regions

Kinds of track

- **Audio track** (`Cmd/Ctrl+Option/Alt+A`) - recordings and audio files.
- **Software instrument track** (`Cmd/Ctrl+Option/Alt+S`) - MIDI regions played by an instrument: Tech Synth, Nemix, Drum Kit, Sampler or a plugin instrument. Choose the instrument with the instrument key on the track header (it shows the instrument's name) or the input slot of its mixer strip.
- **Aux bus** - receives sends from other tracks (for a shared reverb or delay) or is the output of a group of tracks.
- **VCA** (Track > New VCA) - a fader that moves the selected tracks together, keeping their balance. It carries no sound itself.
- **Drummer, Bass Player and Keyboard Player** tracks - session players that play along with your song (see *Session Players and Generate*).

Working with tracks

- **Select** a track by clicking its header; `Up` and `Down` select the track above or below.
- **Rename** - double-click the name.
- **Mute** (`M`), **Solo** (`S`), **Record arm**, **Input monitoring** and **Automation** (`A`) - the M, S, R, I and A keys on the header.
- **Reorder** - drag a header up or down.
- **Colour** - **Track > Track Colours...** (`Option/Alt+C`).
- **Duplicate** - **Track > New Track With Duplicate Settings** (`Cmd/Ctrl+D`): the same instrument, fader, inserts and sends, without the regions.
- **Hide** - **Track > Hide Track** hides tracks you're not working on (they still play). `H` shows or hides the hidden ones.
- **Delete** - `Cmd/Ctrl+Backspace`.
- **MIDI controller** - right-click a track > **Learn MIDI Controller**, then move a knob or fader on your controller. **Controller Assignments...** lists them all.

Track stacks and groups

Select several tracks and choose **Track > Create Track Stack...** (`Cmd/Ctrl+Shift+D`):

- **Folder stack** - one fader, mute and solo on top for all of them. Fold it up to tidy the window.
- **Summing stack** - they play into one aux bus, so you can add effects to all of them together (a drum bus, a vocal bus).

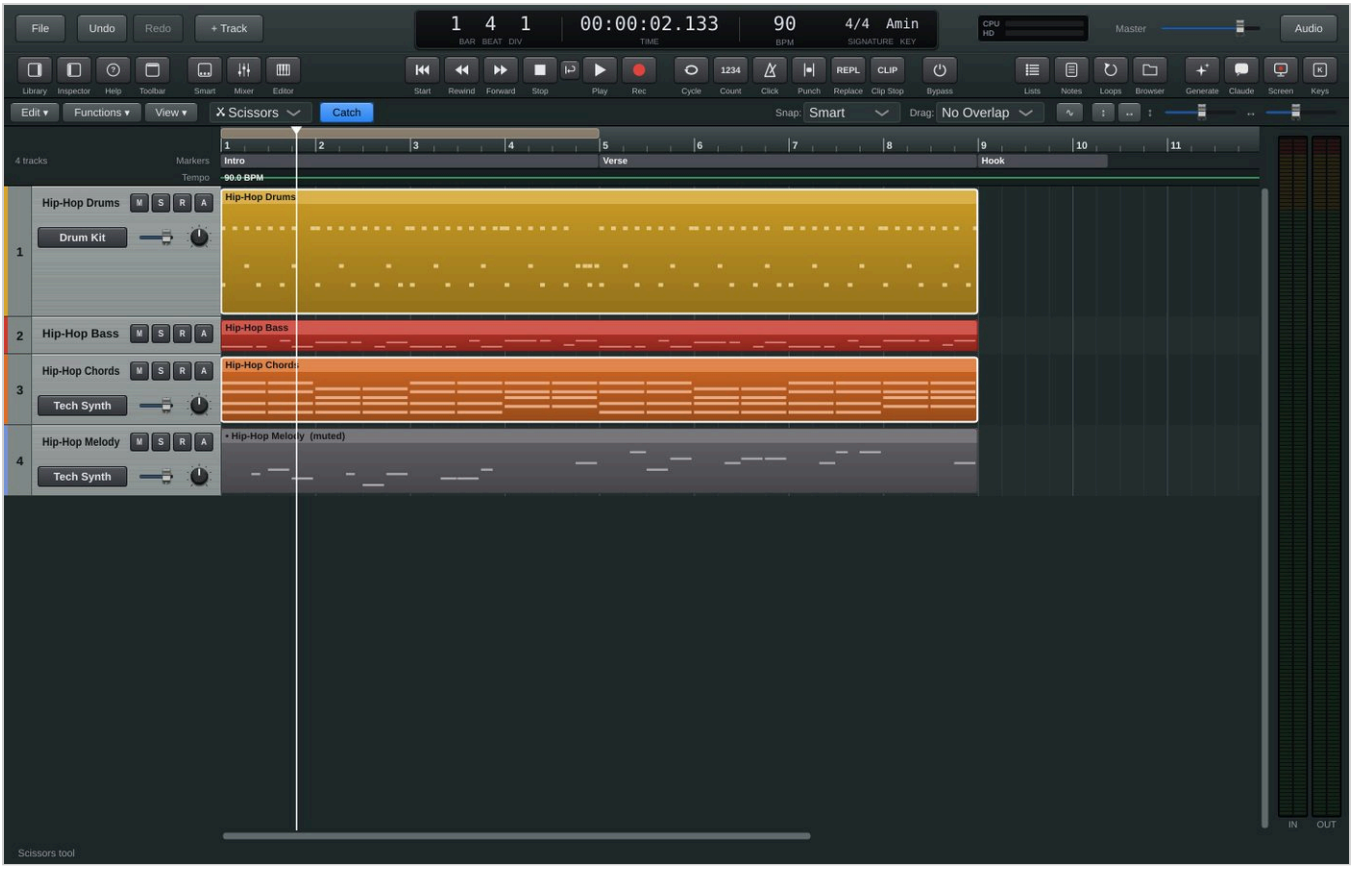
Mix > Group Settings... names groups and chooses what each one links: volume, mute, solo, record. Put tracks in a group from the mixer strip or the Inspector.

Freeze Track (Track menu) renders a track with its instrument and plugins to audio, to save processing. Choose it again to unfreeze and edit.

Regions

A **region** is a piece of audio or MIDI on a track.

- **Move** - drag it. It lands on the grid (see **Snap**); hold `Cmd` (Ctrl on Windows) while dragging to move freely.
- **Copy** - `Option/Alt`-drag, or `Cmd/Ctrl+C` and `Cmd/Ctrl+V` (pastes at the playhead on the selected track).
- **Shorten or lengthen** - drag its left or right edge. `Option/Alt`-drag an edge of an audio region to **time-stretch** it instead.
- **Loop** - `L`, or drag the upper right corner: the region repeats up to where you drag.
- **Repeat** - `Cmd/Ctrl+R` puts a copy right after it.
- **Split** - `Cmd/Ctrl+T` at the playhead, or click with the Scissors.
- **Join** - `Cmd/Ctrl+J` joins the selected regions on a track into one. **Consolidate Clips** (`Cmd/Ctrl+Shift+J`) makes selected audio regions one new file.
- **Nudge** - `Option/Alt+Left/Right` moves by a 16th.
- **Trim to playhead** - `Option/Alt+[` and `Option/Alt+]`.
- **Mute a region** - **Edit > Mute Regions**, or click with the Mute tool.
- **Delete** - `Backspace` or `Delete`.
- **New MIDI region** - double-click an empty spot on an instrument track, or draw with the Pencil.



Fades and clip gain

- **Fades** - drag the small handle at the **top corner** of an audio region (start or end) to fade in or out. The Fade tool does the same anywhere. **Edit > Remove Fades** takes them off.
- **Clip gain** - each audio region has a gain handle: drag it up or down to make the region louder or quieter. Double-click it for 0 dB. **Edit > Region Gain...** sets an exact amount.

Editing a whole section

Set the cycle (drag in the top half of the ruler) over a section, then:

- **Cut Section Between Locators** - removes that stretch of the song from every track and closes the gap.
- **Insert Silence Between Locators** - opens a gap the length of the cycle.
- **Repeat Section Between Locators** - plays the section twice (a copy right after it).
- **Split Regions by Locators** - splits every region at the cycle's start and end.
- `Cmd/Ctrl+U` sets the cycle to the selected region.

The tools

Press `T` for the tool menu, or right-click an empty space in the tracks area.

- **Pointer** - select, move, trim and loop regions.
- **Pencil** - draws a new MIDI region on an instrument track (drag for its length).
- **Eraser** - deletes the region you click.
- **Text** - click a region to rename it.
- **Scissors** - splits a region where you click.
- **Glue** - joins regions.
- **Solo** - hold the mouse on a region to hear its track alone from there.
- **Mute** - mutes or unmutes a region.
- **Zoom** - click or drag to zoom in; `Option/Alt`-click zooms out.
- **Fade** - drag at a region's start or end to make a fade; `Option/Alt`-click removes it.
- **Marquee** - select a stretch of time across tracks; `Delete` removes it, `Cmd/Ctrl+T` splits at its edges.

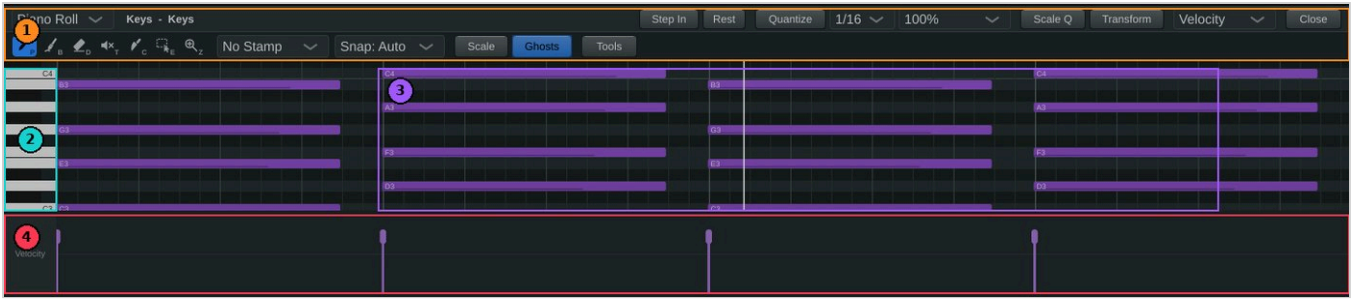
Snap and Drag modes

Snap (`N` turns it on and off): Smart (follows the zoom), Bar, Beat, Division or Off.

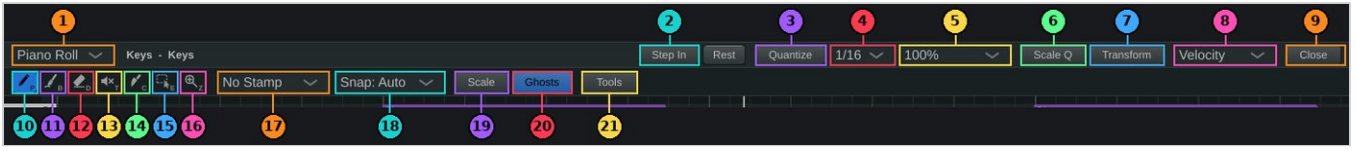
Drag: *Overlap* (regions can lie on top of each other), *No Overlap* (a moved region shortens what it covers), *X-Fade* (they crossfade where they overlap), *Shuffle R* and *Shuffle L* (regions close up the gaps, like beads on a string).

Piano Roll and MIDI

Double-click a MIDI region, or select it and press `P`, to open the **piano roll** under the tracks. `Cmd/Ctrl+4` opens it in a window of its own. Notes are bars: up and down is pitch (the keyboard on the left), left to right is time, and the length of a bar is how long the note plays.



1. **The two bars** - all the piano roll's controls (next picture).
2. **Keyboard** - click a key to hear the note.
3. **Notes** - drag a note to move it, drag its right end to make it longer or shorter.
4. **Velocity lane** - how hard each note is played (taller = louder). Drag a stalk up or down. Choose another lane in the bar: pitch bend, modulation and other controllers.

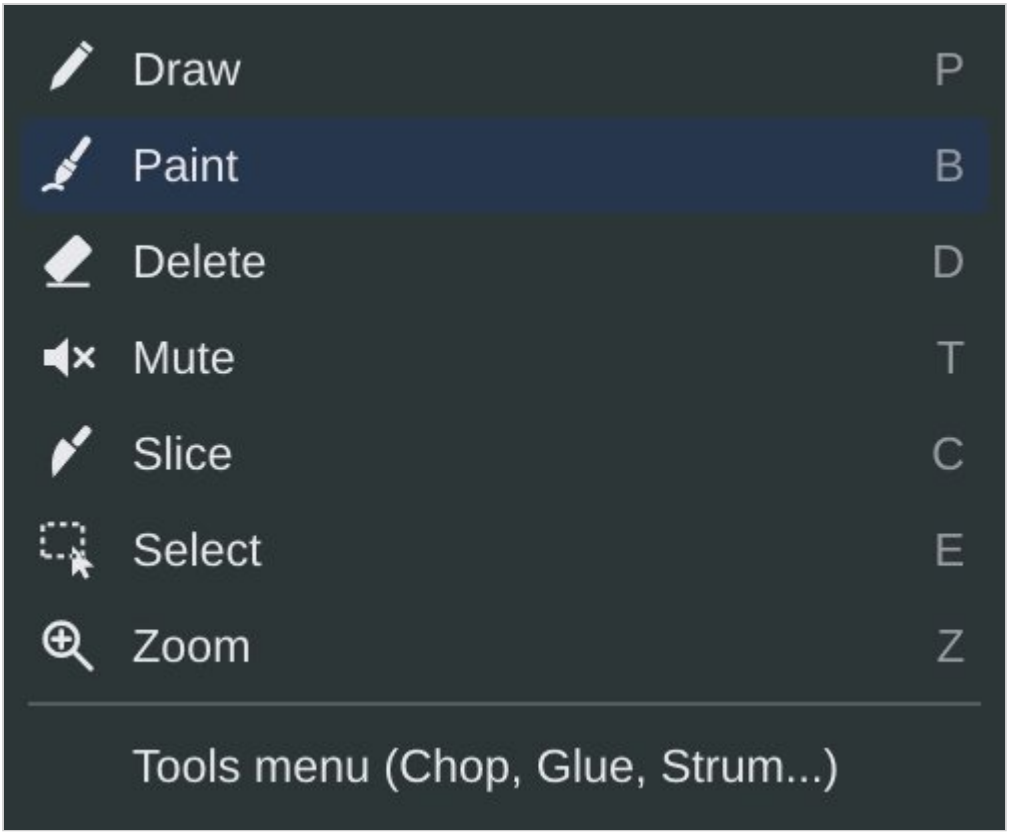


1. **View** - the region as a Piano Roll, as notation (**Score**) or as a grid of steps (**Step Editor**).
2. **Step In** - step input: play notes on your MIDI keyboard to enter them one grid step after another, starting at the playhead (a chord goes in together). **Rest** moves on without a note.
3. **Quantize** - snaps note starts to the grid (`Q`): the selected notes, or all of them.
4. **Quantize grid** - 1/4, 1/8, 1/16, triplets and more.
5. **Quantize strength** - how far notes move toward the grid (100% = exactly on it), and swing.
6. **Scale Q** - Scale Quantize: moves notes outside the song's key onto the nearest note of its scale.
7. **Transform** - MIDI Transform: humanize, fixed velocity, crescendo, legato, double or half speed, transpose, reverse.
8. **Lane** - what the lane under the notes shows and edits (velocity, controllers, pitch bend).
9. **Close** the piano roll.
10. **Draw** tool (`P`) - click to put in a note; it remembers the last note's length.
11. **Paint** tool (`B`) - drag to lay notes along the grid.
12. **Delete** tool (`D`) - click notes to delete them.
13. **Mute** tool - click notes to mute them.
14. **Slice** tool (`C`) - cut notes where you click.
15. **Select** tool (`E`) - drag a box round notes.

16. **Zoom** tool (`Z`) - click to zoom in; `Option/Alt`-click zooms out.
17. **Stamp** - Draw and Paint put in a whole chord (major, minor, 7th...) with the clicked note as its root.
18. **Snap** - where notes go when you draw, move and resize them (*Auto* follows the zoom). Hold `Cmd` while dragging for no snap.
19. **Scale** - highlights the rows of the song's key; the others are darker and the root is marked.
20. **Ghosts** - shows the other regions' notes faintly behind these, to write parts that fit.
21. **Tools** - the note tools menu (see below).

The tool menu

Press `T` in the piano roll for a menu of every tool with its icon and key; the lit one is in use. Press `T` twice for Mute. Right-click (or right-drag) deletes notes whatever the tool.



Editing notes

- **Move** - drag. **Up/Down** arrows move a semitone (`Shift` : an octave), **Left/Right** a grid step.
- **Copy** - `Option/Alt`-drag, or `Cmd/Ctrl+C` / `Cmd/Ctrl+V` (pastes at the playhead). `Cmd/Ctrl+D` duplicates.
- **Select all** - `Cmd/Ctrl+A` ; `Esc` deselects.
- **Zoom** - `Cmd`-scroll zooms in time; `Option`-scroll makes rows taller or shorter.

Note tools

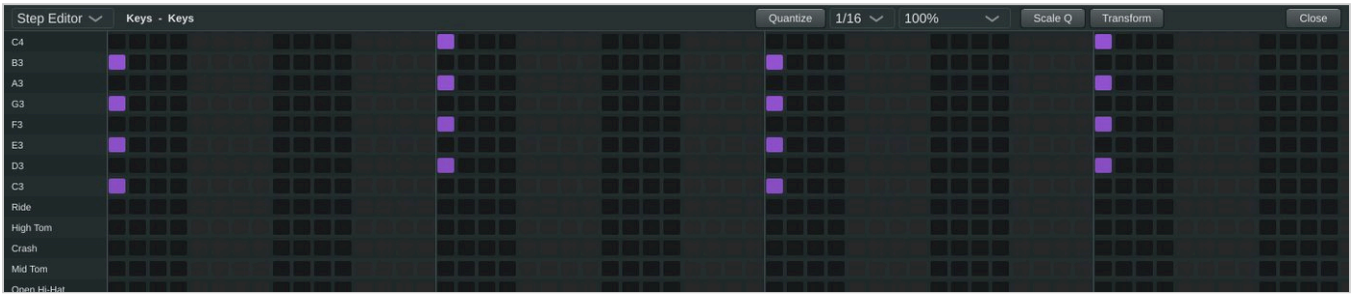
From the **Tools** menu, on the selected notes (or all of them). Each has a key:

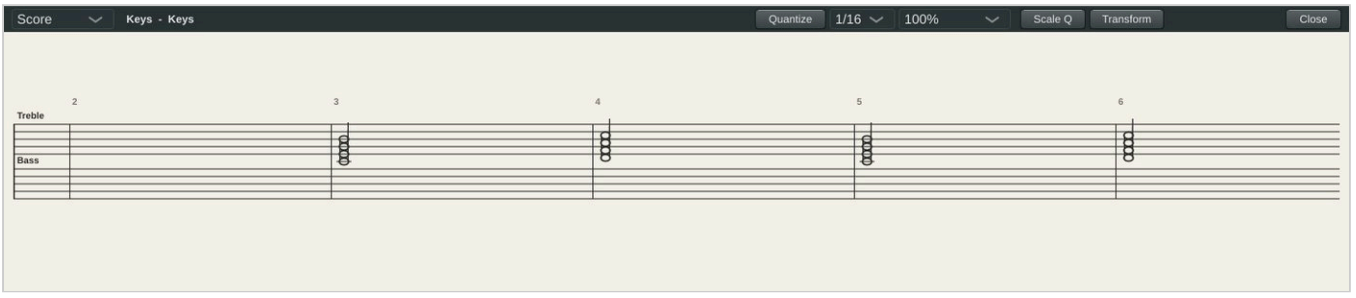
- **Quick Legato** (`Cmd/Ctrl+L`) - each note lasts until the next.
- **Chop** (`Option/Alt+U`) - cuts notes into pieces at the quantize grid (rolls, trap hi-hats).
- **Glue** (`Cmd/Ctrl+G`) - joins notes of the same pitch.
- **Strum...** (`Option/Alt+S`) - spreads a chord's notes out like a guitar strum.
- **Arpeggiate...** (`Option/Alt+A`) - turns chords into arpeggios.
- **Flam** (`Option/Alt+F`) - a quick double hit.
- **Randomize...** (`Option/Alt+R`) - random notes in a range, or small random changes for a human feel.
- **Scale Levels...** (`Option/Alt+X`) - velocity up, down, flattened or ramped.
- **Limit...** (`Option/Alt+K`) - folds notes into a range of pitches.
- **Claw Machine...** (`Option/Alt+C`) - thins notes out (keeps every second, third...).
- **Articulate...** (`Option/Alt+L`) - note lengths: staccato to legato.
- **Mute / Unmute** (`Option/Alt+M`), **Flip Horizontally** (`Option/Alt+Y`, reverse) and **Flip Vertically** (`Shift+Option/Alt+Y`, invert).

Press `?` in the piano roll for a box with every piano roll key. The full list is also in *More > Piano Roll Keys*.

Step Editor and Score

The **Step Editor** shows the region as a grid of steps - one row per drum sound or note - like a drum machine: click a step to put in a hit. The **Score** shows it as notation.





Event List

View > Event List (`Cmd/Ctrl+Option/Alt+D`) lists the selected region's notes and controllers with their exact position, pitch, velocity and length. Type in a cell to change it.

Quantize from the tracks area

Select MIDI regions and press `Q` to quantize them to 16th notes without opening the piano roll.

Audio Editing, Flex and Pitch

The audio editor

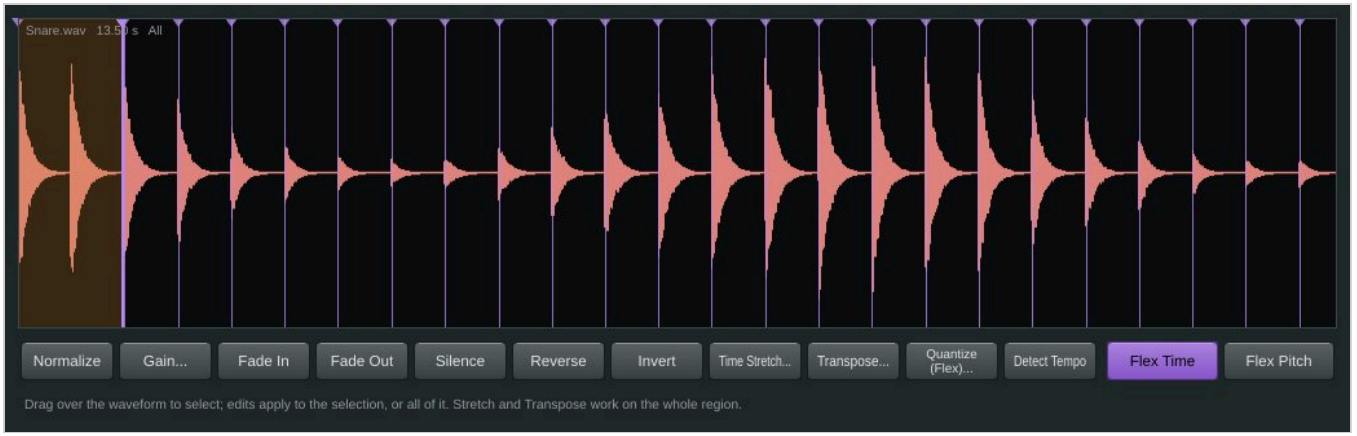
Double-click an audio region (or select it and press `P`) to open the **audio editor** under the tracks. Drag over the waveform to select a part; the buttons work on the selection, or on the whole region if nothing is selected.



- **Normalize** - turns up so the loudest peak is just under full scale.
- **Gain...** - turns the selection up or down by a number of dB.
- **Fade In** and **Fade Out**.
- **Silence** - makes the selection silent.
- **Reverse** - plays the selection backwards.
- **Invert** - flips the polarity (phase).
- **Time Stretch...** - makes the region longer or shorter without changing the pitch (200% = twice as long).
- **Transpose...** - changes the pitch without changing the length.
- **Quantize (Flex)...** - moves every hit onto the grid; the audio between is stretched to fit.
- **Detect Tempo** - finds the tempo this audio was played at, and sets the project to it if you like.

Flex Time

Click **Flex Time** in the audio editor. Markers appear on the hits. Drag a marker sideways to move that moment; the audio on either side stretches to fit. Double-click to add a marker; right-click one to remove it. Use it to tighten a drum or bass part without cutting it up.



Flex Pitch - the pitch editor

Click **Flex Pitch** to see a vocal or solo instrument as notes (blobs) at their pitch, as in a pitch-correction program. Each note shows its name and how many cents it is off.



- **Pointer** - select notes; drag them up or down to change the pitch (`Option/Alt`: in cents) or sideways to move them in time.
- **Scissors** - click a note to split it there. **Glue** - join notes into one.
- **Time Quantize** - choose a grid and click **Q** to move notes onto it, by the **Strength**.
- **Scale Quantize** - choose a key and scale (*Song* = the project's key) and click **Q**: each note goes to the nearest note of the scale.
- **Keep in Scale** - notes stay on the scale's notes whatever you do to them.
- **Pitch Correction** - how far the notes go onto the nearest semitone (0% = as sung, 100% = exactly in tune).
- **Gain** - the level of the selected notes, from -30 to +12 dB.
- **Pitch Drift** - flattens each note's slow drift (scoops and sags); 100% leaves none.
- **Vibrato** - 0% none, 100% as sung, 200% twice as wide.
- **Formant Shift** - the voice's character darker or brighter (in semitones) while the pitch stays.

All of these work on the selected notes, or on every note when none is selected.

AudioSuite - process regions offline

Edit > AudioSuite..., or the **AudioSuite** menu: click **Choose an effect...** (built-in or a plugin), set it, click **Preview** to hear it, choose **Clip** or **Whole File**, and click **Render**. The selected regions are processed into new audio files. Use it to fix one word, reverse a cymbal, or print an effect to save CPU.



Smart tools

- **Edit > Align Audio to Guide...** - lines up the timing of the selected regions with a guide region (the top one): doubled vocals and backing vocals that match the lead exactly.
- **Edit > Split to Stems...** - splits a mixed song into vocals, drums, bass and the rest (or six stems, adding guitar and piano), each on a new track. Great for remixes, practice and sampling.
- **Edit > Detect Song Key...** - listens to the song (or the selected regions) and finds its key, to set as the project's key.
- **Track > Bounce in Place** - the track rendered to a new audio track.
- **Track > Freeze Track** - the track rendered behind the scenes to save CPU; unfreeze to edit again.

Automation

Automation makes a control move by itself while the song plays: the vocal fader up in the chorus, a filter opening over a build, a delay throw on one word.

Showing automation

- Press `A`, or click **A** on the track header, to show the selected track's automation lane. **Right-click A** to choose what to automate: volume, pan, mute, sends, or any control of the track's plugins and instrument.
- Or **right-click any knob** - on the mixer, a plugin or an instrument - and choose **Show Automation**.

Drawing it

- **Click** on the line to make a point; **drag** points to move them.
- **Double-click** a point to remove it.
- With no automation, a dashed line shows the control's current value.
- Right-click a control > **Create Automation Point at Playhead**, or **Delete Automation**.

Recording it (automation modes)

Each track has a mode - on its automation key, the mixer strip, or right-click a control > **Automation Mode**:

- **Off** - the automation is ignored.
- **Read** - the controls follow the automation (the usual mode).
- **Touch** - while you hold a control, your moves are recorded; let go and it returns to the automation.
- **Latch** - like Touch, but after you let go the control stays where you left it until playback stops.
- **Write** - records over everything from where you start. After one pass it turns into Latch, so the next play doesn't erase it.

Choose Touch or Latch, press play, and move the fader or knob with the mouse or your MIDI controller. Then set it back to Read.

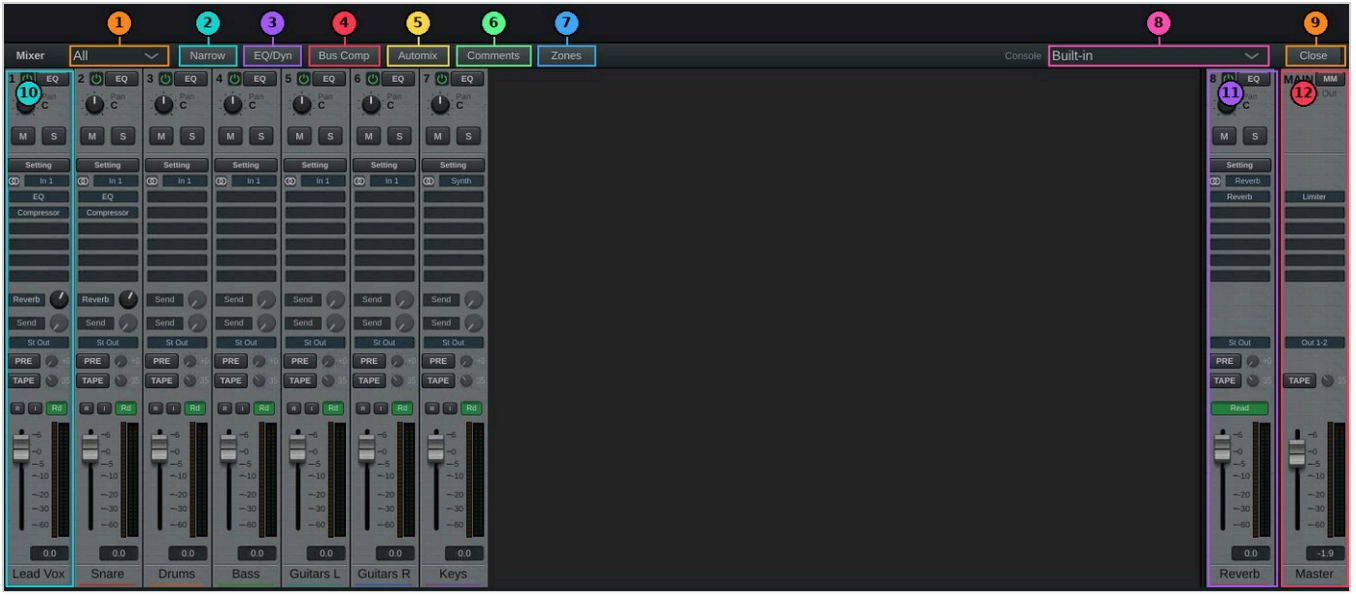
Tips

- **Smart Controls** and **macros** in Nemix can be automated too: one knob that moves several things at once.
- To change the whole level of an automated track, use **clip gain** on its regions or a **Gain** plugin, so the automation stays as it is.

Mix

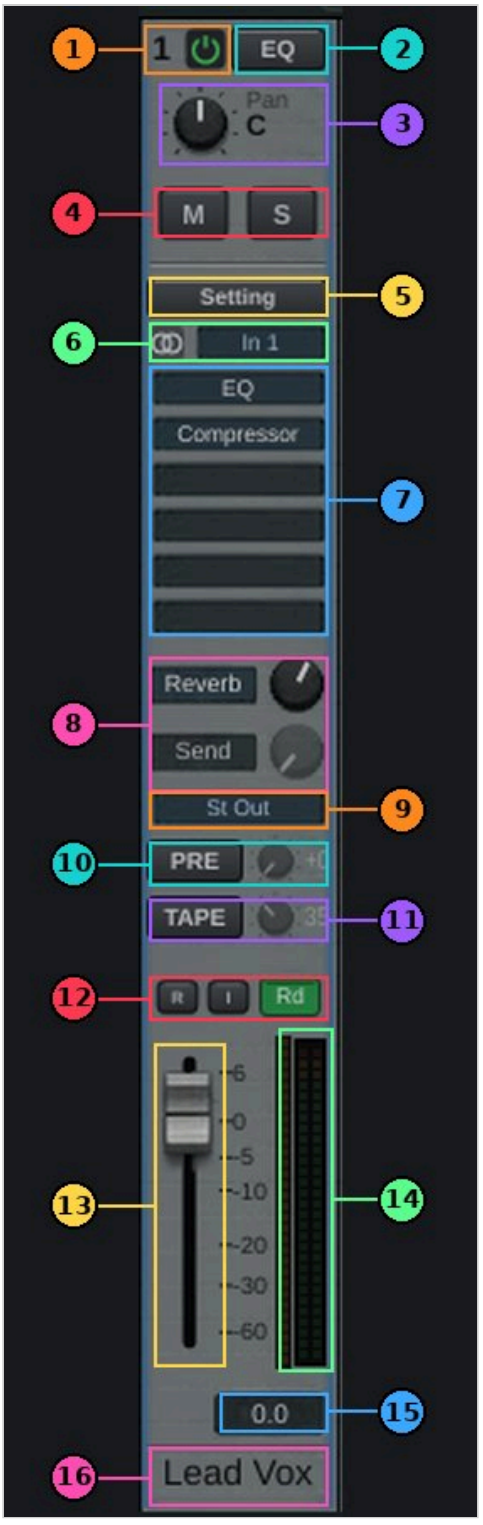
The Mixer

Press `X` to show the mixer under the tracks, or `Cmd/Ctrl+2` for the mixer in its own window (resize it or put it on a second screen). Each track has a **channel strip**; aux buses are to the right, and the **master** is at the far right.



1. **Which strips to show** - every strip, or only one kind (audio, instruments, buses...).
2. **Narrow** - thinner strips, to see more tracks.
3. **EQ/Dyn** - small pictures of each strip's EQ and dynamics.
4. **Bus Comp** - the bus compressor on the master (see *Mixing Tools*).
5. **Automix** - a balanced starting mix (see *Mixing Tools*).
6. **Comments** - a notes field under each strip.
7. **Zones** - put the buses and summing channels on the left or right side of the mixer, where they stay while the tracks scroll (right-click a channel's name for its own zone).
8. **Console** - the channel strip on every channel: the built-in one, or a channel-strip plugin you own.
9. **Close** the mixer.
10. **A track's channel strip** - part by part in the next picture.
11. **Aux (bus) strip** - receives sends; put a reverb or delay here.
12. **Master strip** - everything you hear. Its **MM** key opens the MultiMeter.

A channel strip, part by part



1. **Track number and On/Off** - switch the whole channel off.
2. **EQ** - the channel strip's EQ and dynamics, lit when they're working. Click to open the full channel strip.
3. **Pan** - left and right. Drag up/down; double-click for centre.
4. **Mute and Solo**.
5. **Setting** - copy, paste or reset this channel strip; save it as a patch or load one.
6. **Input** - what the track records or plays (an interface input, or the instrument). The round key beside it switches mono (one circle) and stereo (two).

7. **Inserts** - effects on this track, played top to bottom. Click an empty slot to add an effect, or click a used slot to open it. Right-click a slot to bypass or remove it.
8. **Sends** - send some of the track to an aux bus (for a shared reverb or delay). Click to choose the bus; the knob sets how much (`Option/Alt`-click or double-click for -20 dB).
9. **Output** - where the track goes: the Stereo Out (St Out), a bus, or other outputs of your interface (Output 3-4 and so on).
10. **Preamp (PRE)** - a class-A transformer console preamp. Its gain (0 to +30 dB) turns the channel up, and the harder it's driven the more it colours: thicker lows, warm harmonics, a softer top. Pull the fader down to taste.
11. **Tape** - runs the channel through an analog tape machine: saturation, the low-end bump, the softened top. The knob is the drive (double-click for 35%); right-click TAPE for the tape speed.
12. **Record, Input monitor, Automation mode** - R, I, and the mode (Read, Touch, Latch, Write, Off).
13. **Fader** - the track's level. `Option/Alt`-click for 0 dB.
14. **Level meter** - with the highest peak held. Red means clipping.
15. **Fader level** - in dB.
16. **Track name and colour**.

Your interface's preamps

With an interface whose preamps are set by its maker's control app, and that app installed and running, an audio track's strip also shows the **interface's preamp gain** for its input - drag it up or down; right-click for the high-pass filter, the pad and Auto Gain - and a **48 V** key for phantom power (for condenser mics). It all happens on your computer; nothing goes over the network.

Setting up a reverb or delay bus

1. **Track > New Aux Bus** (or choose a new bus in any strip's **Send** slot).
2. On the bus, add **Studio Reverb** or **Space Designer** (reverb) or **Stereo Delay** (delay), with its mix at 100% wet.
3. On each track, turn up the send knob for that bus: more send, more reverb.

One reverb shared by many tracks sounds more natural, and uses far less CPU, than one on every track.

Signal flow

Mix > Signal Flow shows the selected track's path from its input to the main output, with every stage: input, preamp, channel strip, inserts, sends, fader, pan, tape, the bus or master.

Groups and VCAs

- **Groups** - link the faders, mutes, solos or record keys of several tracks (**Mix > Group Settings...**).

- **VCA** - one fader that moves several tracks, keeping their balance (**Track > New VCA**).
- **Exclusive Solo** (Mix menu) - soloing a track unsolos the others.

The Channel Strip

Every channel has a full console channel strip built in - filters, a four-band EQ, a compressor and a gate/expander - before its insert effects. Click the **EQ** key at the top of a mixer strip to open it.



Its parts, top to bottom

- **Analyzer** - the sound's spectrum with the EQ curve over it. Choose the view (bars or line), the resolution, the range (dB), the speed and how long peaks hold.
- **Input / Output** - the level going in and coming out. **EQ in** and **DYN in** switch the EQ and the dynamics in or out (to compare).
- **Filters** - **HPF** (high pass: removes rumble and low mud below the frequency; 80 Hz is good for vocals and guitars) and **LPF** (low pass: removes harsh highs).
- **EQ** - four bands:
- **HF** (high frequencies, a shelf or a **Bell**) - air and brightness.
- **HMF** (high mids, with frequency and **Q**, the width) - presence, harshness.
- **LMF** (low mids) - body, boxiness, mud.
- **LF** (low frequencies, a shelf or a **Bell**) - weight and boom.
- **Compressor** - **Comp** is the amount (turn up to compress more), **Ratio** (Auto, or a fixed ratio), **Release**, and **Fast** for a quicker attack. It evens out the level so the track sits steady in the mix.
- **Gate / Expander** - **Threshold** (below it, the sound is turned down), **Range** (by how much), **Gate Rel** (how fast it closes), and **Exp** for a gentler expander instead of a gate. Use it to cut noise and spill between notes or hits.
- **Meters** - **IN**, **GR** (gain reduction: how much the compressor and gate are turning down) and **OUT**.

How to use it

1. Start with the **HPF**: raise it until the sound gets thin, then back off a little.
2. **Cut before you boost**: find an ugly frequency by boosting a narrow band and sweeping, then cut it.
3. Add **Comp** until the GR meter moves 3 to 6 dB on the loud parts.
4. Use **EQ in / DYN in** to compare with and without - it should sound better, not just louder.

The console

Console at the top right of the mixer chooses the channel strip on every channel: **Built-in**, or a channel-strip plugin you own (any of your plugins). The same strip then appears on every track, as on a real console. **Mix > Reset Channels...** puts every strip back to a fresh start.

Smart Controls

Smart Controls (`B`) show eight knobs for the selected track's sound under the tracks: its volume, pan, instrument (cutoff, envelope...) and channel strip EQ. **Right-click a knob** to choose another parameter for it - any control of the track's plugins. **Default Mapping** puts them back.


Audio
Keys
M
S
R
A
7
Tech Synth
Keys
IN
OUT
Smart Controls: Snare
Default Mapping
Close
Strip High-pass
80.00 Hz
Strip LF
0.00 dB
Strip LMF
-5.00 dB
Strip HMF
0.00 dB
Strip HF
4.00 dB
Strip Dynamics
4.00
Pan
C
Volume
0.0 dB
No audio device open - click Audio to choose one

Using Effects and Plugins

Audio Tech Pro comes with 95 effects (all described in the **Plugins** tab), and it also runs your own **VST3**, **Audio Unit** (Mac) and **CLAP** plugins.

Adding an effect

Click an empty **insert slot** on a mixer strip (or in the Inspector, or beside the track header with **View > Show Inserts in Edit Window**). The **plugin picker** opens:

- **Search** - type part of a name.
- **Favorites** - click the star beside any plugin to add it here.
- **Maker** - the built-in effects ("Audio Tech Pro"), then your plugins by company.
- **Category** - EQ, Dynamics, Delay, Reverb, Modulation, Distortion, Filter, Amps and Pedals, Pitch, Imaging, Utility, Specialized, Metering, Multi Effects, Legacy.
- **Format** - VST3, AU or CLAP.

Effects play top to bottom. A common order on a vocal: EQ (cleanup) > compressor > de-esser > EQ (tone) > send to reverb.

The plugin window

Every effect opens in its own window, with a bar on top:



- **Pin** - the window stays above the main window while you work.
- **Bypass** - take the plugin out of the sound to compare (`Option/Alt`-click an insert slot does the same).
- **Print** - Print While Recording: the input is recorded through this plugin, so the recording keeps it.
- **< Preset >** - save the plugin's settings, load one, step through presets, or **Set as Default** so every new one starts that way.
- **Side Chain** - another track keys this plugin (its detector hears that track): a compressor ducking the bass under the kick, a gate opened by a drum, a vocoder's carrier.

Right-click any knob for its menu: Show Automation, Create Automation Point, Automation Mode, Set to Default, and Learn MIDI Controller. For third-party plugins, the bar has an **automation** key with every parameter, and a key for the last knob you touched.

Most built-in effects show **meters** (input, output and gain reduction) and an **analyzer** of the sound.

Your own plugins

- **Plug-in Manager...** (in the View and Mix menus) - scans for VST3, Audio Unit and CLAP plugins. Do this once after installing new plugins.
- **Mix > Plugin Formats in Menus** - show all formats, or only VST3 or only AU, so each plugin appears once.
- Plugin **instruments** are chosen from the instrument key on an instrument track's header (or its mixer strip's input slot) > **Plugins**.

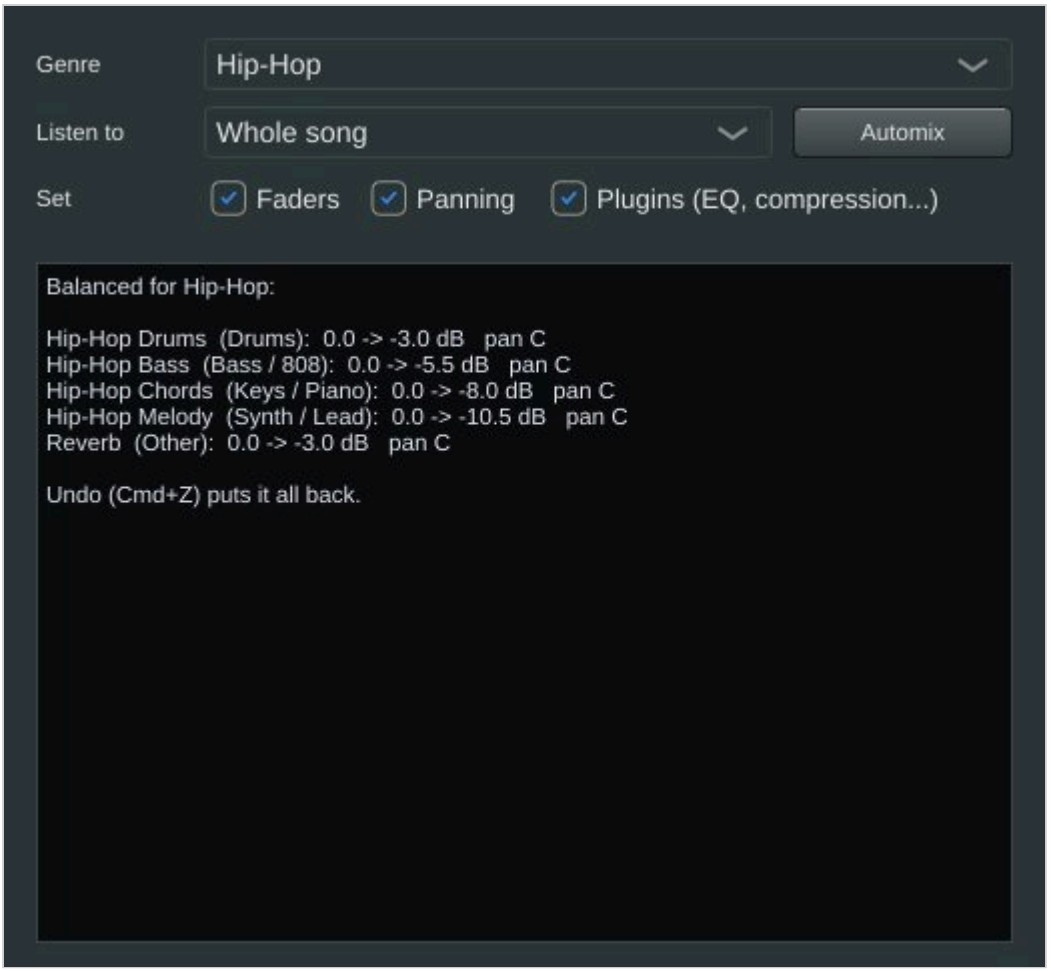
Saving CPU

- **Mix > Suspend Plug-ins When Silent** - plugins on a silent track stop working until sound comes again.
- **Freeze** heavy tracks (Track > Freeze Track), or use a **bus** for reverbs instead of one per track.
- **View > Performance...** shows which tracks use the most.

Mixing Tools

Automix

Mix > Automix... (or **Automix** on the mixer) listens to every track and sets a balanced starting mix for a genre - in one undo step.



- **Genre** - the style to balance for: how loud the kick, bass, vocal and keys sit against each other.
- **Listen to** - the whole song, or just the cycle (a chorus, say: quicker).
- **Faders** - each track's fader set to where its part sits in the genre.
- **Panning** - kick, snare, bass and lead vocal in the middle, doubled guitars left and right, backing vocals spread.
- **Plugins** - adds the plugins each part usually gets in the genre: EQ, compression set from its own level, de-essing, saturation on an 808, tuning on a lead vocal - ready to adjust.

Then fine-tune by ear. `Cmd/Ctrl+Z` puts it all back.

Bypass All (Compare)

The **Bypass** key in the control bar (**Mix > Bypass All**) lets you hear the mix with every channel's processing switched off - preamps, channel strips, inserts, tape - and the master's inserts, bus compressor and tape. Click again for it all back. Use it to check that your processing really makes it better.

The master bus compressor

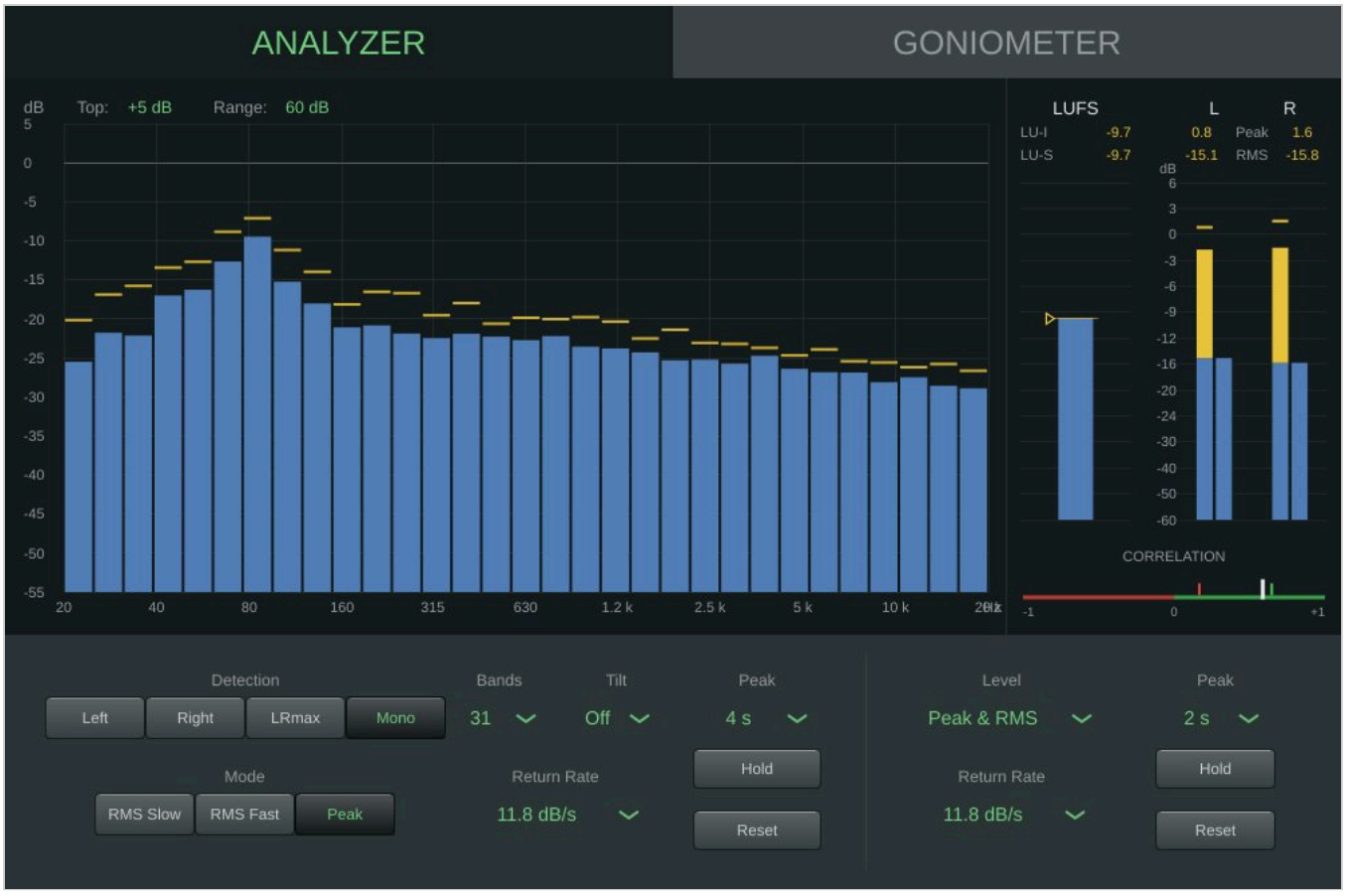
Bus Comp on the mixer opens the master bus compressor: a console-style stereo compressor that "glues" the mix together.



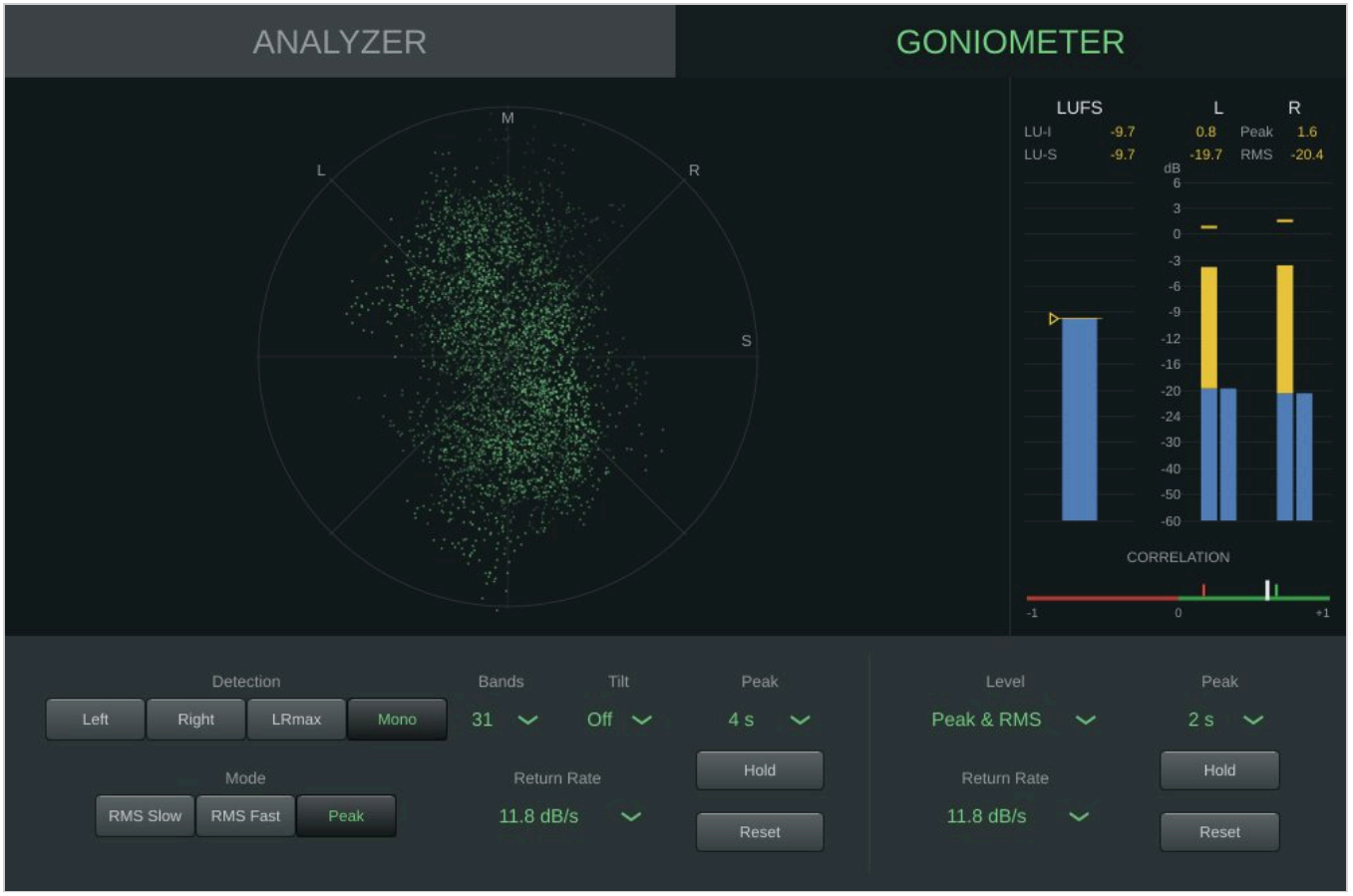
- **Threshold** - lower it until the needle moves 1 to 3 dB on the loud parts.
- **Make-up** - turns the result back up.
- **Attack** and **Release** - slower attack lets the drums punch through; **Auto** release follows the music.
- **Ratio** - 2:1 or 4:1 for glue.
- **Mix** - blends in the uncompressed mix (parallel compression).
- **Grit** and **COLOR** - the console's saturation, from warm weight to a pushed edge.
- **IN** - switches it in or out. You can also choose any compressor plugin you own instead.

MultiMeter

Mix > MultiMeter, or **MM** on the master strip: the master's levels, loudness and stereo image.



- **Analyzer** - 31 or 63 bands. **Tilt** lifts the bars above 1 kHz (4.5 dB/oct makes a balanced mix look flat). **Return Rate** and **Peak** set how fast bars fall and peak marks stay; **Hold** keeps them until **Reset**.
- **LUFS** - Momentary, Short-term and Integrated loudness; **L** and **R** levels as Peak, RMS or both.
- **Goniometer** - the stereo picture: a vertical line is mono, a wide cloud is wide.
- **Correlation** - +1 mono, 0 wide, below 0 out of phase (it will disappear on a mono speaker).



Reset Channels

Mix > Reset Channels... puts every channel (or the selected ones) and the master back to a fresh start: faders, pans, mute and solo, the channel strips, inserts and sends. Useful to start a mix again from scratch.

Signal flow, groups and more

- **Mix > Signal Flow** - the selected track's whole path, stage by stage.
- **Mix > Group Settings...** - what each group links.
- **Mix > Exclusive Solo** - one solo at a time.
- **View > Show Comments in Mixer** - a notes box under each channel.

Mixing tips

- Balance with **faders and pan first**, effects later.
- Check the mix **quietly** and in **mono** (the **Gain** plugin's Mono switch on the master).
- Leave **headroom**: the master should peak around -6 dB before mastering.
- Compare often with **Bypass All** and with a song you like.

Mastering and Exporting

Mastering with the Mastering Assistant

Mastering makes the finished mix sound polished and as loud as other songs.

1. On the **master strip**, add **Mastering Assistant** (Metering category) as the last insert.
2. Play the loudest part of the song (a chorus). It listens to the first 30 seconds you play after **Reanalyze**.
3. It sets a corrective **EQ** (Auto EQ, with how much of it), the **gain** to reach the loudness you choose, and a true peak limiter at -1 dBTP.
4. Choose a **character**: **Clean**, **Valve** (warm), **Punch** (tight, punchy) or **Transparent**.
5. Adjust **Excite** (upper-mid saturation, for presence), **Width** (stereo width above 150 Hz) and the custom **EQ** (low, mid, high).
6. Turn on **Loudness Compensation** to hear it at the original loudness, for a fair comparison.

The window shows the input and output levels, the EQ curve, the loudness against your target, the PLR (peak to loudness ratio - how much punch is left) and the stereo correlation.



Loudness targets

- Streaming (Spotify, Apple Music, YouTube): about **-14 LUFS** integrated, true peak below -1 dBTP.
- Loud genres (trap, EDM, reggaeton): often **-9 to -7 LUFS** - louder, but less punch.

You can also master by hand: **Linear Phase EQ** or **Match EQ**, then **Multipressor**, **Saturator**, and **Limiter** or **Adaptive Limiter** last. Check with the **Loudness Meter** or **MultiMeter**.

Exporting

- **File > Bounce Song to WAV...** (`Cmd/Ctrl+B`) - the song (or the cycle) to a WAV file, with a 2-second tail for reverbs.
- **File > Export All Tracks as Audio Files...** (`Cmd/Ctrl+Shift+E`) - stems: each track and bus to its own WAV, with its plugins, fader and pan.
- **File > Save a Copy As...** - the whole project with its audio, to send.
- **View > Record Screen...** - a video of the app with the mix's sound (see *Screen Recording*).

Instruments

Instruments

An **instrument track** plays its MIDI through an instrument. To choose it, click the **instrument key** on the track header (it shows the instrument's name) or the **input slot** at the top of its mixer strip. The menu offers:

- **Show synth controls / Show plugin window** - opens the current instrument.
- **Tech Synth** - the simple built-in synth.
- **Nemix (wavetable synth)** - the big synth with hundreds of sounds and the recorded instruments. See the next chapter.
- **Drum Kit** - a built-in drum kit.
- **Sampler...** - plays any audio file of yours across the keyboard.
- **Plugins** - your own instrument plugins (VST3, AU, CLAP), by maker.

Tech Synth

A classic one-oscillator synth: **Wave** (Saw, Square, Triangle, Sine), a low-pass filter (**Cutoff**, **Resonance**), an envelope (**Attack**, **Decay**, **Sustain**, **Release**) and **Volume**. Quick basses, leads and pads.

Drum Kit

A drum kit on the General MIDI drum notes (kick on note 36, snare on 38, closed hi-hat on 42 and so on), so any standard drum pattern plays on it. **Tune** (up or down 12 semitones), **Decay** (shorter or longer hits) and **Volume**. The Drummer and the MIDI generator write for these notes too.

Sampler

Choose an audio file (a note, a hit, a vocal chop) with **Choose Sound...**. It plays across the keyboard, pitched from the **Root Key** (the note that plays it as recorded), with **Attack**, **Release** and **Volume**.

The Library - patches

Press `Y` (or the **Library** key) for the Library: **patches** - a track's whole sound (instrument, channel strip, inserts and sends) saved to put on other tracks. Select a track, click a patch, and it's loaded. Save your own from the strip's **Setting** slot.

The Loop Browser

Press `0` for the Loop Browser. Choose a folder of loops and samples, click one to hear it, and drag it onto a track. Loops at another tempo can be fitted to the song's tempo.

Smart Controls

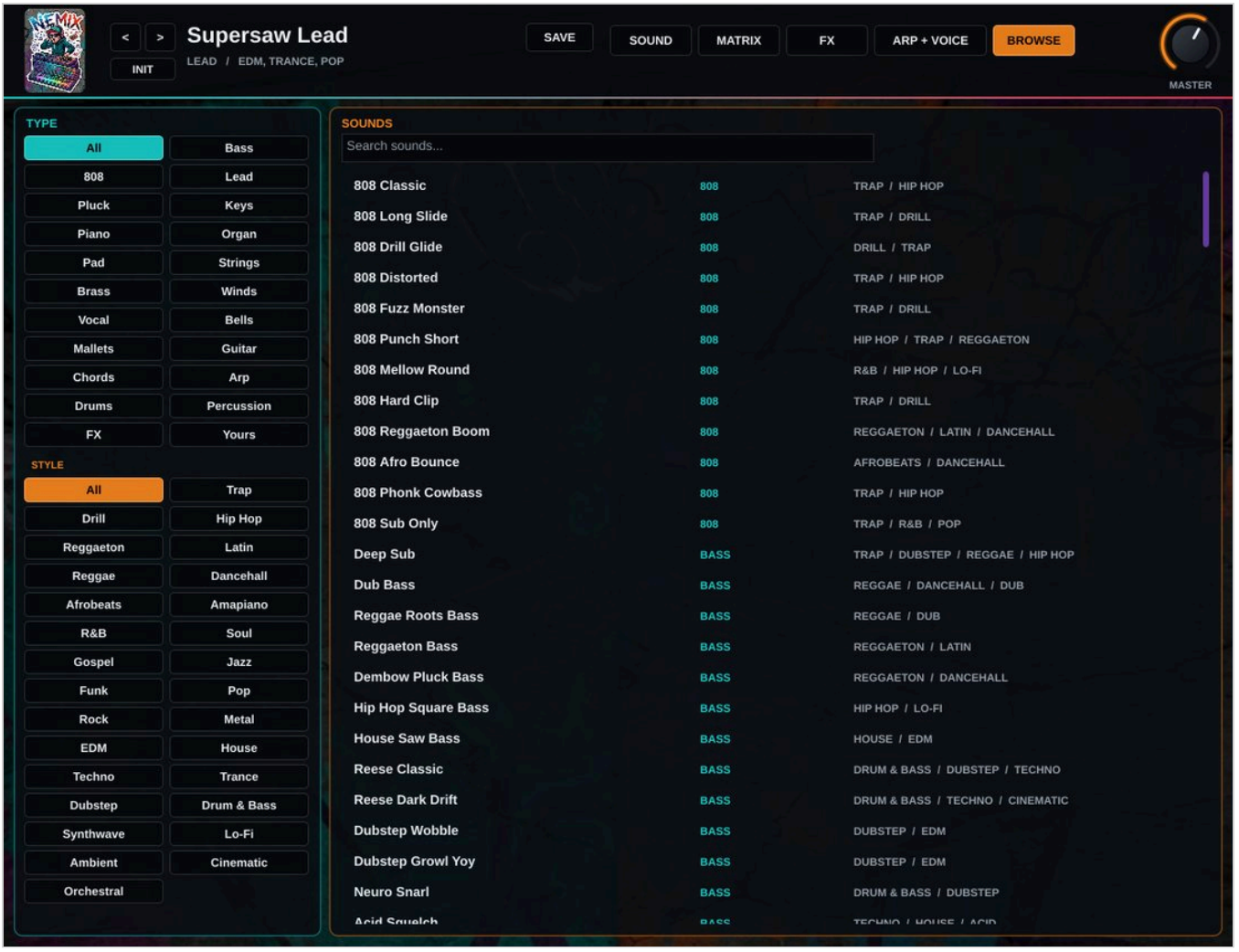
`B` shows eight knobs for the selected track's instrument and channel strip. Right-click a knob to give it any other parameter.

Nemix

Nemix is Audio Tech Pro's big synthesizer: three wavetable oscillators, a layer of **recorded instruments** (pianos, basses, guitars, drums, strings, brass, winds), a sub oscillator and noise, two filters, four envelopes, four LFOs, four macros, a 32-slot modulation matrix, an arpeggiator and eight built-in effects. It comes with hundreds of sounds for every genre - trap, drill, reggaeton, hip hop, R&B, reggae, afrobeats, jazz, rock, EDM, orchestra and more.

To use it, make an instrument track and choose **Nemix (wavetable synth)** from its instrument key. Its window opens with five pages: **SOUND**, **MATRIX**, **FX**, **ARP + VOICE** and **BROWSE**.

Choosing a sound - BROWSE



- **TYPE** - Bass, 808, Lead, Keys, Pluck, Piano, Organ, Pad, Strings, Brass, Winds, Vocal, Bells, Mallets, Guitar, Chords, Arp, Drums, Percussion, FX, and **Yours** (the sounds you saved).
- **STYLE** - the genre: Trap, Drill, Hip Hop, Reggaeton, Latin, Reggae, Dancehall, Afrobeats, Amapiano, R&B, Soul, Gospel, Jazz, Funk, Pop, Rock, Metal, EDM, House, Techno, Trance, Dubstep, Drum & Bass, Synthwave, Lo-Fi, Ambient, Cinematic, Orchestral.

- **Search** - type part of a name.
- Click a sound to load it. The **<** and **>** keys at the top step through sounds, so you can audition them while you play.

The full list of sounds is in the chapter *Nemix Sounds*.

The SOUND page



1. **Logo**.
2. **< >** - the previous and next sound.
3. **INIT** - a plain starting sound, to build your own from scratch.
4. **Sound name**, its type and styles.
5. **SAVE** - save your sound (it appears under **Yours** in BROWSE).
6. **Pages** - SOUND, MATRIX, FX, ARP + VOICE, BROWSE.
7. **MASTER** - Nemix's output volume.
8. **Oscillator A**, 9. **Oscillator B**, 10. **Oscillator C** - three wavetable oscillators (see below).
9. **SAMPLE / SUB / NOISE** - the recorded instruments, a sub oscillator and noise (see below).
10. **Filter 1** and 13. **Filter 2**.
11. **Macros** - four knobs that can move many controls at once.
12. **Envelopes** - ENV 1 to 4.

13. **LFOs** - LFO 1 to 4.
14. **Other modulation sources** - VELO (velocity), NOTE (the key played), WHEEL (mod wheel), AFTER (aftertouch), BEND (pitch bend), RAND (random per note).

The oscillators

Each oscillator plays a **wavetable**: a set of 32 waveforms you move through with **WT POS**. The picture shows the waveform; the menu chooses the table. There are 26: Basic Shapes, Pulse Width, Saw Harmonics, Square Harmonics, Harmonic Sweep, Vowels, FM Bells, Sync Sweep, Fold, Digital, Growl, Organ, Hollow, Bright Stack, Brass, Strings, Electric Piano, Choir, Spectral Noise, Phase Distortion, Reese, Bell Partials, Wobble, Sub Thump, Glass and Chip.

- **ON** - switches the oscillator on.
- **WT POS** - the position in the wavetable. Modulate it (with an LFO or envelope) for moving sounds.
- **UNISON** - up to 16 voices stacked; **DETUNE** spreads their pitch (the famous supersaw); **BLEND** balances the centre voice against the others; **WIDTH** spreads them across the stereo field.
- **OCTAVE**, **SEMI**, **FINE** - the pitch.
- **LEVEL** and **PAN**.
- **WARP** and **AMOUNT** - bends the waveform: Sync, Bend +, Bend -, PWM, Mirror, Quantize, FM, AM and Ring Mod (the last three use the next oscillator).
- **OUT** - which filter it goes to: Filter 1, Filter 2, Both, or Direct (no filter).

Recorded instruments, sub and noise

The **SAMPLE** tab plays real recordings - the reason the acoustic sounds sound real:



1. **ON** - switches the recorded layer on.

2. **Instrument** - choose from 34 recorded instruments, grouped by kind: Grand and Upright Piano, Rhodes and Electric Grand, six electric basses and an upright bass, nylon, steel, clean electric and hollow electric guitars, a rock drum kit, violins, violas, cellos (with pizzicato and tremolo), trumpet, French horn, trombone, tuba, flute, oboe, clarinet, bassoon, glockenspiel, marimba and harp.
3. **LEVEL**, 4. **PAN**, 5. **START** (start later in the recording, to soften the attack).
4. **SEMI** and **FINE** - tune it.
5. **Route** - *Direct* (straight out, as recorded) or through Filter 1, Filter 2 or both, so you can shape real instruments with the synth's filters and envelopes.

Recordings play with velocity layers (soft and hard notes really sound different) and round robins (repeated notes don't sound like a machine gun). The list of instruments and who recorded them is in *Recorded Instruments*.

The **SUB** tab adds a sub oscillator (Sine, Triangle, Saw, Square or Pulse, 0, -1 or -2 octaves) for weight. The **NOISE** tab adds noise (White, Pink, Vinyl, Metallic or Air) for breath, texture and crackle.

The filters

Filter 1 and **Filter 2** each have a type - Low Pass 12/24, Ladder, High Pass 12/24, Band Pass 12/24, Notch, Peak, Comb + and Comb - and Formant (vowel sounds) - with **CUTOFF**, **RESONANCE**, **DRIVE**, **KEY TRACK** (cutoff follows the notes), **MIX** and **PAN**. **Serial** runs Filter 1 into Filter 2; **Parallel** runs them side by side.

Envelopes and LFOs

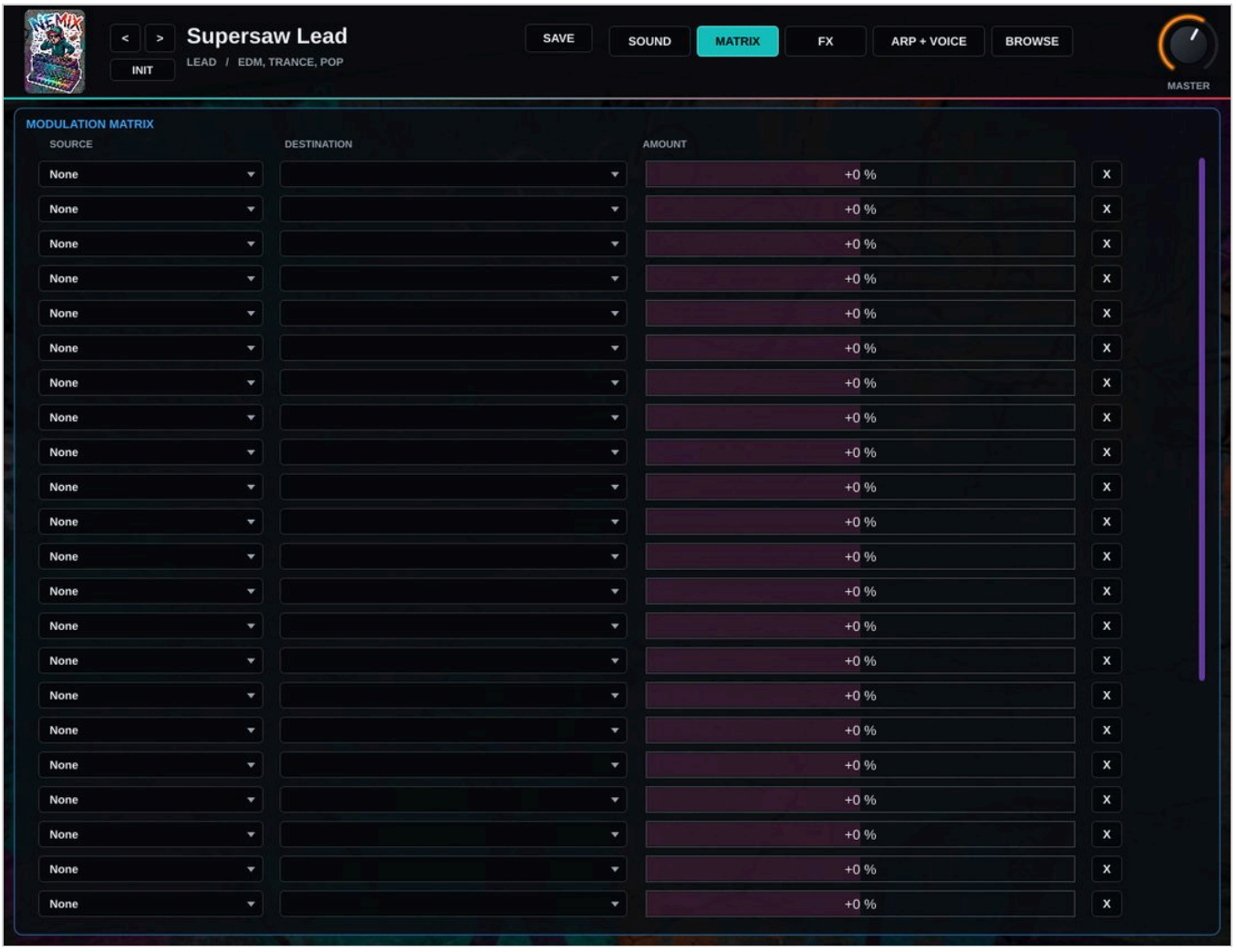
ENV 1 shapes each note's volume; ENV 2 to 4 are free to modulate anything. Each has **ATTACK**, **HOLD**, **DECAY**, **SUSTAIN**, **RELEASE** and **CURVE**.

The **LFOs** wobble controls in a loop. Each has a shape (Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random, Pluck, Ramp, Stairs), a rate (free, or in time with the song from 8 bars to 1/32), a mode (**Free**, **Retrigger** with each note, or **Envelope** - once only), **RATE**, **RISE** (fade in), **PHASE** and **SMOOTH**.

Modulation: drag and drop

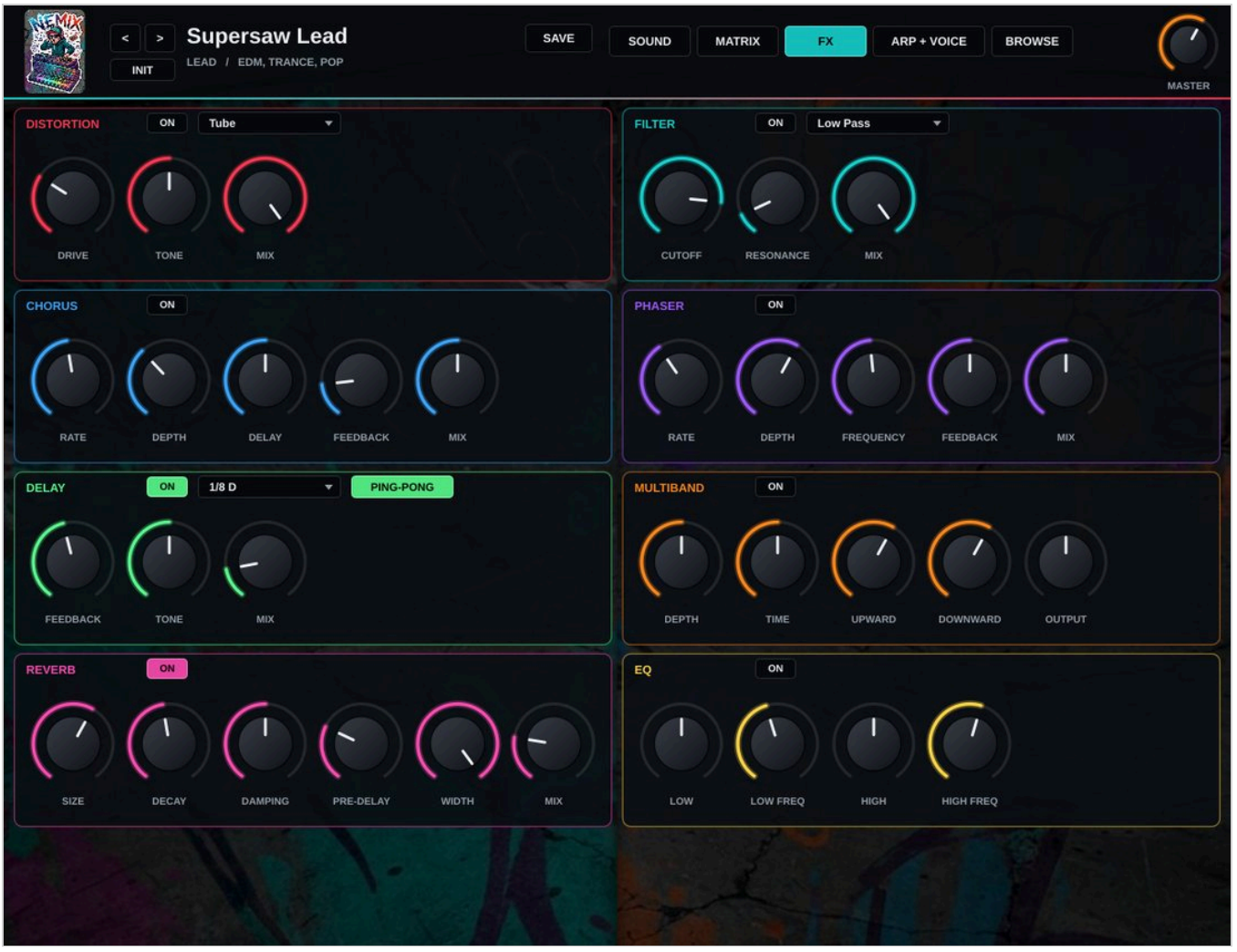
To make anything move, **drag a tile** - ENV 2, LFO 1, M1 (a macro), VELO, WHEEL... - by its dotted handle **onto any knob**. The knob shows its modulation as a coloured arc. **Right-click the knob** to set each modulation's amount (+100% to -100%) or remove it, or to reset the knob. Every connection also appears on the **MATRIX** page, where you can set the amount exactly.

The MATRIX page



32 rows of **SOURCE > DESTINATION > AMOUNT**. Choose a source (envelopes, LFOs, velocity, note, mod wheel, aftertouch, pitch bend, macros, random), a destination (almost any knob), and drag the amount (negative values move it the other way). **X** clears a row.

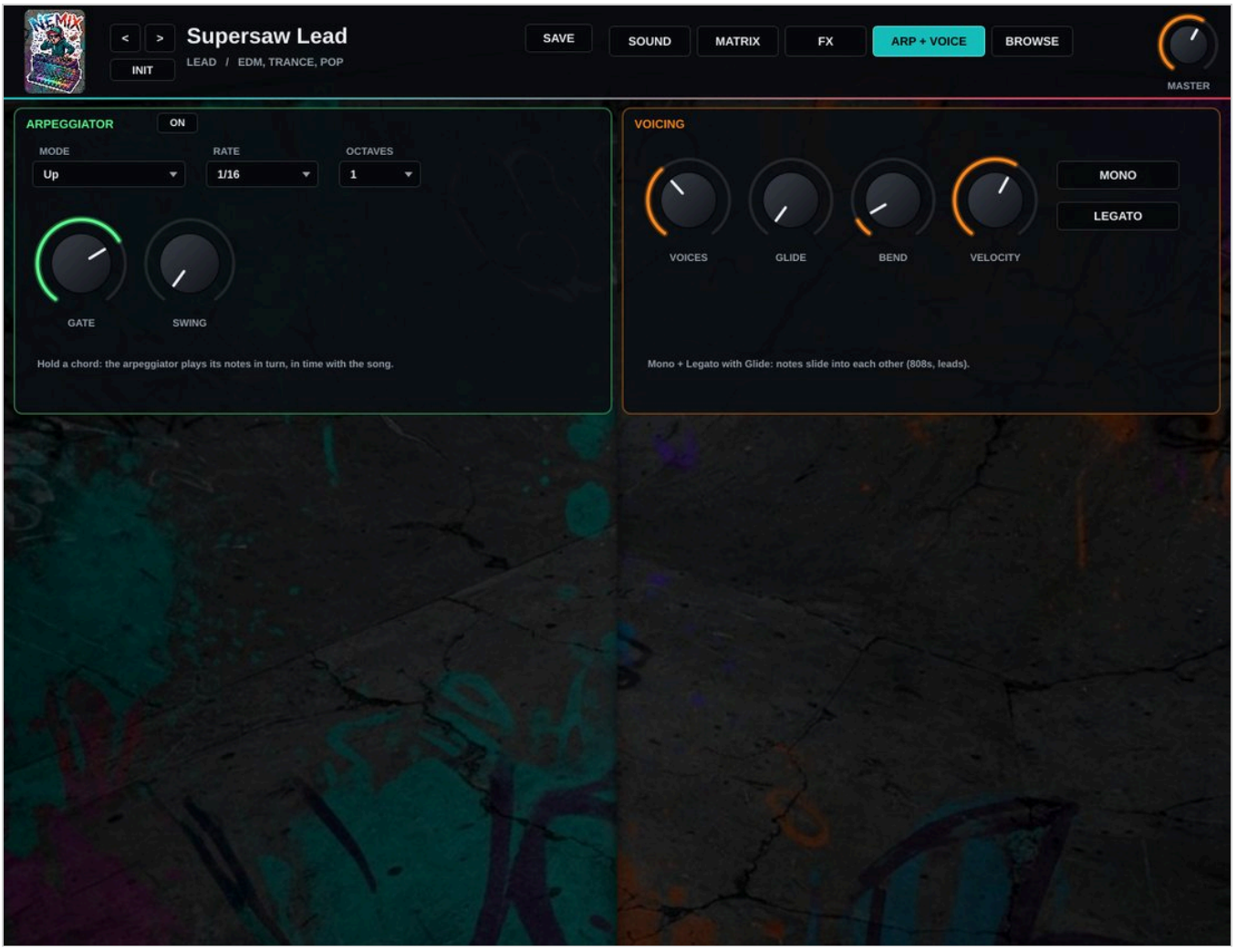
The FX page



Eight effects, each with **ON**:

- **DISTORTION** - Tube, Soft Clip, Hard Clip, Diode, Fold, Bitcrush, Downsample or Asymmetric; DRIVE, TONE, MIX.
- **FILTER** - low, high or band pass; CUTOFF, RESONANCE, MIX.
- **CHORUS** - RATE, DEPTH, DELAY, FEEDBACK, MIX.
- **PHASER** - RATE, DEPTH, FREQUENCY, FEEDBACK, MIX.
- **DELAY** - a time in note values (1/8 D, 1/4...), **PING-PONG** left and right; FEEDBACK, TONE, MIX.
- **MULTIBAND** - multiband compression for loud, upfront sounds: DEPTH, TIME, UPWARD, DOWNWARD, OUTPUT.
- **REVERB** - SIZE, DECAY, DAMPING, PRE-DELAY, WIDTH, MIX.
- **EQ** - LOW and HIGH shelves with their frequencies.

The ARP + VOICE page



ARPEGGIATOR - hold a chord and it plays the notes in turn, in time with the song. **MODE** (Up, Down, Up/Down, Random, As Played, Chord), **RATE**, **OCTAVES** (1 to 4), **GATE** (note length) and **SWING**.

VOICING - **VOICES** (how many notes at once), **GLIDE** (notes slide into each other), **BEND** (pitch bend range), **VELOCITY** (how much velocity changes the sound), **MONO** and **LEGATO**. Mono + Legato + Glide gives the sliding 808s and leads of trap and drill.

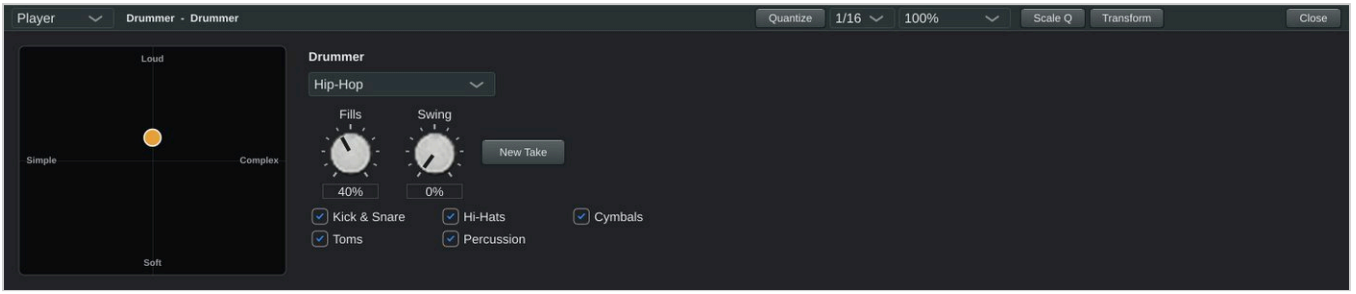
Recipes

- **Trap 808**: BROWSE > 808 > "808 Long Slide". Play overlapping notes for slides; raise GLIDE for longer ones.
- **Realistic piano ballad**: "Grand Piano", then on FX turn on REVERB with SIZE about halfway.
- **Supersaw lead**: "Supersaw Lead"; raise UNISON and DETUNE for a wider, thicker sound.
- **Your own pluck**: INIT, set ENV 1 SUSTAIN to 0 and DECAY short, drag ENV 2 onto Filter 1 CUTOFF.

Session Players and Generate

Session players

Track > New Drummer Track, **New Bass Player Track** and **New Keyboard Player Track** add a musician who plays along with your song, in its tempo and key. Select the player's region to show its editor under the tracks:

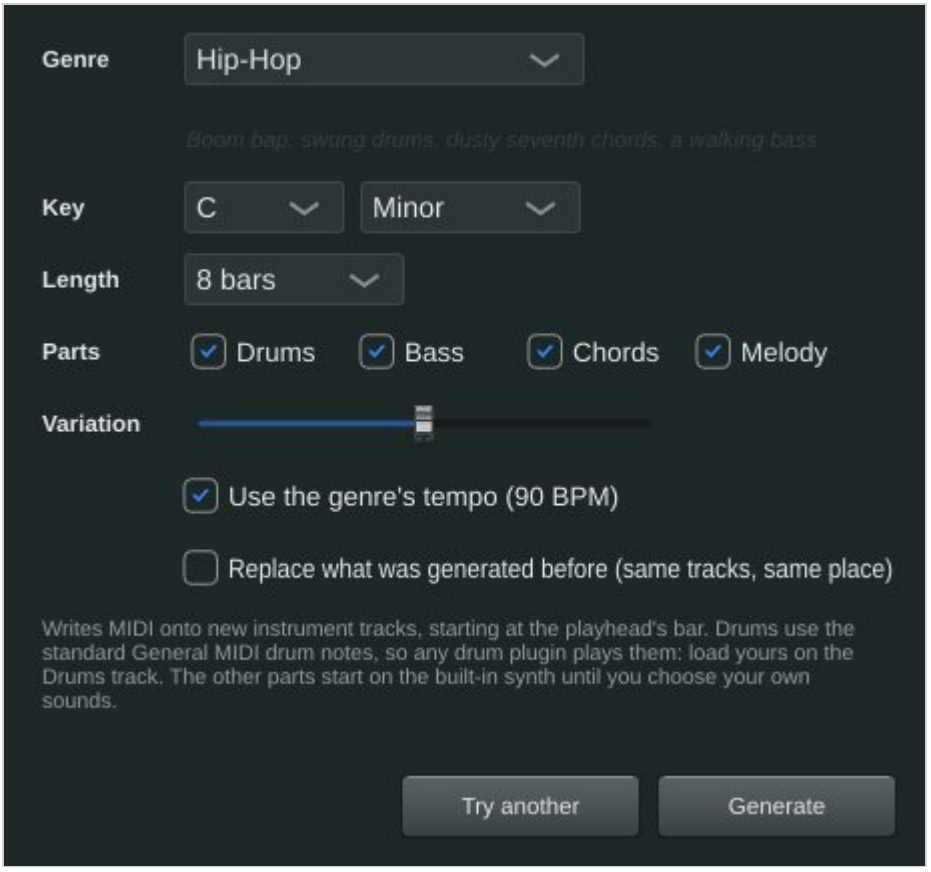


- **Genre** - Hip-Hop, Trap, Drill, Pop, R&B, Lo-fi, House, Techno, Afrobeats, Reggaeton or Rock.
- **The square** - drag the dot: left is **Simple**, right is **Complex**; up is **Loud**, down is **Soft**.
- **Fills** - how often it plays a fill. **Swing** - how much it swings.
- **Drummer parts** - switch **Kick & Snare**, **Hi-Hats**, **Cymbals**, **Toms** and **Percussion** on or off (play just the hats, say).
- **New Take** - the same settings, played a new way.

Make a region longer or shorter, or copy it to other parts of the song, and the player fills it. Different regions can have different settings: quieter in the verse, busier in the chorus. The bass and keyboard players follow the song's key.

Generate MIDI

Track > Generate MIDI..., or the **Generate** key, writes a whole idea - drums, bass, chords and a melody - in a genre's style.



- **Genre** - Hip-Hop (boom bap: swung drums, dusty seventh chords, a walking bass), Trap, Drill, Pop, R&B, Lo-fi, House, Techno, Afrobeats, Reggaeton and Rock.
- **Key** - root and Major or Minor (it starts in the song's key if it has one).
- **Length** - how many bars.
- **Parts** - Drums, Bass, Chords, Melody.
- **Variation** - low: a steady loop; high: more fills, rolls and changes.
- **Use the genre's tempo** - sets the project to the genre's usual tempo.
- **Replace what was generated before** - each part's new notes replace the last ones, on the same tracks.
- **Generate** - adds a new set of parts at the playhead's bar. **Try another** - replaces the parts you just made with a new take.

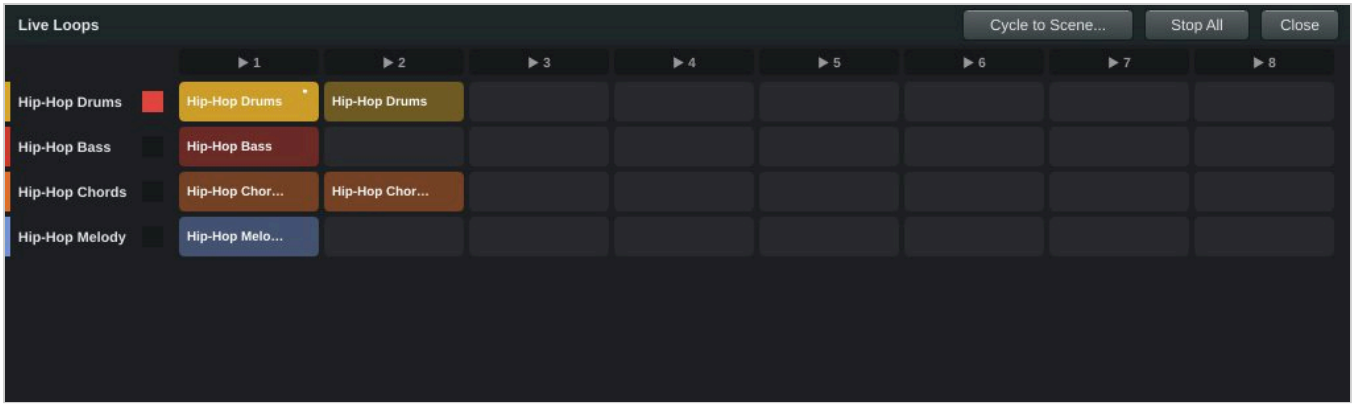
Drums use the standard General MIDI drum notes, so any drum instrument plays them. Then choose your own sounds - in Nemix, for example - and edit the notes in the piano roll.

The Claude assistant can help too

Ask Claude "add a reggaeton beat and a bass line in A minor" and it writes and places the parts. See *The Claude Assistant*.

Live Loops

View > Live Loops shows a grid of loops: one row per track, one column per **scene**. Use it to try arrangements, jam, or perform live.



- **Click a cell** to start its loop on the next bar. It loops until you stop it, and meanwhile it replaces that track's timeline. Click it again to stop it at the next bar.
- **The numbers on top** start a whole scene - every loop in that column.
- **An empty cell** takes the selected region when you click it. Right-click a cell to **Put the selected region here**, **Clear cell**, or **Copy cell to the tracks area at the playhead**.
- **Cycle to Scene...** - copies what's on the tracks inside the cycle into a scene's cells: a quick way to fill a scene with your verse or chorus.
- **Stop All** - stops every loop.

A way to work

1. Make a 4- or 8-bar idea on several tracks and set the cycle over it.
2. **Cycle to Scene** into scene 1.
3. Change the parts (a new drum pattern, a different bass) and capture them into scene 2, 3...
4. Play scenes in different orders to find the arrangement, then copy the cells you like into the tracks area.

Nemix Sounds

Every sound that comes with Nemix, by type, with the styles it's made for. Sounds marked **(recorded)** play the recorded instruments. Find them in Nemix's **BROWSE** page by type, style or name.

808

- **808 Classic** - Trap, Hip Hop
- **808 Long Slide** - Trap, Drill
- **808 Drill Glide** - Drill, Trap
- **808 Distorted** - Trap, Hip Hop
- **808 Fuzz Monster** - Trap, Drill
- **808 Punch Short** - Hip Hop, Trap, Reggaeton
- **808 Mellow Round** - R&B, Hip Hop, Lo-Fi
- **808 Hard Clip** - Trap, Drill
- **808 Reggaeton Boom** - Reggaeton, Latin, Dancehall
- **808 Afro Bounce** - Afrobeats, Dancehall
- **808 Phonk Cowbass** - Trap, Hip Hop
- **808 Sub Only** - Trap, R&B, Pop

Bass

- **Deep Sub** - Trap, Dubstep, Reggae, Hip Hop
- **Dub Bass** - Reggae, Dancehall, Dub
- **Reggae Roots Bass** - Reggae, Dub
- **Reggaeton Bass** - Reggaeton, Latin
- **Dembow Pluck Bass** - Reggaeton, Dancehall
- **Hip Hop Square Bass** - Hip Hop, Lo-Fi
- **House Saw Bass** - House, EDM
- **Reese Classic** - Drum & Bass, Dubstep, Techno
- **Reese Dark Drift** - Drum & Bass, Techno, Cinematic
- **Dubstep Wobble** - Dubstep, EDM
- **Dubstep Growl Yoy** - Dubstep, EDM
- **Neuro Snarl** - Drum & Bass, Dubstep
- **Acid Squelch** - Techno, House, Acid
- **Funk Finger Bass** - Funk, Soul, R&B, Pop
- **Moog-Style Bass** - Funk, Synthwave, Pop
- **Synthwave Pulse Bass** - Synthwave, Pop, 80s
- **Upright Jazz Bass (synth)** - Jazz, Soul, Lo-Fi
- **Rock Pick Bass (synth)** - Rock, Pop, Punk
- **Afro Groove Bass** - Afrobeats, Amapiano, Dancehall

- **Amapiano Log Drum** - Amapiano, Afrobeats
- **Techno Rumble** - Techno
- **Trap Sub Saw** - Trap, Drill
- **Drill Slide Bass** - Drill
- **Pop Synth Bass** - Pop, Dance
- **Wavy Chip Bass** - Chiptune, Hip Hop
- **Finger Bass** (recorded) - Pop, Funk, Soul, R&B, Reggae
- **Finger Bass Funk** (recorded) - Funk, Soul, Disco
- **Reggae Dub Finger Bass** (recorded) - Reggae, Dub, Dancehall
- **Pick Bass Rock** (recorded) - Rock, Punk, Pop
- **Motown Pick Bass** (recorded) - Soul, Funk, Pop
- **Jazz Bass** (recorded) - Jazz, Funk, R&B, Pop
- **Jazz Bass Warm R&B** (recorded) - R&B, Neo Soul, Gospel
- **Mirror Bass** (recorded) - Pop, Rock, Funk
- **Studio Bass** (recorded) - Pop, R&B, Hip Hop, Latin
- **Reggaeton Live Bass** (recorded) - Reggaeton, Latin, Afrobeats
- **Killer Finger Bass** (recorded) - Funk, Rock, Fusion
- **Upright Jazz Bass** (recorded) - Jazz, Blues, Lo-Fi, Swing
- **Upright Bass Lo-Fi** (recorded) - Lo-Fi, Hip Hop, Jazz

Lead

- **Supersaw Lead** - EDM, Trance, Pop
- **Trance Hoover Lead** - Trance, EDM, Rave
- **Sync Scream** - EDM, Electro, Dubstep
- **Trap Bell Lead** - Trap, Drill, Hip Hop
- **Drill Flute Lead** - Drill, Trap
- **Hip Hop Whistle** - Hip Hop, G-Funk, West Coast
- **G-Funk Portamento Lead** - Hip Hop, G-Funk, West Coast
- **Acid Lead** - Techno, Acid, House
- **Chiptune Lead** - Chiptune, Pop
- **Reggaeton Square Lead** - Reggaeton, Latin, Pop
- **Dancehall Synth Lead** - Dancehall, Reggae, Afrobeats
- **Synthwave Lead** - Synthwave, 80s, Pop
- **Screech Growl Lead** - Dubstep, EDM
- **Jazz Fusion Lead** - Jazz, Fusion, Funk
- **Rock Synth Guitar Lead** - Rock, Metal
- **R&B Silk Lead** - R&B, Soul, Pop
- **Pop Pluck Lead** - Pop, EDM
- **Vowel Talker** - Electro, Funk, EDM
- **Retro Square Lead** - 80s, Synthwave, Pop

Pluck

- **Trap Pluck** - Trap, Hip Hop
- **House Pluck** - House, EDM, Pop
- **Future Bass Pluck** - Future Bass, EDM, Pop
- **Harp Pluck (synth)** - Orchestral, Cinematic, Pop
- **Lo-Fi Pluck** - Lo-Fi, Hip Hop, Chill
- **Bubble Pluck** - House, Pop, Afrobeats
- **Glass Pluck** - Ambient, Pop, Cinematic
- **Afro Pluck** - Afrobeats, Amapiano
- **Drill Plucked Bell** - Drill, Trap
- **Koto-ish Pluck** - World, Cinematic, Trap

Mallets

- **Marimba (synth)** - Afrobeats, Pop, Latin, Orchestral
- **Kalimba (synth)** - Afrobeats, Lo-Fi, Ambient
- **Steel Drum (synth)** - Reggae, Dancehall, Latin, Afrobeats
- **Celesta (synth)** - Orchestral, Cinematic, Pop
- **Vibraphone (synth)** - Jazz, Lo-Fi, Soul
- **Glockenspiel (synth)** - Orchestral, Pop, Cinematic
- **Xylophone (synth)** - Orchestral, Pop, Latin
- **Glockenspiel** (recorded) - Orchestral, Pop, Cinematic, Trap
- **Trap Glock** (recorded) - Trap, Drill, Hip Hop
- **Marimba** (recorded) - Afrobeats, Latin, Pop, Orchestral
- **Afro Marimba** (recorded) - Afrobeats, Amapiano, Dancehall

Guitar

- **Nylon Guitar Pluck (synth)** - Latin, Reggaeton, Pop, Jazz
- **Steel Guitar Pluck (synth)** - Pop, Country, Rock, Reggae
- **Muted Guitar Chop (synth)** - Reggae, Funk, Afrobeats
- **Rock Power Chord (synth)** - Rock, Metal, Punk
- **Grunge Chug (synth)** - Rock, Metal
- **Nylon Guitar** (recorded) - Latin, Pop, Jazz, Bossa Nova, Reggaeton
- **Spanish Guitar Reggaeton** (recorded) - Reggaeton, Latin, Pop
- **Bossa Nylon** (recorded) - Bossa Nova, Jazz, Latin
- **Steel Acoustic Guitar** (recorded) - Pop, Folk, Country, Rock
- **Acoustic Pop Shimmer** (recorded) - Pop, R&B, Ambient
- **Clean Electric Guitar** (recorded) - Pop, Rock, Funk, R&B
- **Clean Chorus R&B Guitar** (recorded) - R&B, Neo Soul, Gospel, Pop
- **Funk Clean Guitar** (recorded) - Funk, Disco, Soul
- **Afrobeats Clean Guitar** (recorded) - Afrobeats, Amapiano, Highlife
- **Reggae Skank Guitar** (recorded) - Reggae, Dub, Ska

- **Crunch Guitar** (recorded) - Rock, Blues, Indie
- **Rock Drive Guitar** (recorded) - Rock, Metal, Punk
- **Hollow Jazz Guitar** (recorded) - Jazz, Blues, Soul
- **Lo-Fi Guitar** (recorded) - Lo-Fi, Hip Hop, Chill

Chords

- **Reggae Skank Chop** - Reggae, Dub, Ska
- **House Chord Stab** - House, Deep House, Garage
- **Rave Stab** - Rave, Techno, Breakbeat
- **Trance Gate Chords** - Trance, EDM
- **Neo Soul Chords** - Neo Soul, R&B, Jazz
- **Reggaeton Chord Stab** - Reggaeton, Latin, Pop
- **Dub Chord Echo** - Dub, Reggae, Dub Techno
- **Future Bass Supersaw Chords** - Future Bass, EDM, Pop
- **Jazz Comp Keys** - Jazz, Bossa Nova, Lo-Fi
- **Afrobeats Chord Pluck** - Afrobeats, Amapiano, Pop

Keys

- **Electric Piano** - R&B, Soul, Jazz, Pop
- **Lo-Fi Keys** - Lo-Fi, Hip Hop, Chill
- **Tremolo EP** - Soul, R&B, Gospel
- **Wurly Grit** - Soul, Rock, Lo-Fi
- **Clav Funk (synth)** - Funk, Soul
- **Toy Piano** - Pop, Lo-Fi, Cinematic
- **Rhodes Classic** (recorded) - Soul, R&B, Jazz, Neo Soul
- **Rhodes Tremolo Soul** (recorded) - Soul, R&B, Gospel, Neo Soul
- **Rhodes Lo-Fi** (recorded) - Lo-Fi, Hip Hop, Chill
- **Rhodes Phaser Funk** (recorded) - Funk, Jazz, Fusion
- **Electric Grand 80s** (recorded) - 80s, Pop, Synthwave
- **Electric Grand Ballad** (recorded) - Pop, R&B

Piano

- **Synth Piano** - Pop, EDM, House
- **Felt Piano (synth)** - Lo-Fi, Ambient, Cinematic
- **House Piano Stab** - House, Dance
- **Grand Piano** (recorded) - Pop, R&B, Gospel, Jazz, Cinematic, Orchestral
- **Grand Piano Concert Hall** (recorded) - Orchestral, Cinematic, Ambient
- **Grand Piano Ballad** (recorded) - Pop, R&B, Soul, Gospel
- **Bright Pop Piano** (recorded) - Pop, EDM, House, Reggaeton
- **Lo-Fi Grand** (recorded) - Lo-Fi, Hip Hop, Chill
- **Trap Piano Dark** (recorded) - Trap, Drill, Hip Hop

- **Upright Piano** (recorded) - Pop, Folk, Jazz, Soul
- **Upright Jazz Club** (recorded) - Jazz, Blues, Soul
- **Honky Tonk Upright** (recorded) - Rock, Country, Blues

Organ

- **Gospel Organ** - Gospel, Soul, R&B
- **Jazz Organ** - Jazz, Soul, Funk
- **Rock Organ Drive** - Rock, Blues
- **Church Organ (synth)** - Gospel, Cinematic, Orchestral
- **Reggae Organ Bubble** - Reggae, Dub

Pad

- **Warm Analog Pad** - Pop, R&B, Ambient
- **Dark Cinema Pad** - Cinematic, Trap, Ambient
- **Glass Pad** - Ambient, Pop, Cinematic
- **Ambient Wash** - Ambient, Cinematic, Lo-Fi
- **Sweep Pad** - Trance, EDM, Synthwave
- **R&B Velvet Pad** - R&B, Soul, Neo Soul
- **Lo-Fi Tape Pad** - Lo-Fi, Hip Hop, Chill
- **Trap Dark Pad** - Trap, Drill, Hip Hop
- **Gospel Pad** - Gospel, Soul, R&B
- **Synthwave Brass Pad** - Synthwave, 80s
- **Evolving Vowel Pad** - Ambient, Cinematic, Electronic

Vocal

- **Choir Pad** - Gospel, Cinematic, Orchestral, Ambient
- **Vocal Aahs (synth)** - Pop, R&B, Gospel

Strings

- **String Ensemble (synth)** - Orchestral, Cinematic, Pop, Disco
- **Violin Section (synth)** - Orchestral, Cinematic, Latin
- **Cello Section (synth)** - Orchestral, Cinematic
- **Pizzicato (synth)** - Orchestral, Cinematic, Pop
- **Staccato Strings (synth)** - Cinematic, Orchestral, Trap
- **Violin Section** (recorded) - Orchestral, Cinematic, Pop, R&B
- **Viola Section** (recorded) - Orchestral, Cinematic
- **Cello Section** (recorded) - Orchestral, Cinematic, Trap
- **Cinematic Cellos** (recorded) - Cinematic, Trap, Drill
- **Tremolo Violins** (recorded) - Cinematic, Orchestral, Trap
- **Pizzicato Violins** (recorded) - Orchestral, Pop, Cinematic

- **Pizzicato Cellos** (recorded) - Orchestral, Cinematic, Pop
- **Drill Pizzicato** (recorded) - Drill, Trap
- **Harp** (recorded) - Orchestral, Cinematic, Pop
- **Harp Dream** (recorded) - Ambient, Cinematic, R&B

Brass

- **Orchestra Brass (synth)** - Orchestral, Cinematic, Trap
- **French Horns (synth)** - Orchestral, Cinematic
- **Trumpet Lead (synth)** - Jazz, Latin, Salsa, Ska
- **Horn Stabs (synth)** - Funk, Soul, Hip Hop, Ska
- **Synth Brass** - 80s, Synthwave, Pop
- **Trumpet** (recorded) - Jazz, Latin, Orchestral, Ska
- **Trumpet Staccato** (recorded) - Latin, Salsa, Funk, Ska
- **Salsa Brass Stabs** (recorded) - Salsa, Latin, Reggaeton
- **French Horn** (recorded) - Orchestral, Cinematic
- **Epic Horns** (recorded) - Cinematic, Trap, Orchestral
- **Trombone** (recorded) - Jazz, Orchestral, Ska
- **Tuba** (recorded) - Orchestral, Cinematic, Banda

Winds

- **Flute (synth)** - Orchestral, Latin, Drill, Jazz
- **Pan Flute (synth)** - World, Latin, Ambient
- **Clarinet (synth)** - Jazz, Orchestral
- **Oboe (synth)** - Orchestral, Cinematic
- **Sax-ish Lead (synth)** - Jazz, Soul, Funk, R&B
- **Flute** (recorded) - Orchestral, Jazz, Latin, Pop
- **Drill Flute (real)** (recorded) - Drill, Trap, Hip Hop
- **Oboe** (recorded) - Orchestral, Cinematic
- **Clarinet** (recorded) - Jazz, Orchestral, Klezmer
- **Bassoon** (recorded) - Orchestral, Cinematic

Percussion

- **Timpani Roll (synth)** - Orchestral, Cinematic
- **Conga (synth)** - Latin, Afrobeats, Salsa
- **Shaker** - Latin, Pop, Afrobeats
- **Perc Wood Block** - Latin, Afrobeats, Reggaeton

Bells

- **Music Box** - Cinematic, Lo-Fi, Pop
- **Church Bell** - Cinematic, Trap, Orchestral

- **Trap Bells Hard** - Trap, Drill
- **Digital Bell Keys** - Pop, EDM, Future Bass

Arp

- **Trap Bell Arp** - Trap, Drill
- **Synthwave Arp** - Synthwave, 80s
- **Trance Run Arp** - Trance, EDM
- **Lo-Fi Dream Arp** - Lo-Fi, Chill, Ambient
- **Reggaeton Arp** - Reggaeton, Latin
- **Afro Marimba Arp** - Afrobeats, Amapiano
- **Chiptune Arp** - Chiptune, Pop
- **Techno Sequence** - Techno, Minimal
- **Ambient Pluck Arp** - Ambient, Cinematic
- **Harp Glissando Arp** - Orchestral, Cinematic

FX

- **White Noise Riser** - EDM, House, Trap
- **Downlifter** - EDM, Trap, Cinematic
- **Laser Zap** - Dancehall, Reggae, Dubstep
- **Dub Siren** - Reggae, Dub, Dancehall
- **Impact Boom** - Cinematic, Trap, EDM
- **Air Sweep** - Ambient, Cinematic, EDM
- **Vinyl Crackle Bed** - Lo-Fi, Hip Hop, Chill
- **Sci-Fi Bubbles** - Electronic, Ambient
- **Tape Stop Drop** - Trap, Hip Hop

Drums

- **Kick Deep** - Hip Hop, Trap, House
- **Kick Punch** - House, Techno, EDM
- **Snare Synth** - Hip Hop, Trap, Pop
- **Clap** - Trap, House, Reggaeton
- **Hi-Hat Closed** - Trap, Drill, Hip Hop, House
- **Hi-Hat Open** - House, Disco, Trap
- **Cowbell** - Latin, Reggaeton, Trap, Disco
- **Rimshot** - Reggaeton, Dancehall, Hip Hop
- **Tom Synth** - 80s, Rock, Pop
- **Rock Kit** (recorded) - Rock, Pop, Indie, Funk
- **Rock Kit Room** (recorded) - Rock, Indie, Pop
- **Rock Kit Crushed** (recorded) - Rock, Hip Hop, Trap
- **Live Kit Tight Pop** (recorded) - Pop, R&B, Gospel, Latin
- **Lo-Fi Live Kit** (recorded) - Lo-Fi, Hip Hop, Chill

Recorded Instruments

The recorded instruments of the **Nemix Library**, played by Nemix's SAMPLE layer. Each note was recorded by real players on real instruments, at several strengths, and many with alternating takes (round robins).

Piano

- **Grand Piano** - Salamander Grand Piano by Alexander Holm (CC-BY 3.0)
- **Upright Piano** - Upright Piano KW, FreePats (CC0)

Keys

- **Electric Piano (Rhodes)** - Rhodes Mark I by Jeff Learman (CC0)
- **Electric Grand** - Electric grand by Greg Sullivan (CC0)

Bass

- **Finger Bass** - Electric Bass YR, FreePats (CC0)
- **Pick Bass** - Electric Bass YR, FreePats (CC0)
- **Jazz Bass** - Growlybass, Karoryfer Samples (CC0)
- **Mirror Bass** - Fashionbass, Karoryfer Samples (CC0)
- **Studio Bass** - Swagbass, Karoryfer Samples (CC0)
- **Killer Finger Bass** - CC0 finger bass from the Discord SFZ GM bank
- **Upright Bass (Jazz)** - Versilian Studios VSCO 2 CE (CC0)

Guitar

- **Nylon Guitar** - Spanish Classical Guitar, FreePats (CC0)
- **Steel Acoustic Guitar** - Martin HD28 by Jeff Learman (CC0)
- **Clean Electric Guitar** - FSBS Electric Guitar Clean, FreePats (CC0)
- **Hollow Electric Guitar** - Emilyguitar, Karoryfer Samples (CC0)

Drums

- **Rock Kit** - Big Rusty Drums, Karoryfer Samples (CC0)

Strings

- **Violin Section** - Versilian Studios VSCO 2 CE (CC0)
- **Viola Section** - Versilian Studios VSCO 2 CE (CC0)
- **Cello Section** - Versilian Studios VSCO 2 CE (CC0)

- **Pizzicato Violins** - Versilian Studios VSCO 2 CE (CC0)
- **Pizzicato Cellos** - Versilian Studios VSCO 2 CE (CC0)
- **Tremolo Violins** - Versilian Studios VSCO 2 CE (CC0)
- **Harp** - Versilian Studios VSCO 2 CE (CC0)

Brass

- **Trumpet** - Versilian Studios VSCO 2 CE (CC0)
- **French Horn** - Versilian Studios VSCO 2 CE (CC0)
- **Trombone** - Versilian Studios VSCO 2 CE (CC0)
- **Tuba** - Versilian Studios VSCO 2 CE (CC0)
- **Trumpet Staccato** - Versilian Studios VSCO 2 CE (CC0)

Winds

- **Flute** - Versilian Studios VSCO 2 CE (CC0)
- **Oboe** - Versilian Studios VSCO 2 CE (CC0)
- **Clarinet** - Versilian Studios VSCO 2 CE (CC0)
- **Bassoon** - Versilian Studios VSCO 2 CE (CC0)

Mallets

- **Glockenspiel** - Versilian Studios VSCO 2 CE (CC0)
- **Marimba** - Versilian Studios VSCO 2 CE (CC0)

Credits

The recordings are free to use. The Salamander Grand Piano is by Alexander Holm (Creative Commons Attribution 3.0); the others are in the public domain (CC0), from Versilian Studios (VSCO 2 CE), Karoryfer Samples, FreePats and other free libraries. The full list is in the Nemix Library folder (CREDITS.txt).

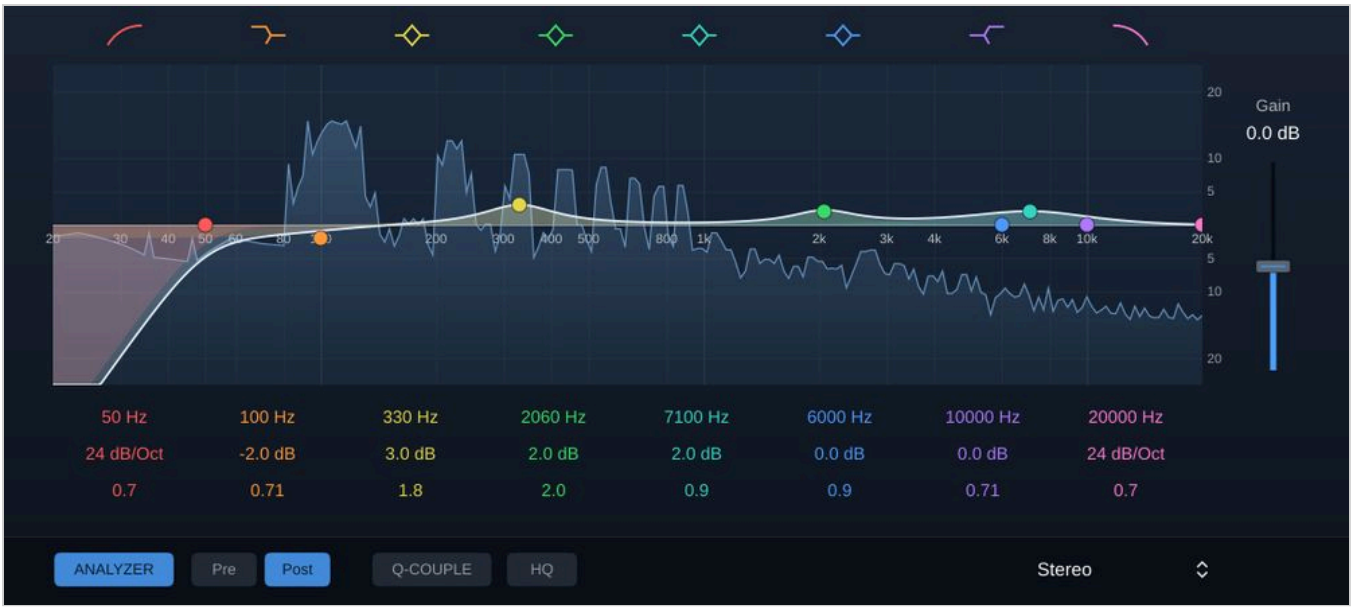
Plugins

EQ Effects

EQ (equalization) turns frequency ranges up or down: less mud, more air, a clearer vocal. **How to use it:** cut narrow bands to remove problems, boost wide bands gently to add character. Cut lows below what an instrument needs with a low cut (80-100 Hz on vocals and guitars). Use the analyzer to see where the energy is, but decide with your ears.

In this chapter: **Channel EQ**, **Single Band EQ**, **Console EQ**, **Graphic EQ**, **Tube EQ**, **Linear Phase EQ**, **Match EQ**.

Channel EQ



Channel EQ: the eight bands on both channels or just one side (Left, Right, Mid or Side); HQ runs them at twice the rate (no cramping near the top); the analyzer reads before or after.

> **How to use it:** Start most mixes here: a low cut, then small moves. Click the analyzer to see the sound while you EQ.

Controls

Low Cut (10 to 20000 Hz) · **Low Shelf** (10 to 20000 Hz) · **Shelf Gain** (-24 to 24 dB) · **Low Mid** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **Mid** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **High Mid** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **High Shelf** (10 to 20000 Hz) · **Shelf Gain** (-24 to 24 dB) · **High Cut** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **High Mid 2** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **Low Cut Slope** (6 dB/Oct, 12 dB/Oct, 18 dB/Oct, 24 dB/Oct, 36 dB/Oct, 48 dB/Oct) · **Low Cut Q** (0.1 to 100) · **Shelf Q** (0.1 to 2) · **Shelf Q** (0.1 to 2) · **High Cut Slope** (6 dB/Oct, 12 dB/Oct, 18 dB/Oct, 24 dB/Oct, 36 dB/Oct, 48 dB/Oct) · **High Cut Q** (0.1 to 100) · **Band 1** (Off, On) · **Band 2** (Off, On) · **Band 3** (Off, On) · **Band 4** (Off, On) · **Band 5** (Off, On) · **Band 6** (Off, On) · **Band**

7 (Off, On) · **Band 8** (Off, On) · **Analyzer** (Off, On) · **Analyzer Pre/Post** (Pre, Post) · **Q-Couple** (Off, On) · **HQ** (Off, On) · **Processing** (Stereo, Left, Right, Mid, Side)

Single Band EQ



One band of EQ, of the type you choose: low cut, high cut, low shelf, high shelf, bell, notch or band pass, with its frequency, gain, Q and (for the cuts) a 12 or 24 dB/oct slope.

Controls

Type (Low Cut, High Cut, Low Shelf, High Shelf, Bell, Notch, Band Pass) · **Frequency** (20 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 10) · **Slope** (12 dB/oct, 24 dB/oct)

Console EQ



A British-console style EQ: a stepped high pass, a low shelf (four frequencies), a mid bell (six frequencies) and a 12 kHz high shelf, with a drive that adds the circuit's colour, and output.

Controls

High Pass (Off, 50 Hz, 80 Hz, 160 Hz, 300 Hz) · **Low Freq** (35 Hz, 60 Hz, 110 Hz, 220 Hz) · **Low** (-16 to 16 dB) · **Mid Freq** (360 Hz, 700 Hz, 1.6 kHz, 3.2 kHz, 4.8 kHz, 7.2 kHz) · **Mid** (-18 to 18 dB) · **High (12k)** (-16 to 16 dB) · **Drive** (0 to 10) · **Output** (-18 to 18 dB)

Graphic EQ



A ten-band graphic EQ, one slider per octave from 31 Hz to 16 kHz, and output. Quick broad tone shaping.

Controls

31 (-12 to 12 dB) · **63** (-12 to 12 dB) · **125** (-12 to 12 dB) · **250** (-12 to 12 dB) · **500** (-12 to 12 dB) · **1k** (-12 to 12 dB) · **2k** (-12 to 12 dB) · **4k** (-12 to 12 dB) · **8k** (-12 to 12 dB) · **16k** (-12 to 12 dB) · **Output** (-12 to 12 dB)

Tube EQ



A passive tube program EQ: low boost and low attenuation at the same frequency (boost both for the famous tight low end), a high boost with its bandwidth, a separate high attenuation, and output through the tube stage.

Controls

Low Freq (20 Hz, 30 Hz, 60 Hz, 100 Hz) · **Low Boost** (0 to 10) · **Low Atten** (0 to 10) · **High Freq** (3 kHz, 4 kHz, 5 kHz, 8 kHz, 10 kHz, 12 kHz, 16 kHz) · **High Boost** (0 to 10) · **Bandwidth** (0 to 10) · **Atten Sel** (5 kHz, 10 kHz, 20 kHz) · **High Atten** (0 to 10) · **Output** (-12 to 12 dB)

Linear Phase EQ



Linear Phase EQ: the same eight bands as the Channel EQ (low and high cut with slopes up to 48 dB/oct, two shelves, four bells), but as a linear-phase filter, so nothing is shifted in time - the choice for mastering and for parallel or multi-mic sources. It adds a little latency.

> **How to use it:** Use it on the master or on drum buses, where phase shifts between tracks would smear the sound.

Controls

Low Cut (10 to 20000 Hz) · **Low Shelf** (10 to 20000 Hz) · **Shelf Gain** (-24 to 24 dB) · **Low Mid** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **Mid** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **High Mid** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **High Shelf** (10 to 20000 Hz) · **Shelf Gain** (-24 to 24 dB) · **High Cut** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **High Mid 2** (10 to 20000 Hz) · **Gain** (-24 to 24 dB) · **Q** (0.1 to 100) · **Low Cut Slope** (6 dB/Oct, 12 dB/Oct, 18 dB/Oct, 24 dB/Oct, 36 dB/Oct, 48 dB/Oct) · **Low Cut Q** (0.1 to 100) · **Shelf Q** (0.1 to 2) · **Shelf Q** (0.1 to 2) · **High Cut Slope** (6 dB/Oct, 12 dB/Oct, 18 dB/Oct, 24 dB/Oct, 36 dB/Oct, 48 dB/Oct) · **High Cut Q** (0.1 to 100) · **Band 1** (Off, On) · **Band 2** (Off, On) · **Band 3** (Off, On) · **Band 4** (Off, On) · **Band 5** (Off, On) · **Band 6** (Off, On) · **Band 7** (Off, On) · **Band 8** (Off, On) · **Analyzer** (Off, On) · **Analyzer Pre/Post** (Pre, Post) · **Q-Couple** (Off, On) · **HQ** (Off, On) · **Processing** (Stereo, Left, Right, Mid, Side)

Match EQ



Match EQ: learns the average spectrum of a reference (a song you like - the Template) and of your track (Current), then sets a filter that makes yours sound like it. Apply sets how much (negative values do the opposite), Smoothing how fine the curve is, Phase linear or minimum; four bands of your own on top.

> **How to use it:** Load a reference song as the Template, play your mix to learn Current, then set Apply to 30-60% for a gentle match.

Controls

Apply (-100 to 200 %) · **Smoothing** (0 to 100 %) · **Phase** (Linear, Minimum) · **Analyzer** (Off, On) · **Clear Template** (Off, On) · **Clear Current** (Off, On) · **Gain** (-24 to 24 dB) · **Template File** (0 to 100000)

Each band (4, 1 to 4): Freq, Gain, Q, On/Off

The template curve (31, 20 to 20.2k): Level

The current curve (31, 20 to 20.2k): Level

Dynamics Effects

Dynamics effects control the difference between loud and quiet. **How to use them:** a compressor evens a vocal or bass so it sits steady - lower the threshold until the gain reduction meter shows 3-6 dB on the loud parts, then raise the make-up gain. Slow attack keeps the punch; fast attack tames peaks. A gate cuts noise between notes; a de-esser tames harsh S sounds; a limiter stops peaks going over a ceiling (last on the master).

In this chapter: **Compressor**, **Limiter (simple)**, **Studio Compressor**, **Multipressor**, **DeEsser 2**, **Noise Gate**, **Expander**, **Enveloper**, **Transient Shaper**, **Limiter**, **Adaptive Limiter**.

Compressor

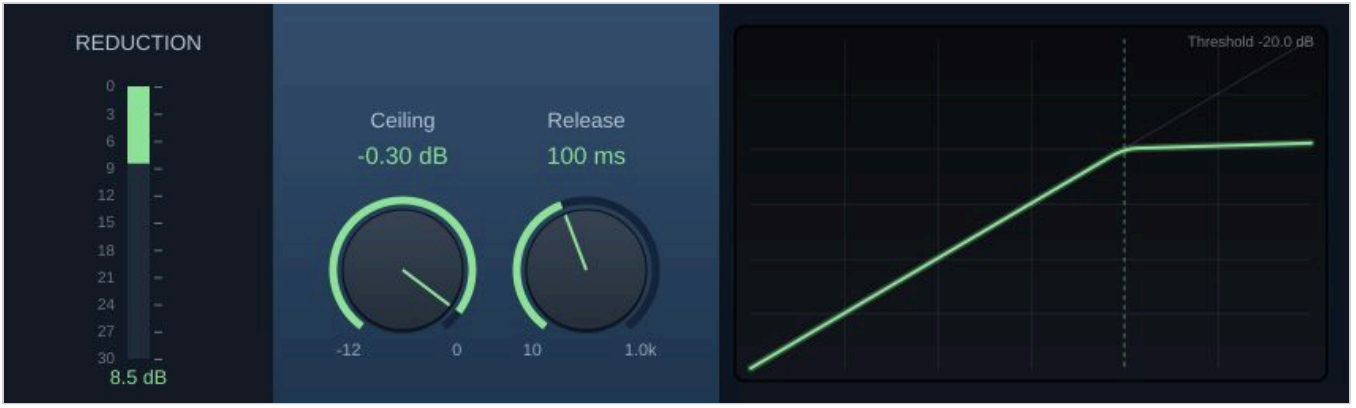


The simple compressor: threshold, ratio, attack, release and make-up gain - all a vocal or a bass usually needs. The meter shows the gain reduction.

Controls

Threshold (-60 to 0 dB) · **Ratio** (1 to 20 :1) · **Attack** (0.1 to 100 ms) · **Release** (10 to 1000 ms) · **Makeup** (0 to 24 dB)

Limiter (simple)



The simple limiter: nothing goes past the ceiling; release sets how quickly it lets go.

Controls

Ceiling (-12 to 0 dB) · **Release** (10 to 1000 ms)

Studio Compressor



Studio Compressor: seven circuits (Platinum Digital, Studio VCA, Studio FET, Classic VCA, Vintage VCA, Vintage FET, Vintage Opto), each with its own detector, timing, knee and colour. Threshold, ratio, attack, release, knee, make-up and Auto Gain; a side-chain filter (LP, BP, HP, ParEQ, HS) with Listen, Max or Sum and Peak or RMS detection; on the way out a limiter, Soft/Hard/Clip distortion and a dry/wet Mix for parallel compression.

> **How to use it:** Vintage Opto for smooth vocals, Vintage FET for aggressive drums and rap vocals, Classic VCA for punchy drums and buses. Use Mix at 50% for parallel compression.

Controls

Circuit (Platinum Digital, Studio VCA, Studio FET, Classic VCA, Vintage VCA, Vintage FET, Vintage Opto) · **Threshold** (-60 to 0 dB) · **Ratio** (1 to 30 :1) · **Attack** (0.05 to 200 ms) · **Release** (5 to 3000 ms) · **Knee** (0 to 1) · **Make Up** (-30 to 30 dB) · **Auto Gain** (Off, 0 dB, -12 dB) · **Filter Freq** (20 to 20000 Hz) · **Mix** (0 to 100 %) · **Distortion** (Off, Soft, Hard, Clip) · **Output Gain** (-30 to 30 dB) · **Limiter** (Off, On) · **Limiter Threshold** (-20 to 0 dB) · **Input Gain** (-30 to 30 dB) · **Auto Release** (Off, On) · **Detection** (Max, Sum) · **Peak/RMS** (Peak, RMS) · **Filter** (Off, On) · **Filter Mode** (LP, BP, HP, ParEQ, HS) · **Q** (0.1 to 10) · **Filter Gain** (-24 to 24 dB) · **Listen** (Off, On)

Multipressor



Multipressor: four bands split by three crossovers (phase-matched), each with its own compression (threshold, ratio), downward expansion (threshold, ratio, reduction), peak/RMS detection time, attack, release, make-up gain, on/off and solo; auto gain, lookahead and output.

> **How to use it:** On a mix or a bus: compress just the band that jumps out (boomy lows, harsh highs) and leave the rest alone.

Controls

Crossover 1 (30 to 1000 Hz) · **Crossover 2** (200 to 5000 Hz) · **Crossover 3** (1000 to 16000 Hz) · **Auto Gain** (Off, On) · **Lookahead** (0 to 20 ms) · **Out** (-24 to 24 dB) · **Low Comp Threshold** (-60 to 0 dB) · **Low Comp Ratio** (1 to 20 :1) · **Low Exp Threshold** (-100 to 0 dB) · **Low Exp Ratio** (1 to 10 :1) · **Low Exp Reduction** (0 to 60 dB) · **Low Peak/RMS** (0 to 100 ms) · **Low Attack** (0.1 to 200 ms) · **Low Release** (10 to 2000 ms) · **Low Gain** (-24 to 24 dB) · **Low Band** (Off, On) · **Low Solo** (Off, On) · **Low Mid Comp Threshold** (-60 to 0 dB) · **Low Mid Comp Ratio** (1 to 20 :1) · **Low Mid Exp Threshold** (-100 to 0 dB) · **Low Mid Exp Ratio** (1 to 10 :1) · **Low Mid Exp Reduction** (0 to 60 dB) · **Low Mid Peak/RMS** (0 to 100 ms) · **Low Mid Attack** (0.1 to 200 ms) · **Low Mid Release** (10 to 2000 ms) · **Low Mid Gain** (-24 to 24 dB) · **Low Mid Band** (Off, On) · **Low Mid Solo** (Off, On) · **High Mid Comp Threshold** (-60 to 0 dB) · **High Mid Comp Ratio** (1 to 20 :1) · **High Mid Exp Threshold** (-100 to 0 dB) · **High Mid Exp Ratio** (1 to 10 :1) · **High Mid Exp Reduction** (0 to 60 dB) · **High Mid Peak/RMS** (0 to 100 ms) · **High Mid Attack** (0.1 to 200 ms) · **High Mid Release** (10 to 2000 ms) · **High Mid Gain** (-24 to 24 dB) · **High Mid Band** (Off, On) · **High Mid Solo** (Off, On) · **High Comp Threshold** (-60 to 0 dB) · **High Comp Ratio** (1 to 20 :1) · **High Exp Threshold** (-100 to 0 dB)

· **High Exp Ratio** (1 to 10 :1) · **High Exp Reduction** (0 to 60 dB) · **High Peak/RMS** (0 to 100 ms) · **High Attack** (0.1 to 200 ms) · **High Release** (10 to 2000 ms) · **High Gain** (-24 to 24 dB) · **High Band** (Off, On) · **High Solo** (Off, On)

DeEsser 2



DeEsser 2: a detection filter (high shelf or peak) at Frequency; Relative mode compares the band with the whole signal (steady whatever the level), Absolute with a fixed threshold; Max Reduction caps it; Split turns down just the band, Wide the whole signal; Filter Solo plays the band.

> **How to use it:** Find the S frequency (5-8 kHz for most voices) with Filter Solo, then lower the threshold until the S sounds are calm but not lisped.

Controls

Frequency (2000 to 12000 Hz) · **Threshold** (-60 to 0 dB) · **Max Reduction** (0 to 24 dB) · **Range** (Split, Wide) · **Filter Solo** (Off, On) · **Mode** (Relative, Absolute) · **Filter** (Shelf, Peak)

Noise Gate



Noise Gate: Gate (closes below the threshold) or Ducker (turns down while the key is loud); threshold, reduction, hysteresis, attack, hold, release, lookahead; the key's filter (band pass or band reject between its low and high cutoff) with Monitor to hear it.

> **How to use it:** Set the threshold just above the noise between notes. On a kick or snare mic, it cuts the spill of the other drums.

Controls

Threshold (-80 to 0 dB) · **Reduction** (-100 to 0 dB) · **Attack** (0.01 to 100 ms) · **Hold** (0 to 500 ms) · **Release** (5 to 2000 ms) · **Hysteresis** (0 to 12 dB) · **Low Cutoff** (20 to 2000 Hz) · **High Cutoff** (500 to 20000 Hz) · **Mode** (Gate, Ducker) · **Lookahead** (0 to 20 ms) · **Filter** (Off, On) · **Characteristics** (Bandpass, Band Reject) · **Monitor** (Off, On)

Expander



Expander: above the threshold the signal is made louder still (upward expansion, by the ratio) - or, Downward, below it quieter, down to the range; knee, attack, release, Peak or RMS detection, gain with Auto Gain, and an output clip (Off, Soft, Hard, Clip).

Controls

Threshold (-80 to 0 dB) · **Ratio** (1 to 10 :1) · **Attack** (0.1 to 100 ms) · **Release** (10 to 2000 ms) · **Knee** (0 to 12 dB) · **Range** (-80 to 0 dB) · **Gain** (-24 to 24 dB) · **Direction** (Upward, Downward) · **Auto Gain** (Off, On) · **Peak/RMS** (Peak, RMS) · **Output Clip** (Off, Soft, Hard, Clip)

Enveloper

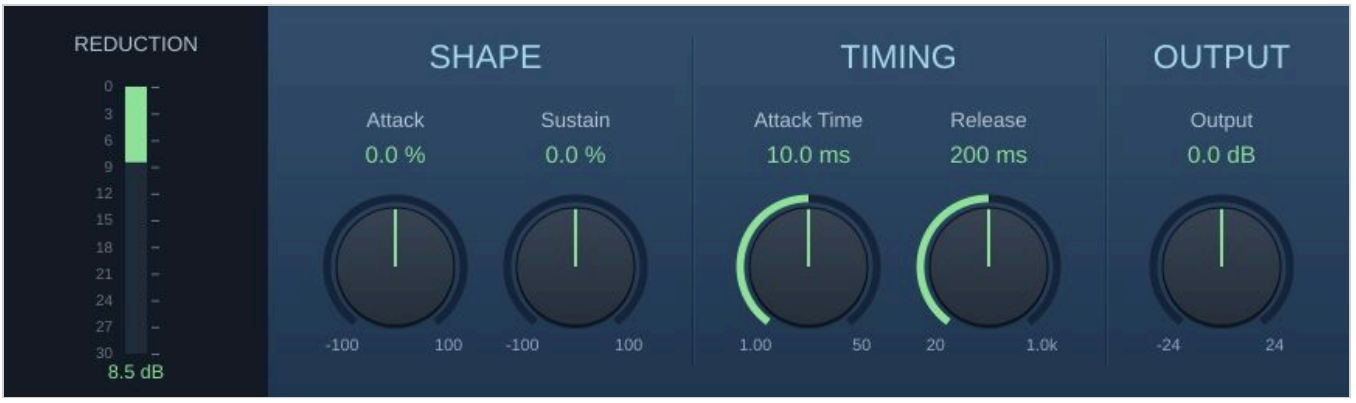


Enveloper: turns the attack of each sound up or down, and its release up or down, whatever its level: the threshold sets what counts, Lookahead reads ahead, Release Time how long the tail lasts, Out Level the result.

Controls

Threshold (-100 to 0 dB) · **Attack Gain** (-100 to 100 %) · **Attack Time** (1 to 200 ms) · **Release Gain** (-100 to 100 %) · **Release Time** (20 to 5000 ms) · **Lookahead** (0 to 20 ms) · **Out Level** (-24 to 24 dB)

Transient Shaper



More or less attack and sustain, whatever the level (two envelope followers, compared).

> **How to use it:** More attack makes drums snap; less sustain makes them tight and dry.

Controls

Attack (-100 to 100 %) · **Sustain** (-100 to 100 %) · **Attack Time** (1 to 50 ms) · **Release** (20 to 1000 ms) · **Output** (-24 to 24 dB)

Limiter



Limiter: looks ahead so peaks are caught before they arrive - nothing goes over the output level. Gain in, release, lookahead; Legacy (with an optional soft knee) or Precision (with optional true peak detection, catching the peaks between samples); a Warm character option.

> **How to use it:** Last on the master: set the output to -1 dB and raise the gain until it's as loud as you want, listening for the punch going away.

Controls

Gain (0 to 24 dB) · **Output Level** (-12 to 0 dB) · **Release** (1 to 1000 ms) · **Character** (Clean, Warm) · **Lookahead** (0.1 to 10 ms) · **Mode** (Legacy, Precision) · **Soft Knee** (Off, On) · **True Peak Detection** (Off, On)

Adaptive Limiter



Adaptive Limiter: raises the level and rounds the peaks off like a well-driven analog stage, never above the ceiling; its release adapts (quick for transients, slower for dense music). Gain, Out Ceiling, Lookahead, Remove DC Offset, True Peak Detection.

> **How to use it:** A warmer, rounder limiter for the master: raise Gain, keep Out Ceiling at -1 dB.

Controls

Gain (0 to 24 dB) · **Out Ceiling** (-12 to 0 dB) · **Lookahead** (0.5 to 20 ms) · **Remove DC Offset** (Off, On) · **True Peak Detection** (Off, On)

Delay Effects

Delays repeat the sound: echoes, slapback, ping-pong, rhythmic patterns. **How to use them:** put a delay on an aux bus and send to it; set the time to a note value (1/4, 1/8 dotted) so the echoes follow the tempo; use low and high cut on the echoes so they don't clutter the mix.

In this chapter: **Delay**, **Stereo Delay**, **Tape Delay**, **Modulation Delay**, **Echo**, **Sample Delay**, **Delay Designer**.

Delay



The simple delay: a time, how much of each echo comes back (feedback) and the dry/wet mix.

Controls

Time (1 to 2000 ms) · **Feedback** (0 to 0.95) · **Mix** (0 to 1)

Stereo Delay



Stereo Delay: each side its own input (Off, Left, Right, L+R, L-R), time or note with deviation, feedback with phase, crossfeed to the other side with phase; the routing presets (Straight, Crossfeed, 90/10, 10/90, Ping Pong, Pan, Rotate) or Customized; low and high cut on the echoes; output level per side, width, dry/wet mix.

> **How to use it:** A classic vocal throw: 1/4 on the left and 1/8 dotted on the right, feedback 20-30%, low and high cut on.

Controls

Sync (Off, On) · **Left Time** (1 to 4000 ms) · **Right Time** (1 to 4000 ms) · **Left Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Right Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Left Feedback** (0 to 100 %) · **Right Feedback** (0 to 100 %) · **Crossfeed L to R** (0 to 100 %) · **Low Cut** (10 to 2000 Hz) · **High Cut** (1000 to 20000 Hz) · **Mix** (0 to 100 %) · **Width** (0 to 100 %) · **Crossfeed R to L** (0 to 100 %) · **Left Feedback Phase** (Normal, Inverted) · **Right Feedback Phase** (Normal, Inverted) · **Crossfeed Phase** (Normal, Inverted) · **Left Input** (Off, Left, Right, L + R, L - R) · **Right Input** (Off, Left, Right, L + R, L - R) · **Left Deviation** (-50 to 50 %) · **Right Deviation** (-50 to 50 %) · **Left Output** (0 to 150 %) · **Right Output** (0 to 150 %) · **Routing** (Customized, Straight, Crossfeed, 90/10, 10/90, Ping Pong L/R, Pan L/R, Rotate L/R)

Tape Delay



Tape Delay: time or note with deviation, feedback, Freeze, low and high cut in the loop, the tape's LFO and flutter (each a rate and an intensity) evened out by Smoothing, a clip threshold for the tape's saturation, Clean or Diffuse head, spread, dry and wet.

> **How to use it:** Warm, wobbly echoes for reggae, dub and lo-fi. Raise feedback to over 100% briefly for a runaway dub echo.

Controls

Tempo Sync (Off, On) · **Delay Time** (1 to 4000 ms) · **Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Deviation** (-50 to 50 %) · **Feedback** (0 to 110 %) · **Freeze** (Off, On) · **Low Cut** (10 to 2000 Hz) · **High Cut** (1000 to 20000 Hz) · **Smoothing** (0 to 100 %) · **Clip Threshold** (0 to 100 %) · **Spread** (0 to 100 %) · **Tape Head Mode** (Clean, Diffuse) · **LFO Rate** (0.05 to 10 Hz) · **LFO Intensity** (0 to 100 %) · **Flutter Rate** (0.5 to 20 Hz) · **Flutter Intensity** (0 to 100 %) · **Dry** (0 to 100 %) · **Wet** (0 to 100 %)

Modulation Delay



Modulation Delay: a short delay swept by two LFOs (left and right rates, or linked, mixed), with feedback; De-Warble keeps the pitch steady (the sweep moves between two fixed taps instead of bending the delay), Constant Mod keeps the sweep's width the same at any rate, D-Mode's spatial filtering; stereo phase and its distribution, a low/high cut filter, volume modulation, mix.

Controls

Time (0.5 to 50 ms) · **Feedback** (-95 to 95 %) · **LFO 1 Rate** (0.02 to 10 Hz) · **Mod Intensity** (0 to 100 %) · **Phase** (-180 to 180 °) · **Output Mix** (0 to 100 %) · **De-Warble** (Off, On) · **Constant Mod** (Off, On) · **D-Mode** (Off, On) · **LFO 2 Rate** (0.02 to 10 Hz) · **Link L & R** (Off, On) · **LFO Mix** (0 to 100 %) · **Distribution** (Circular, Left/Right, Random) · **Filter** (Off, On) · **Low Cut** (20 to 2000 Hz) · **High Cut** (1000 to 20000 Hz) · **Volume Mod** (0 to 100 %)

Echo



A simple echo: time (or a note), repeats and a colour knob from dark to bright.

Controls

Sync (Off, On) · **Time** (1 to 4000 ms) · **Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Repeats** (0 to 100 %) · **Color** (-100 to 100 %) · **Wet** (0 to 100 %) · **Dry** (0 to 100 %)

Sample Delay

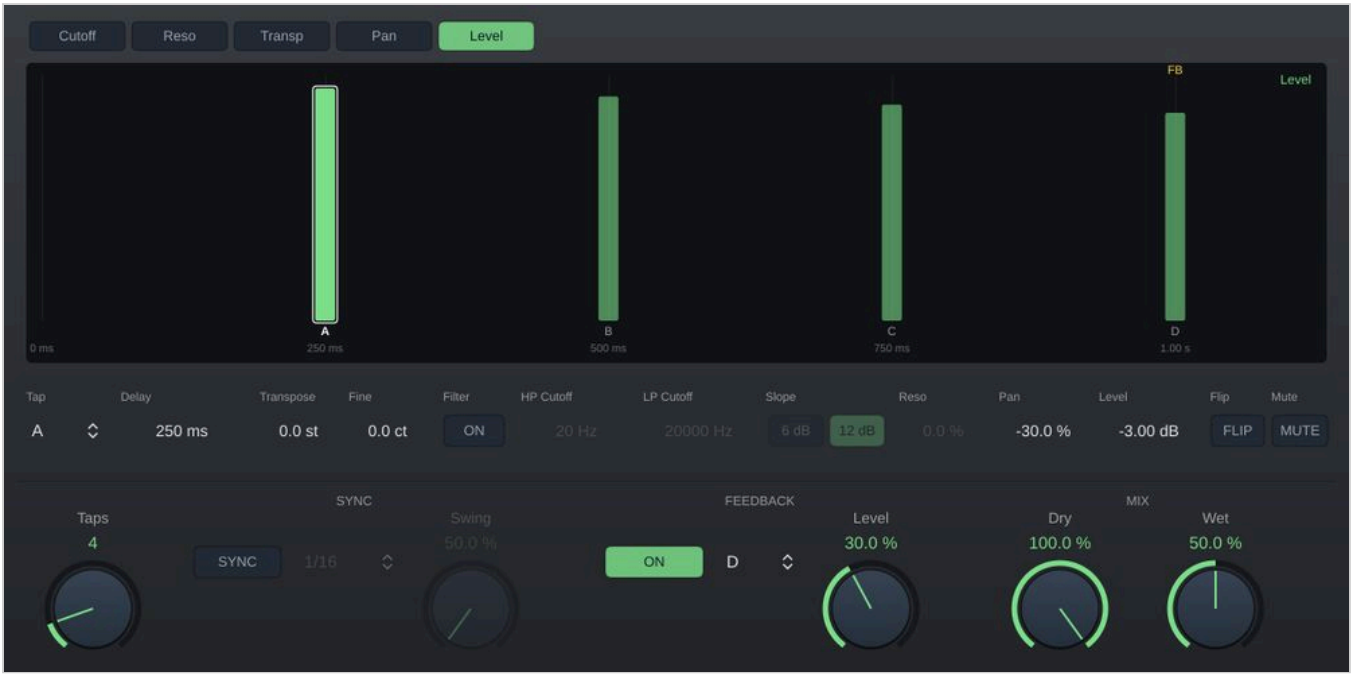


Delays each side by a number of samples (to line up mics, or for the Haas widening trick).

Controls

Left (0 to 10000 smp) · **Right** (0 to 10000 smp) · **Link L & R** (Off, On)

Delay Designer



Delay Designer: up to 26 taps (A to Z), each with its own time, level, pan, transposition, filter, flip and mute; taps can snap to a tempo grid with swing, and one tap feeds back.

> **How to use it:** Draw taps for rhythmic delays, each with its own pan, pitch and filter - great for glitchy and spacious effects.

Controls

Tap (A, B, C, D, E, F, G, H, I, J, ...) · **Taps** (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...) · **Sync** (Off, On) · **Grid** (1/4, 1/8, 1/16, 1/32, 1/4 T, 1/8 T, 1/16 T, 1/32 T) · **Swing** (50 to 100 %) · **Feedback** (Off, On) · **Feedback Tap** (A, B, C, D, E, F, G, H, I, J, ...) · **Feedback Level** (0 to 100 %) · **Dry** (0 to 100 %) · **Wet** (0 to 100 %) · **View** (Cutoff, Reso, Transp, Pan, Level)

Each tap (26, A to Z): Delay, Level, Pan, Transpose, Fine, Filter, HP Cutoff, LP Cutoff, Slope, Reso, Flip, Mute

Reverb Effects

Reverb puts the sound in a space - a room, a hall, a plate. **How to use it:** put one or two reverbs on aux buses and send tracks to them. Short rooms add life without distance; long halls and plates make big, smooth tails. Predelay (20-60 ms) keeps a vocal clear in front of its reverb; cut the reverb's lows to avoid mud.

In this chapter: **Reverb**, **Studio Reverb**, **Plate Reverb**, **Spring Reverb**, **Space Designer**, **PlatinumVerb**, **EnVerb**, **SilverVerb**, **Room Simulator**.

Reverb



The simple reverb: room size, damping (darker tail), stereo width and mix.

Controls

Size (0 to 1) · **Damping** (0 to 1) · **Width** (0 to 1) · **Mix** (0 to 1)

Studio Reverb



Studio Reverb: an algorithmic reverb with eight room types (Room, Chamber, Concert Hall, Theater, Synth Hall, Digital, Dark Room, Dense Room). Size, decay, predelay (or a note), attack, distance, density, modulation, width and the early/late balance; a four-band decay EQ (how long each range rings), Freeze, a mono maker for the lows and a five-band output EQ.

> **How to use it:** Choose a Room for drums, a Plate or Chamber for vocals, a Hall for strings and pads. Turn down the low decay for a cleaner mix.

Controls

Room Type (Room, Chamber, Concert Hall, Theater, Synth Hall, Digital, Dark Room, Dense Room, Smooth Space, Vocal Hall, ...) · **Size** (0 to 100 %) · **Decay** (0.1 to 20 s) · **Predelay** (0 to 2000 ms) · **Low Cut** (10 to 1000 Hz) · **High Cut** (1000 to 20000 Hz) · **Density** (0 to 100 %) · **Mod Depth** (0 to 100 %) · **Width** (0 to 200 %) · **Wet** (0 to 100 %) · **Freeze** (Off, On) · **Attack** (0 to 100 %) · **Distance** (0 to 100 %) · **Dry** (0 to 100 %) · **Predelay Sync** (Off, On) · **Predelay Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Decay Sync** (Off, On) · **Decay Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Low Freq** (20 to 1000 Hz) · **Low Ratio** (0.1 to 1.9 x) · **Low Mid Freq** (100 to 5000 Hz) · **Low Mid Ratio** (0.1 to 1.9 x) · **Low Mid Q** (0.1 to 4) · **High Mid Freq** (500 to 15000 Hz) · **High Mid Ratio** (0.1 to 1.9 x) · **High Mid Q** (0.1 to 4) · **High Freq** (1000 to 20000 Hz) · **High Ratio** (0.1 to 1.9 x) ·

Quality (Low, High, Ultra) · **Mod Speed** (0.05 to 10 Hz) · **Mod Source** (Sine, Random, Noise) · **Smoothing** (0 to 100 %) · **Early/Late Mix** (0 to 100 %) · **Mono Maker** (Off, On) · **Mono Maker Freq** (20 to 1000 Hz) · **Output EQ** (Off, On) · **EQ Low Cut** (Off, On) · **EQ Low Shelf** (Off, On) · **Low Shelf Freq** (20 to 2000 Hz) · **Low Shelf Gain** (-24 to 24 dB) · **EQ Low Mid** (Off, On) · **Low Mid Freq** (50 to 10000 Hz) · **Low Mid Gain** (-24 to 24 dB) · **Low Mid Q** (0.1 to 10) · **EQ High Mid** (Off, On) · **High Mid Freq** (200 to 18000 Hz) · **High Mid Gain** (-24 to 24 dB) · **High Mid Q** (0.1 to 10) · **EQ High Shelf** (Off, On) · **High Shelf Freq** (1000 to 20000 Hz) · **High Shelf Gain** (-24 to 24 dB) · **EQ High Cut** (Off, On) · **View** (Main, Details)

Plate Reverb



The classic plate (Jon Dattorro's 1997 design): a figure-of-eight tank with modulated diffusers.

> **How to use it:** The smooth, bright reverb of countless vocal and snare sounds.

Controls

Pre-Delay (0 to 200 ms) · **Decay** (0 to 100 %) · **Size** (0 to 100 %) · **Bandwidth** (0 to 100 %) · **Damping** (0 to 100 %) · **Modulation** (0 to 100 %) · **Low Cut** (10 to 1000 Hz) · **Width** (0 to 100 %) · **Mix** (0 to 100 %)

Spring Reverb



A spring tank: a chain of all-passes spreads each sound into the spring's chirp, round a feedback loop.

Controls

Springs (1, 2, 3) · **Tension** (0 to 100 %) · **Decay** (0 to 100 %) · **Dispersion** (0 to 100 %) · **Tone** (1000 to 10000 Hz) · **Drip** (0 to 100 %) · **Mix** (0 to 100 %)

Space Designer



Convolution with an impulse response made from the knobs (built again, off the audio thread, when they move).

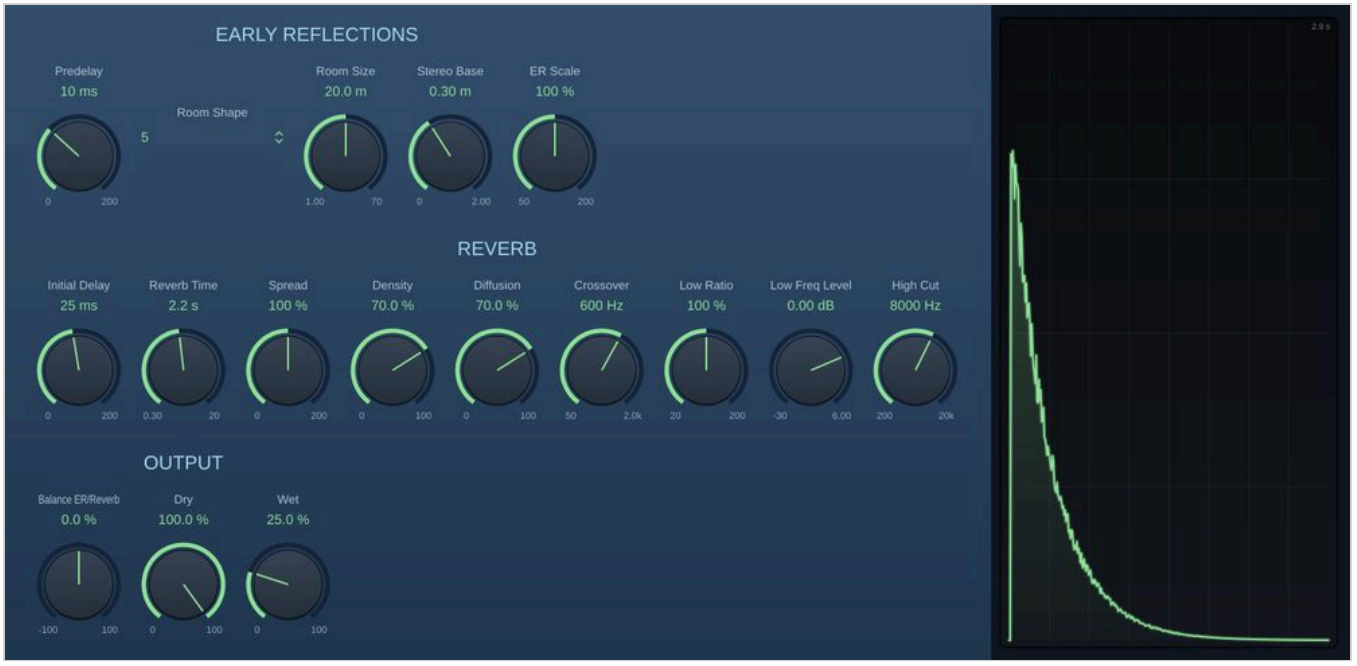
> **How to use it:** Real spaces from impulse responses: load your own IR (Load IR...) or shape one with the knobs.

Controls

Length (0.1 to 20 s) · **Pre-Delay** (0 to 500 ms) · **Early Reflections** (0 to 100 %) · **Bright Start** (1000 to 20000 Hz) · **Bright End** (200 to 20000 Hz) · **Density** (0 to 100 %) · **Spread** (0 to 200 %) · **Reverse** (Off, On) · **Reverb** (0 to 100 %) · **Low Cut** (10 to 1000 Hz) · **IR Mode** (Sampled IR, Synthesized IR) · **Quality** (Lo-Fi, Low, Medium, High) · **Size** (25 to 200 %) · **Volume Env** (Off, On) · **Init Level** (0 to 100 %) · **Attack** (0 to 2000 ms) · **Decay** (0.05 to 20 s) · **End Level** (0 to 100 %) · **Decay Shape** (Lin, Exp) · **Filter Env** (Off, On) · **Filter Type** (LP 6dB, LP 12dB, BP, HP) · **Resonance** (0 to 100 %) · **Init Freq** (50 to 20000 Hz) · **Filter Attack** (0 to 2000 ms) · **Break Level** (50 to 20000 Hz) · **Filter Decay** (0.05 to 20 s) · **End Freq** (50 to 20000 Hz) · **Density Env** (Off, On) · **Init Density** (0 to 100 %) · **Ramp Time** (0 to 2000 ms) · **End Density** (0 to 100 %) · **Reflection Shape** (0 to 100 %) · **Output EQ** (Off, On) · **EQ Low Cut** (Off, On) · **Low Cut Freq** (20 to 2000 Hz) · **EQ Low Shelf** (Off, On) · **Low Shelf Freq** (20

to 2000 Hz) · **Low Shelf Gain** (-24 to 24 dB) · **EQ Low Mid** (Off, On) · **Low Mid Freq** (50 to 10000 Hz) · **Low Mid Gain** (-24 to 24 dB) · **Low Mid Q** (0.1 to 10) · **EQ High Mid** (Off, On) · **High Mid Freq** (200 to 18000 Hz) · **High Mid Gain** (-24 to 24 dB) · **High Mid Q** (0.1 to 10) · **EQ High Shelf** (Off, On) · **High Shelf Freq** (1000 to 20000 Hz) · **High Shelf Gain** (-24 to 24 dB) · **EQ High Cut** (Off, On) · **High Cut Freq** (1000 to 20000 Hz) · **Input** (Stereo, Mono, XStereo) · **Direct** (-60 to 6 dB) · **X-Over** (0 to 1000 Hz) · **View** (Volume Env, Filter Env, Density Env, Output EQ)

PlatinumVerb



PlatinumVerb: early reflections from a room of 3 to 7 corners, its size, the stereo base of the two microphones and an ER scale; then the tail - initial delay, spread, two bands (crossover, low ratio, low level), high cut, density, diffusion, reverb time - and the balance.

Controls

Predelay (0 to 200 ms) · **Room Shape** (3, 4, 5, 6, 7) · **Room Size** (1 to 70 m) · **Stereo Base** (0 to 2 m) · **ER Scale** (50 to 200 %) · **Initial Delay** (0 to 200 ms) · **Spread** (0 to 200 %) · **Crossover** (50 to 2000 Hz) · **Low Ratio** (20 to 200 %) · **Low Freq Level** (-30 to 6 dB) · **High Cut** (200 to 20000 Hz) · **Density** (0 to 100 %) · **Diffusion** (0 to 100 %) · **Reverb Time** (0.3 to 20 s) · **Balance ER/Reverb** (-100 to 100 %) · **Dry** (0 to 100 %) · **Wet** (0 to 100 %)

EnVerb



EnVerb: a reverb shaped by an envelope - attack, decay to the sustain level, hold, then release - started by each new phrase; the dry signal can be delayed; density, spread, high cut, a crossover with the low band's level, dry and wet.

Controls

Attack (0 to 2000 ms) · **Decay** (0 to 3000 ms) · **Sustain** (0 to 100 %) · **Hold** (0 to 10000 ms) · **Release** (0.05 to 20 s) · **Dry Signal Delay** (0 to 500 ms) · **Density** (0 to 100 %) · **Spread** (0 to 200 %) · **High Cut** (200 to 20000 Hz) · **Crossover** (50 to 2000 Hz) · **Low Freq Level** (-30 to 6 dB) · **Dry** (0 to 100 %) · **Wet** (0 to 100 %)

SilverVerb



SilverVerb: predelay, reflectivity, size, density/time, low and high cut, and an LFO (on/off, rate, left-right phase, intensity) that wobbles the tail; dry and wet.

Controls

Predelay (0 to 200 ms) · **Reflectivity** (0 to 100 %) · **Size** (0 to 100 %) · **Density/Time** (0 to 100 %) · **Low Cut** (10 to 2000 Hz) · **High Cut** (1000 to 20000 Hz) · **Modulation** (Off, On) · **Rate** (0.05 to 10 Hz) · **Phase** (0 to 360 °) · **Intensity** (0 to 100 %) · **Dry** (0 to 100 %) · **Wet** (0 to 100 %)

Room Simulator



Room Simulator: a classic digital room algorithm (Classic or Modern). Reverb time, room size, density, predelay, the first reflections' level, delay and spread, high cut, a bass band (gain and crossover), correlation patterns for each side, Freeze, and the dry, early and reverb levels.

Controls

Model (Classic, Modern) · **Reverb Time** (0.07 to 400 s) · **Freeze** (Off, On) · **Add** (Off, On) · **Clear** (Off, On) · **Mode** (Complex, Medium, Simple) · **Enhance** (Off, On) · **Low** (0.1 to 10 x) · **Low Xover** (50 to 1000 Hz) · **High** (0.1 to 10 x) · **High Xover** (1000 to 16000 Hz) · **Parameters** (Primary, Secondary, Extended) · **Room Size** (1 to 1000000 m³) · **Reverb Density** (0 to 200 %) · **Reverb Delay** (0 to 200 ms) · **Dry Level** (-60 to 0 dB) · **1st Reflection Level** (-60 to 0 dB) · **Reverb Level** (-60 to 0 dB) · **1st Reflection Delay** (0 to 200 ms) · **1st Reflection Spread** (0 to 50 ms) · **Reverb High Cut** (1000 to 20000 Hz) · **Bass Gain** (-12 to 12 dB) · **Bass Crossover** (50 to 1000 Hz) · **Reverb Correlation** (0 to 150 %) · **Dry Source** (L R, R L) · **1st Reflection Source** (L R, R L) · **L Correlation Pattern** (Off, Pattern 1, Pattern 2, Pattern 3, Pattern 4, Pattern 5, Pattern 6, Pattern 7, Pattern 8) · **R Correlation Pattern** (Off, Pattern 1, Pattern 2, Pattern 3, Pattern 4, Pattern 5, Pattern 6, Pattern 7, Pattern 8) · **Subsonic** (Off, On)

Modulation Effects

Modulation effects move the sound in time: chorus and ensemble thicken, flanger and phaser sweep, tremolo and auto pan pulse, the rotor cabinet spins like an organ speaker. **How to use them:** start with the rate slow and the mix low, then raise until it's an effect you hear on purpose.

In this chapter: **Chorus**, **Ensemble**, **Flanger**, **Microphaser**, **Phaser**, **Tremolo**, **Auto Pan**, **Rotor Cabinet**, **Ringshifter**, **Scanner Vibrato**, **Vibrato**.

Chorus



Chorus: rate, intensity, delay, voices, spread, feedback, D-Mode, mix.

> **How to use it:** On clean guitars, electric pianos and pads for width and shimmer.

Controls

Rate (0.02 to 8 Hz) · **Intensity** (0 to 100 %) · **Delay** (3 to 40 ms) · **Voices** (1, 2, 3, 4) · **Spread** (0 to 100 %) · **Feedback** (0 to 80 %) · **Mix** (0 to 100 %) · **D-Mode** (Off, On)

Ensemble



Ensemble: up to eight voices modulated by LFO 1, LFO 2 and a random source (each on/off, rate, intensity), spread across the stereo field, their phases offset, volume compensation, mix.

Controls

Voices (2, 3, 4, 5, 6, 7, 8) · **Stereo Spread** (0 to 200 %) · **Output Mix** (0 to 100 %) · **LFO 1** (Off, On) · **LFO 1 Rate** (0.05 to 10 Hz) · **LFO 1 Intensity** (0 to 100 %) · **LFO 2** (Off, On) · **LFO 2 Rate** (0.05 to 20 Hz) · **LFO 2 Intensity** (0 to 100 %) · **Random** (Off, On) · **Random Rate** (0.05 to 10 Hz) · **Random Intensity** (0 to 100 %) · **Phase** (0 to 360 °) · **Volume Compensation** (0 to 100 %)

Flanger



Flanger: a very short delay swept by an LFO (rate or a note value), with intensity, feedback for the jet sound, stereo phase and mix.

Controls

Sync (Off, On) · **Rate** (0.02 to 10 Hz) · **Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Delay** (0.1 to 10 ms) · **Intensity** (0 to 100 %) · **Feedback** (-100 to 100 %) · **Stereo Phase** (0 to 180 °) · **Mix** (0 to 100 %)

Microphaser



Microphaser: a small, simple phaser - rate (or synced), intensity, feedback.

Controls

Sync (Off, On) · **Rate** (0.02 to 10 Hz) · **Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Intensity** (0 to 100 %) · **Feedback** (0 to 100 %)

Phaser

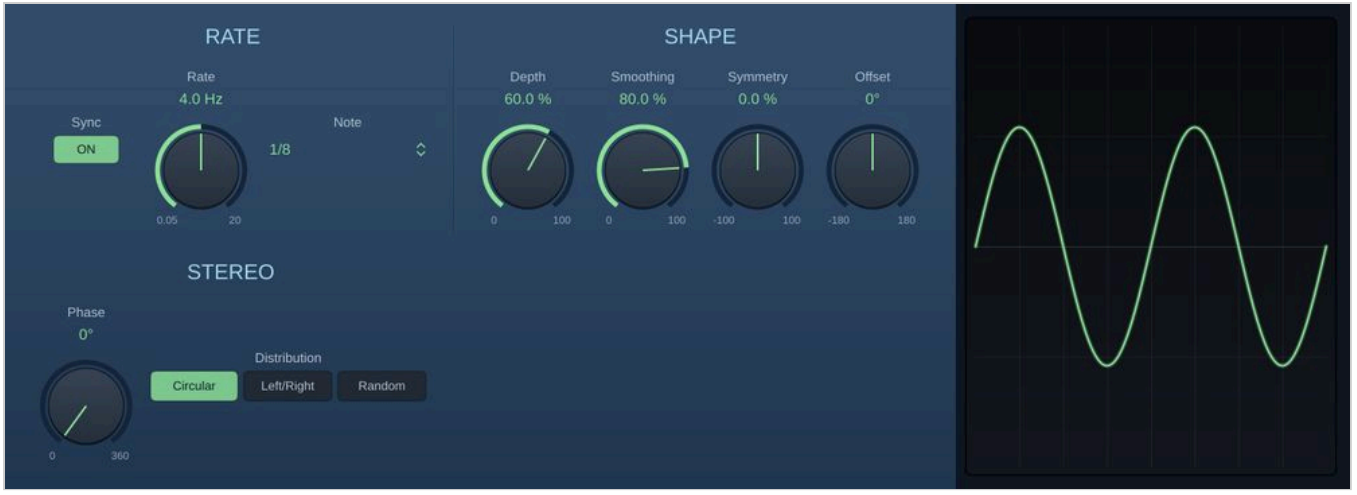


Phaser: 4 to 12 stages (odd numbers: comb filtering), the sweep between Floor and Ceiling by two LFOs (each its rate or a note, mixed), or by the input's envelope; the stereo phase and its distribution; feedback level, Warmth, a low/high cut filter, output mix.

Controls

Stages (4, 5, 6, 7, 8, 9, 10, 11, 12) · **LFO 1 Sync** (Off, On) · **LFO 1 Rate** (0.02 to 10 Hz) · **LFO 1 Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Ceiling** (200 to 18000 Hz) · **Floor** (20 to 5000 Hz) · **Level** (-100 to 100 %) · **Phase** (0 to 360 °) · **Output Mix** (0 to 100 %) · **Sweep Mode** (LFO, Envelope) · **LFO 2 Rate** (0.02 to 10 Hz) · **LFO 2 Sync** (Off, On) · **LFO 2 Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **LFO Mix** (0 to 100 %) · **Distribution** (Circular, Left/Right, Random) · **Warmth** (Off, On) · **Filter** (Off, On) · **Low Cut** (20 to 2000 Hz) · **High Cut** (1000 to 20000 Hz)

Tremolo



Tremolo: rate (or a note), depth, the wave's smoothing and symmetry, its offset, and the left-right phase with its distribution.

Controls

Sync (Off, On) · **Rate** (0.05 to 20 Hz) · **Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Depth** (0 to 100 %) · **Smoothing** (0 to 100 %) · **Phase** (0 to 360 °) · **Symmetry** (-100 to 100 %) · **Offset** (-180 to 180 °) · **Distribution** (Circular, Left/Right, Random)

Auto Pan



Auto Pan: equal-power balance swung left and right in a chosen shape.

Controls

Sync (Off, On) · **Rate** (0.05 to 20 Hz) · **Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Depth** (0 to 100 %) · **Shape** (Sine, Triangle, Square, Saw, Random)

Rotor Cabinet



Rotor Cabinet: Slow, Brake or Fast; the cabinet (Wood, Proline, Single, Split, and their "& Horn IR" versions); acceleration and maximum rate; motor control (Normal, Inverse, 910, Sync); the horn's and drum's microphones (Dynamic, Condenser or Mid-Side), front or rear, distance, angle, horn/drum balance, and the horn's and drum's width.

> **How to use it:** On organs: switch between Slow and Fast during the song (automate it) for the classic Leslie sound.

Controls

Rotation (Slow, Brake, Fast) · **Max Rate** (4 to 9 Hz) · **Acceleration** (0.2 to 6 s) · **Balance** (0 to 100 %) · **Doppler** (0 to 100 %) · **Drive** (0 to 100 %) · **Horn Width** (0 to 100 %) · **Cabinet** (Wood, Proline, Single, Split, Wood & Horn IR, Proline & Horn IR, Split & Horn IR) · **Motor Control** (Normal, Inv, 910, Sync) · **Upper Mic** (Dynamic, Condenser, Mid-Side) · **Lower Mic** (Dynamic, Condenser, Mid-Side) · **Mic Position** (Front, Rear) · **Distance** (0 to 100 %) · **Angle** (0 to 180 °) · **Drum Width** (0 to 100 %)

Ringshifter



Ringshifter: a frequency shifter (one shift, or opposite shifts left and right) or a ring modulator (by its sine oscillator, or by the side chain). The oscillator's frequency, moved by the envelope follower and the LFO; a delay (or synced) with its level; the envelope follower (sensitivity, attack, decay) and the LFO (symmetry, smoothing, rate or synced); then dry/wet (also moved by the envelope and LFO), feedback and stereo width.

Controls

Mode (Single Freq Shift, Dual Freq Shift, OSC Ring Mod, Side Chain Ring Mod) · **Frequency** (-5000 to 5000 Hz) · **Freq Env Follow** (-100 to 100 %) · **Freq LFO** (-100 to 100 %) · **Delay Time** (0 to 2000 ms) · **Delay Sync** (Off, On) · **Delay Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Delay Level** (0 to 100 %) · **Env Follower** (Off, On) · **Sensitivity** (0 to 100 %) · **Attack** (0.1 to 500 ms) · **Decay** (1 to 2000 ms) · **LFO** (Off, On) · **Symmetry** (-100 to 100 %) · **Smooth** (0 to 100 %) · **LFO Rate** (0.02 to 20 Hz) · **LFO Sync** (Off, On) · **LFO Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Dry/Wet** (0 to 100 %) · **Feedback** (0 to 100 %) · **Stereo Width** (0 to 200 %) · **Dry/Wet Env Follow** (-100 to 100 %) · **Dry/Wet LFO** (-100 to 100 %)

Scanner Vibrato



Scanner Vibrato: V1-V3 vibrato (only the scanned line heard) or C0-C3 chorus (scanned and dry together); depth, stereo phase, left and right rates.

Controls

Type (V1, V2, V3, C0, C1, C2, C3) · **Depth** (0 to 100 %) · **Stereo Phase** (0 to 360 °) · **Rate Left** (2 to 10 Hz) · **Rate Right** (2 to 10 Hz)

Vibrato



Pitch wobble: a short delay swept, heard alone (no dry).

Controls

Sync (Off, On) · **Rate** (0.1 to 14 Hz) · **Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Depth** (0 to 100 cents) · **Stereo Phase** (0 to 180 °) · **Mix** (0 to 100 %)

Distortion Effects

Distortion adds harmonics: warmth and weight at low drive, grit and aggression at high drive. **How to use it:** a saturator on a bass or 808 makes it heard on small speakers; a little on drums adds punch; use Mix to blend in the clean sound (parallel distortion).

In this chapter: **Overdrive**, **Distortion**, **Distortion II**, **Phase Distortion**, **Fuzz**, **Bitcrusher**, **Clip Distortion**, **Saturator**.

Overdrive



A pedal-style overdrive: mids pushed into a soft, slightly uneven clip, then a tone control.

Controls

Drive (0 to 60 dB) · **Tone** (500 to 12000 Hz) · **Output** (-30 to 12 dB) · **Mix** (0 to 100 %) · **Level Compensation** (Off, On)

Distortion



Harder distortion: lots of gain into a sharper clip.

Controls

Drive (0 to 60 dB) · **Tone** (500 to 12000 Hz) · **Output** (-30 to 12 dB) · **Mix** (0 to 100 %) · **Level Compensation** (Off, On)

Distortion II



Distortion II: pre gain, drive and tone (a high shelf before and after) through one of five circuits - Growl (a two-stage tube amp), Bity (an overdriven blues amp), Nasty (hard), Class AB soft and hard (transistor crossover distortion) - then output, mix and level compensation.

Controls

PreGain (-24 to 24 dB) · **Drive** (0 to 100 %) · **Tone** (0 to 100 %) · **Type** (Growl, Bity, Nasty, Class AB Soft, Class AB Hard) · **Output Level** (-30 to 12 dB) · **Mix** (0 to 100 %) · **Level Compensation** (Off, On)

Phase Distortion



Phase Distortion: the input's lows (a resonant low pass at Cutoff) bend a short delay on the signal itself, so its own bass moves its phase; Intensity, the delay's reach, Phase Reverse (the right channel moves the other way), Monitor (hear just the filter), mix.

Controls

Monitor (Off, On) · **Cutoff** (20 to 2000 Hz) · **Resonance** (0 to 100 %) · **Intensity** (0 to 100 %) · **Phase Reverse** (Off, On) · **Delay** (0.1 to 10 ms) · **Mix** (0 to 100 %)

Fuzz



Fuzz: a biased, nearly square clip with a gate for the hiss between notes.

Controls

Fuzz (0 to 60 dB) · **Bias** (0 to 100 %) · **Tone** (500 to 12000 Hz) · **Gate** (-100 to -20 dB) · **Output** (-30 to 12 dB)

Bitcrusher



Fewer bits and a lower sample rate (aliasing on purpose), with a drive and fold/clip/wrap at the top.

> **How to use it:** Lo-fi and chiptune: fewer bits for grit, a lower rate for aliasing.

Controls

Drive (0 to 24 dB) · **Resolution** (1 to 24 bit) · **Downsampling** (1 to 50 x) · **Mode** (Fold, Clip, Wrap) · **Mix** (0 to 100 %) · **Output** (-24 to 12 dB) · **Clip Level** (-24 to 0 dB)

Clip Distortion



Clip Distortion: input gain, drive into an asymmetric clip (Symmetry) through its clip filter, then a high shelf (frequency and gain) and a low pass, mix and output.

Controls

Drive (0 to 48 dB) · **Symmetry** (-100 to 100 %) · **Clip Filter** (200 to 20000 Hz) · **LP Filter** (500 to 20000 Hz) · **Mix** (0 to 100 %) · **Output Gain** (-30 to 12 dB) · **Input Gain** (-24 to 24 dB) · **High Shelving** (500 to 16000 Hz) · **Gain** (-24 to 24 dB)

Saturator



Saturator: five models - Retro Tube, Modern Tube, Magnetic (tape), Squeeze (pushed compression) and Analog Preamp - each with two styles (clean or colourful; soft or hard press),

drive, Bypass Below (the lows left untouched under a frequency), level in and out, mix, and low and high cut filters (slope, resonance, before or after the saturation).

> **How to use it:** On a bass or 808: Magnetic or Retro Tube, Bypass Below at 80 Hz so the sub stays clean, drive until it's heard on a phone speaker.

Controls

Model (Retro Tube, Modern Tube, Magnetic, Squeeze, Analog Preamp) · **Drive** (0 to 100 %) · **Style** (Clean, Colorful) · **Mix** (0 to 100 %) · **Level Out** (-24 to 12 dB) · **Level In** (-24 to 24 dB) · **Bypass Below** (Off, On) · **Bypass Below Freq** (20 to 500 Hz) · **Low Cut** (Off, On) · **Low Cut Slope** (6 dB/Oct, 12 dB/Oct, 24 dB/Oct) · **Low Cut Freq** (20 to 2000 Hz) · **Low Cut Resonance** (0 to 100 %) · **Low Cut Pre/Post** (Pre, Post) · **High Cut** (Off, On) · **High Cut Slope** (6 dB/Oct, 12 dB/Oct, 24 dB/Oct) · **High Cut Freq** (1000 to 20000 Hz) · **High Cut Resonance** (0 to 100 %) · **High Cut Pre/Post** (Pre, Post)

Filter Effects

Filters remove frequencies above or below a cutoff, often moving: sweeps, wahs, rhythmic gating. **How to use them:** automate the cutoff for builds and drops, or sync the movement to the song.

In this chapter: **AutoFilter**, **Auto Wah**, **Fuzz-Wah**, **Spectral Gate**, **Formant Filterbank**, **Step Filter**.

AutoFilter



AutoFilter: a resonant filter (HP, BP, LP, or a 4-pole low pass of 6 to 24 dB/oct) with Fatness and stereo Spread; its cutoff moved by an ADSR envelope (started above the threshold, scaled by Dynamic) and by an LFO (six waves, pulse width, rate or synced with phase, stereo phase, retrigger); distortion before and after the filter; dry signal and main out.

> **How to use it:** Sync the LFO to 1/4 or 1/8 for a rhythmic filter; automate the cutoff for a

> build.

Controls

Filter Type (HP, BP, LP, LP 6, LP 12, LP 18, LP 24) · **Cutoff** (30 to 18000 Hz) · **Resonance** (0 to 100 %) · **LFO Sync** (Off, On) · **LFO Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **LFO Rate** (0.02 to 20 Hz) · **LFO to Cutoff** (-100 to 100 %) · **Envelope to Cutoff** (-100 to 100 %) · **Attack** (0.5 to 2000 ms) · **Release** (10 to 5000 ms) · **Pre Filter Distortion** (0 to 100 %) · **Main Out** (-24 to 12 dB) · **Stereo Phase** (0 to 360 °) · **Filter** (Off, On) · **Fatness** (0 to 100 %) · **Spread** (0 to 2 oct) · **Envelope** (Off, On) · **Threshold** (-60 to 0 dB) · **Dynamic** (0 to 100 %) · **Decay** (1 to 5000 ms) · **Sustain** (0 to 100 %) · **LFO** (Off, On) · **Sync Phase** (0 to 360 °) · **Retrigger** (Off, On) · **Waveform** (Saw Down, Saw Up, Triangle, Pulse, S&H, Smooth) · **Pulse Width** (2 to 98 %) · **Distortion** (Off, On) · **Post Filter Distortion** (0 to 100 %) · **Pre Mode** (Soft, Hard, Tube) · **Post Mode** (Soft, Hard, Tube) · **Dry Signal** (0 to 100 %)

Auto Wah



An envelope wah: the louder you play, the higher the peak sweeps.

Controls

Sensitivity (0 to 36 dB) · **Range Low** (100 to 1000 Hz) · **Range High** (500 to 6000 Hz) · **Resonance** (0 to 100 %) · **Attack** (0.5 to 100 ms) · **Release** (10 to 1000 ms) · **Mode** (Band Pass, Low Pass) · **Mix** (0 to 100 %)

Fuzz-Wah



Fuzz-Wah: a wah (Classic, Retro, Modern, Opto 1 and 2, Resonant LP/HP, Peak) moved by the playing's level (depth, attack, release, range) with relative Q and auto gain; a compressor (ratio); a fuzz (gain, tone); and the order the three come in.

Controls

Wah (Off, On) · **Wah Type** (Classic Wah, Retro Wah, Modern Wah, Opto Wah 1, Opto Wah 2, Resonant LP, Resonant HP, Peak) · **Auto Gain** (Off, On) · **Relative Q** (0 to 100 %) · **Depth** (0 to 100 %) · **Attack** (0.5 to 200 ms) · **Release** (10 to 2000 ms) · **Range Low** (100 to 1000 Hz) · **Range High** (600 to 5000 Hz) · **Compressor** (Off, On) · **Ratio** (1 to 10 :1) · **Fuzz** (Off, On) · **Fuzz Gain** (0 to 48 dB) · **Tone** (0 to 100 %) · **Order** (Wah > Fuzz, Fuzz > Wah) · **Output** (-24 to 12 dB)

Spectral Gate



Spectral Gate: a band (centre frequency and bandwidth, both wobbled at Speed by CF and BW Modulation) split from the rest; the band's parts above the threshold (Super Energy) and below it (Sub Energy) at their own levels; the original lows and highs either side mixed back; gain.

Controls

Threshold (-80 to 0 dB) · **Speed** (0 to 10 Hz) · **CF Modulation** (0 to 100 %) · **BW Modulation** (0 to 100 %) · **Center Freq** (50 to 15000 Hz) · **Bandwidth** (50 to 10000 Hz) · **Low Level** (0 to 100 %) · **Super Energy** (0 to 200 %) · **Sub Energy** (0 to 200 %) · **High Level** (0 to 100 %) · **Gain** (-24 to 24 dB)

Formant Filterbank



Formant Filterbank: two banks of up to 20 band-pass filters between the lowest and highest frequency, each band's level set in bank A and bank B, crossfaded (Fade AB); formant shift, resonance, slope, the lowest band band-pass or high-pass and the highest band-pass or low-pass; a Shift LFO and a Fade LFO (rate, wave, intensity); overdrive, level, stereo width.

Controls

Low Frequency (20 to 2000 Hz) · **High Frequency** (1000 to 18000 Hz) · **Formant Shift** (-100 to 100 %) · **Bands** (5, 6, 7, 8, 9, 10, 11, 12, 13, 14, ...) · **Lowest** (Bandpass, Highpass) · **Highest** (Bandpass, Lowpass) · **Resonance** (0 to 100 %) · **Slope** (12 dB, 24 dB) · **Fade AB** (0 to 100 %) · **Shift LFO Intensity** (0 to 100 %) · **Shift LFO Rate** (0.02 to 20 Hz) · **Shift LFO Wave** (Saw Down, Saw Up, Triangle, Square, S&H, Smooth) · **Fade LFO Intensity** (0 to 100 %) · **Fade LFO Rate** (0.02 to 20 Hz) · **Fade LFO Wave** (Saw Down, Saw Up, Triangle, Square, S&H, Smooth) · **Overdrive** (Off, On) · **Level** (-24 to 12 dB) · **Stereo Width** (0 to 100 %)

Bank A (20, 1 to 20): Level

Bank B (20, 1 to 20): Level

Step Filter



Eight steps of cutoff, one per note value, gliding or jumping, in time with the song.

Controls

Step 1 (0 to 100 %) · **Step 2** (0 to 100 %) · **Step 3** (0 to 100 %) · **Step 4** (0 to 100 %) · **Step 5** (0 to 100 %) · **Step 6** (0 to 100 %) · **Step 7** (0 to 100 %) · **Step 8** (0 to 100 %) · **Step Length** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...) · **Range Low** (30 to 2000 Hz) · **Range High** (500 to 18000 Hz) · **Resonance** (0 to 100 %) · **Glide** (0 to 200 ms) · **Mix** (0 to 100 %)

Amps and Pedals

Guitar and bass amps, cabinets, microphones and stomp boxes. **How to use them:** record your guitar or bass clean through a DI (the instrument input of your interface), then choose the amp and pedals afterwards - you can change the sound at any time.

In this chapter: **Amp Designer**, **Bass Amp Designer**, **Pedalboard**.

Amp Designer



Amp Designer: 24 amps, 25 cabinets (and Direct), an EQ type for the tone stack, gain, bass, mids, treble, presence, master; spring and studio reverbs (ten types, level); tremolo or vibrato (depth, speed or synced); seven microphones, where they point and how close; output.

> **How to use it:** Choose the amp, then the cabinet and microphone - moving the mic changes the tone as much as the amp's knobs.

Controls

Amp (Small Tweed Combo, Large Tweed Combo, Mini Tweed Combo, Large Black Panel Combo, Silver Panel Combo, Mini Black Panel Combo, Small Brown Panel Combo, Blues Blaster Combo, Vintage British Stack, Modern British Stack, ...) · **Gain** (0 to 100) · **Bass** (0 to 10) · **Mids** (0 to 10) · **Treble** (0 to 10) · **Presence** (0 to 10) · **Master** (0 to 10) · **Cabinet** (Tweed 1 x 12, Tweed 4 x 10, Tweed 1 x 10, Black Panel 4 x 10, Silver Panel 2 x 12, Black Panel 1 x 10, Brown Panel 1 x 12, Brown Panel 1 x 15, Vintage British 4 x 12, Modern British 4 x 12, ...) · **Mic Position** (0 to 100 %) · **Output** (-24 to 12 dB) · **Mix** (0 to 100 %) · **EQ** (Vintage, British, Modern, Boutique) · **Mic Distance** (0 to 100 %) · **Mic** (Condenser 87, Condenser 414, Dynamic 20, Dynamic 57,

Dynamic 421, Dynamic 609, Ribbon 121) · **Reverb** (Off, On) · **Reverb Type** (Vintage Spring, Simple Spring, Mellow Spring, Bright Spring, Dark Spring, Resonant Spring, Boutique Spring, Sweet Reverb, Rich Reverb, Warm Reverb) · **Reverb Level** (0 to 100 %) · **Effects** (Off, On) · **Trem/Vib** (Tremolo, Vibrato) · **Depth** (0 to 100 %) · **Speed** (0.1 to 12 Hz) · **Sync/Free** (Free, Sync) · **Speed Note** (1/64, 1/32, 1/16T, 1/16, 1/16., 1/8T, 1/8, 1/8., 1/4T, 1/4, ...)

Bass Amp Designer



Bass Amp Designer: three amps (Classic, Flip Top, Modern), eight cabinets (and the two Direct outs), three microphones with their position; channel I/II, Bright, gain, bass/mids/treble with the Low, 1-2-3 and High switches, master; a compressor (hard/soft, amount, gain); an extra EQ (graphic, or two parametric bands), before or after the compressor; a D.I. box (boost, HF cut, six tone curves) blended with the amp; output.

Controls

Gain (0 to 100) · **Bass** (0 to 10) · **Mids** (0 to 10) · **Treble** (0 to 10) · **Master** (0 to 10) · **Cabinet** (Modern Cabinet 15", Modern Cabinet 10", Modern Cabinet 6", Classic Cabinet 8 x 10", Flip Top Cabinet 1 x 15", Modern 3 Way, Direct (PowerAmp Out), Direct (PreAmp Out)) · **Mic Position** (0 to 100 %) · **Blend** (0 to 100 %) · **DI Level** (-24 to 12 dB) · **Mix** (0 to 100 %) · **Output** (-24 to 12 dB) · **Amp** (Classic Amp, Flip Top Amp, Modern Amp) · **Mic** (Condenser 87, Dynamic 20, Dynamic 421) · **Mic Distance** (0 to 100 %) · **Channel** (I, II) · **Bright** (Off, On) · **EQ** (Off, On) · **Low** (Normal, Deep) · **Mid Freq** (1, 2, 3) · **High** (Normal, Bright) · **Compressor** (Off, On) · **Hard/Soft** (Hard, Soft) · **Comp** (0 to 100 %) · **Comp Gain** (-12 to 12 dB) · **Additional EQ** (Off, On) · **EQ Type** (Graphic, Parametric) · **Pre/Post** (Pre, Post) · **40** (-15 to 15 dB) · **80** (-15 to 15 dB) · **160** (-15 to 15 dB) · **320** (-15 to 15 dB) · **640** (-15 to 15 dB) · **1.25k** (-15 to 15 dB) · **2.5k** (-15 to 15 dB) · **LoMid Gain** (-15 to 15 dB) · **LoMid kHz** (0.05 to 1 kHz) · **LoMid Q** (0.3 to 4) · **HiMid Gain** (-15 to 15 dB) · **HiMid kHz** (0.5 to 8 kHz) · **HiMid Q** (0.3 to 4) · **DI Boost** (0 to 24 dB) · **HF Cut** (Off, On) · **DI Tone** (Off, On) · **DI Tone Curve** (1, 2, 3, 4, 5, 6) · **View** (Amp, Pedals, D.I. Box)

Pedalboard



Pedalboard: a chain of up to six stomp boxes, in order, each switched on or off with its three knobs. 23 pedals: fuzz, octave fuzz, overdrive, tube drive, distortion, grinder, bit crusher, compressor, chorus, flanger, phaser, microphaser, tremolo, vibrato, scanner vibrato, rotary, auto wah, fuzz wah, tape delay, echo, spring reverb, ring shifter and pitch.

> **How to use it:** Stack a compressor, overdrive, chorus and delay like a guitarist's pedal board.

Controls

Pedal 1 (Fuzz, Octave Fuzz, Overdrive, Tube Drive, Distortion, Grinder, Bit Crusher, Compressor, Chorus, Flanger, ...) · **Pedal 1 On** (Off, On) · **Pedal 1 Knob 1** (0 to 100 %) · **Pedal 1 Knob 2** (0 to 100 %) · **Pedal 1 Knob 3** (0 to 100 %) · **Pedal 2** (Fuzz, Octave Fuzz, Overdrive, Tube Drive, Distortion, Grinder, Bit Crusher, Compressor, Chorus, Flanger, ...) · **Pedal 2 On** (Off, On) · **Pedal 2 Knob 1** (0 to 100 %) · **Pedal 2 Knob 2** (0 to 100 %) · **Pedal 2 Knob 3** (0 to 100 %) · **Pedal 3** (Fuzz, Octave Fuzz, Overdrive, Tube Drive, Distortion, Grinder, Bit Crusher, Compressor, Chorus, Flanger, ...) · **Pedal 3 On** (Off, On) · **Pedal 3 Knob 1** (0 to 100 %) · **Pedal 3 Knob 2** (0 to 100 %) · **Pedal 3 Knob 3** (0 to 100 %) · **Pedal 4** (Fuzz, Octave Fuzz, Overdrive, Tube Drive, Distortion, Grinder, Bit Crusher, Compressor, Chorus, Flanger, ...) · **Pedal 4 On** (Off, On) · **Pedal 4 Knob 1** (0 to 100 %) · **Pedal 4 Knob 2** (0 to 100 %) · **Pedal 4 Knob 3** (0 to 100 %) · **Pedal 5** (Fuzz, Octave Fuzz, Overdrive, Tube Drive, Distortion, Grinder, Bit Crusher, Compressor, Chorus, Flanger, ...) · **Pedal 5 On** (Off, On) · **Pedal 5 Knob 1** (0 to 100 %) · **Pedal 5 Knob 2** (0 to 100 %) · **Pedal 5 Knob 3** (0 to 100 %) · **Pedal 6** (Fuzz, Octave Fuzz, Overdrive, Tube Drive, Distortion, Grinder, Bit Crusher, Compressor, Chorus, Flanger, ...) · **Pedal 6 On** (Off, On) · **Pedal 6 Knob 1** (0 to 100 %) · **Pedal 6 Knob 2** (0 to 100 %) · **Pedal 6 Knob 3** (0 to 100 %) · **Mix** (0 to 100 %) · **Output** (-24 to 12 dB)

Pitch Effects

Pitch effects tune, shift and transform voices and instruments. **How to use them:** Pitch Correction on a lead vocal (set the song's key and a response fast for the modern hard-tuned sound, slow for natural); Vocal Transformer for deep or high character voices; Pitch Shifter for octaves and harmonies.

In this chapter: **Pitch Correction**, **Vocal Transformer**, **Pitch Shifter**.

Pitch Correction



Pitch correction: each note pulled to the nearest note of the key, as fast as the Response says.

> **How to use it:** Set the key and scale of the song. A fast response gives the hard-tuned effect of modern trap and reggaeton; a slower one sounds natural.

Controls

Key (C, C#, D, D#, E, F, F#, G, G#, A, ...) · **Scale/Chord** (Chromatic, Major, Minor, Major Pentatonic, Minor Pentatonic, Harmonic Minor, Melodic Minor, Dorian, Phrygian, Lydian, ...) · **Response** (1 to 400 ms) · **Detune** (-50 to 50 cents) · **Amount** (0 to 100 %) · **Bypass All** (Off, On) · **Pitch Range** (Normal, Low) · **Tolerance** (0 to 100 cents) · **Reference** (-50 to 50 cents) · **Keyboard** (Edit Scale, Bypass Notes) · **C in Scale** (Off, On) · **C# in Scale** (Off, On) · **D in**

Scale (Off, On) · **D# in Scale** (Off, On) · **E in Scale** (Off, On) · **F in Scale** (Off, On) · **F# in Scale** (Off, On) · **G in Scale** (Off, On) · **G# in Scale** (Off, On) · **A in Scale** (Off, On) · **A# in Scale** (Off, On) · **B in Scale** (Off, On) · **Bypass C** (Off, On) · **Bypass C#** (Off, On) · **Bypass D** (Off, On) · **Bypass D#** (Off, On) · **Bypass E** (Off, On) · **Bypass F** (Off, On) · **Bypass F#** (Off, On) · **Bypass G** (Off, On) · **Bypass G#** (Off, On) · **Bypass A** (Off, On) · **Bypass A#** (Off, On) · **Bypass B** (Off, On) · **Mode** (Studio, Live)

Vocal Transformer



Pitch and formants moved on their own (a deeper voice that isn't slowed, a higher one that isn't a chipmunk), or a robot.

> **How to use it:** Pitch down with formants kept for a deep voice; robot mode for effects.

Controls

Pitch (-24 to 24 st) · **Formant** (-12 to 12 st) · **Mode** (Natural, Robot) · **Pitch Base** (C1, C#1, D1, D#1, E1, F1, F#1, G1, G#1, A1, ...) · **Mix** (0 to 100 %) · **Tracking** (-100 to 100 %) · **Glide** (0 to 500 ms) · **Grain Size** (Auto, Small, Normal, Large) · **Formants** (Process Always, Keep Unvoiced) · **Detune** (-50 to 50 cents)

Pitch Shifter



Shifts anything (chords, drums) by two crossfading delay taps sweeping through a window.

Controls

Semitones (-24 to 24 st) · **Cents** (-100 to 100) · **Window** (20 to 150 ms) · **Mix** (0 to 100 %) · **Delay** (0 to 100 ms) · **Timing** (Manual, Drums, Speech, Vocals, Pitch Tracking) · **Crossfade** (5 to 100 %) · **Stereo Link** (Normal, Invert) · **Latency Comp** (Off, On)

Imaging Effects

Imaging effects place sound in the stereo field: wider, narrower, moved left or right. **How to use them:** widen pads, backing vocals and effects; keep bass, kick and lead vocal in the centre. Check the mix in mono.

In this chapter: **Stereo Spread**, **Direction Mixer**, **Spreader**.

Stereo Spread



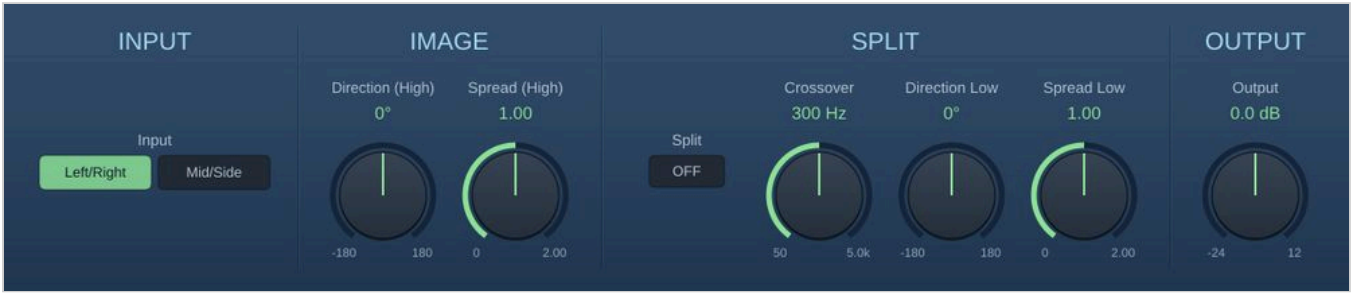
Widens by sending alternate frequency bands left and right (mono-compatible: it only adds side).

> **How to use it:** On pads and backing vocals; keep it off bass and kick.

Controls

Bands (2, 3, 4, 5, 6, 7, 8) · **Lower Intensity** (0 to 100 %) · **Upper Intensity** (0 to 100 %) · **Lower Freq** (80 to 2000 Hz) · **Upper Freq** (1000 to 18000 Hz) · **Mix** (0 to 100 %)

Direction Mixer



Direction Mixer: an LR or MS input, the direction its centre points to (degrees: 90 hard left, -90 hard right, beyond that turning back) and its spread (0 mono, 1 as is, 2 only the sides); Split gives the lows and highs (either side of the crossover) their own direction and spread.

Controls

Input (Left/Right, Mid/Side) · **Spread (High)** (0 to 2) · **Direction (High)** (-180 to 180 °) · **Output** (-24 to 12 dB) · **Split** (Off, On) · **Crossover** (50 to 5000 Hz) · **Direction Low** (-180 to 180 °) · **Spread Low** (0 to 2)

Spreader



Wider: each side a few milliseconds apart and a few cents apart, gently moving.

Controls

Delay (0 to 30 ms) · **Pitch Spread** (0 to 30 cents) · **Rate** (0.05 to 3 Hz) · **Width** (0 to 200 %) · **Mix** (0 to 100 %)

Multi Effects

Multi-effects chain several effects, moved by step sequencers, LFOs and XY pads - for sound design, builds, glitches and performance.

In this chapter: **Phat FX**, **Step FX**, **Remix FX**, **Beat Breaker**.

Phat FX



Phat FX: six effects in an order you choose - a band pass, a filter (ladder, classic and phat types), a distortion (eight types, in three stages), modulation, a bass enhancer and a compressor - with input, output, mix and a soft or hard limiter. Two LFOs, an envelope follower and an XY pad move them. Makes drums, basses and synths bigger and dirtier.

> **How to use it:** Makes a bass, drum loop or synth bigger, dirtier and more aggressive.

Controls

Input (-24 to 24 dB) · **Output** (-24 to 24 dB) · **Mix** (0 to 100 %) · **Limiter** (Off, Soft, Hard) · **Order 1** (Bandpass, Filter, Distortion, Mod FX, Bass Enhancer, Compressor) · **Order 2** (Bandpass, Filter, Distortion, Mod FX, Bass Enhancer, Compressor) · **Order 3** (Bandpass, Filter, Distortion, Mod FX, Bass Enhancer, Compressor) · **Order 4** (Bandpass, Filter, Distortion, Mod FX, Bass Enhancer, Compressor) · **Order 5** (Bandpass, Filter, Distortion, Mod FX, Bass Enhancer, Compressor) · **Order 6** (Bandpass, Filter, Distortion, Mod FX, Bass Enhancer,

Compressor) · **Bandpass** (Off, On) · **Bandpass Type** (12 dB, 24 dB) · **Low** (20 to 2000 Hz) · **High** (200 to 20000 Hz) · **Low Res** (0 to 100 %) · **High Res** (0 to 100 %) · **Reject Mix** (0 to 100 %) · **Filter** (Off, On) · **Filter Type** (Ladder LP 24, Classic LP 12, Classic BP 12, Classic HP 12, Phat LP 24, Phat HP 24, Notch, Peak) · **Cutoff** (20 to 20000 Hz) · **Res** (0 to 100 %) · **Drive** (0 to 100 %) · **Filter Mix** (0 to 100 %) · **Distortion** (Off, On) · **Type 1** (Bitcrusher, Diode, Dirt, Downsampler, Exciter, Grit, Scream, Soft Saturation, Squared, Tube, ...) · **Amount 1** (0 to 100 %) · **Type 2** (Bitcrusher, Diode, Dirt, Downsampler, Exciter, Grit, Scream, Soft Saturation, Squared, Tube, ...) · **Amount 2** (0 to 100 %) · **Type 3** (Bitcrusher, Diode, Dirt, Downsampler, Exciter, Grit, Scream, Soft Saturation, Squared, Tube, ...) · **Amount 3** (0 to 100 %) · **Mod FX** (Off, On) · **Mod FX Type** (Chorus, Wide Chorus, Ensemble, Flanger) · **Mod Mix** (0 to 100 %) · **Mod Rate** (0.05 to 10 Hz) · **Bass Enhancer** (Off, On) · **Bass Type** (Warm, Punch, Sub) · **Bass Amount** (0 to 100 %) · **Tune** (30 to 200 Hz) · **Compressor** (Off, On) · **Compressor Type** (Classic, Clip, Platinum, Studio VCA, Studio FET, Classic VCA, Vintage VCA, Vintage FET, Vintage Opto) · **Comp Amount** (0 to 100 %) · **Comp Release** (10 to 1000 ms) · **LFO 1** (Off, On) · **LFO 1 Waveform** (Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random) · **LFO 1 Target** (Off, Input, Output, Mix, Bandpass Low, Bandpass High, Reject Mix, Filter Cutoff, Filter Res, Filter Drive, ...) · **LFO 1 Rate** (0.05 to 20 Hz) · **LFO 1 Sync** (Off, On) · **LFO 1 Sync Rate** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **LFO 1 Depth** (-100 to 100 %) · **LFO 2** (Off, On) · **LFO 2 Waveform** (Sine, Triangle, Saw Up, Saw Down, Square, Sample & Hold, Smooth Random) · **LFO 2 Target** (Off, Input, Output, Mix, Bandpass Low, Bandpass High, Reject Mix, Filter Cutoff, Filter Res, Filter Drive, ...) · **LFO 2 Rate** (0.05 to 20 Hz) · **LFO 2 Sync** (Off, On) · **LFO 2 Sync Rate** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **LFO 2 Depth** (-100 to 100 %) · **Envelope Follower** (Off, On) · **Envelope Target** (Off, Input, Output, Mix, Bandpass Low, Bandpass High, Reject Mix, Filter Cutoff, Filter Res, Filter Drive, ...) · **Attack** (0.1 to 200 ms) · **Release** (5 to 2000 ms) · **Envelope Depth** (0 to 1000 %) · **XY Pad X** (0 to 1) · **XY Pad Y** (0 to 1) · **X** (Off, On) · **Y** (Off, On) · **X Target 1** (Off, Input, Output, Mix, Bandpass Low, Bandpass High, Reject Mix, Filter Cutoff, Filter Res, Filter Drive, ...) · **X Target 2** (Off, Input, Output, Mix, Bandpass Low, Bandpass High, Reject Mix, Filter Cutoff, Filter Res, Filter Drive, ...) · **X Depth 1** (-100 to 100 %) · **X Depth 2** (-100 to 100 %) · **Y Target 1** (Off, Input, Output, Mix, Bandpass Low, Bandpass High, Reject Mix, Filter Cutoff, Filter Res, Filter Drive, ...) · **Y Target 2** (Off, Input, Output, Mix, Bandpass Low, Bandpass High, Reject Mix, Filter Cutoff, Filter Res, Filter Drive, ...) · **Y Depth 1** (-100 to 100 %) · **Y Depth 2** (-100 to 100 %)

Step FX



Step FX: a filter, distortion, modulation, delay and reverb, moved by three step sequencers in time with the song (gate, cutoff, resonance, amounts) and by an XY pad you can play live. Rhythmic gating and moving filters for loops, pads and builds.

> **How to use it:** On a pad or a loop for rhythmic gating, or automate the XY pad for builds.

Controls

Input (-24 to 24 dB) · **Mix** (0 to 100 %) · **Output** (-24 to 24 dB) · **Step Modulator** (Step 1, Step 2, Step 3) · **Filter** (Off, On) · **Filter Type** (Ladder LP 24, Classic LP 12, Classic BP 12, Classic HP 12, Phat LP 24, Phat HP 24, Notch, Peak) · **Cutoff** (20 to 20000 Hz) · **Res** (0 to 100 %) · **Filter Mix** (0 to 100 %) · **Distortion** (Off, On) · **Exciter** (0 to 100 %) · **Dirt** (0 to 100 %) · **Mod FX** (Off, On) · **Mod Delay** (0.5 to 30 ms) · **Mod Rate** (0.05 to 10 Hz) · **Mod Depth** (0 to 100 %) · **Mod Feedback** (-100 to 100 %) · **Mod Mix** (0 to 100 %) · **Delay** (Off, On) · **Delay Filter** (LP, BP, HP, LP + LoCut) · **Delay Feedback** (0 to 100 %) · **Delay Cutoff** (100 to 20000 Hz) · **Delay Mix** (0 to 100 %) · **Delay Sync** (Off, On) · **Time L** (1 to 2000 ms) · **Time R** (1 to 2000 ms) · **Time L (Sync)** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **Time R (Sync)** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **Reverb** (Off, On) · **Reverb Type** (Room, Hall, Plate) · **Reverb Time** (0.2 to 10 s) · **Reverb Mix** (0 to 100 %) · **XY Pad X** (0 to 1) · **XY Pad Y** (0 to 1) · **X** (Off, On) · **Y** (Off, On) · **X Target 1** (Off, Gate (Mix), Gate (Add), Filter Cutoff, Filter Res, Filter Mix, Exciter, Dirt, Mod FX Rate, Mod FX Depth, ...) · **X Target 2** (Off, Gate (Mix), Gate (Add), Filter Cutoff, Filter Res, Filter Mix, Exciter, Dirt, Mod FX Rate, Mod FX Depth, ...) · **X Depth 1** (-100 to 100 %) · **X Depth 2** (-100 to 100 %) · **Y Target 1** (Off, Gate (Mix), Gate

(Add), Filter Cutoff, Filter Res, Filter Mix, Exciter, Dirt, Mod FX Rate, Mod FX Depth, ...) · **Y Target 2** (Off, Gate (Mix), Gate (Add), Filter Cutoff, Filter Res, Filter Mix, Exciter, Dirt, Mod FX Rate, Mod FX Depth, ...) · **Y Depth 1** (-100 to 100 %) · **Y Depth 2** (-100 to 100 %) · **Step Modulator 1** (Off, On) · **Step 1 Target** (Off, Gate (Mix), Gate (Add), Filter Cutoff, Filter Res, Filter Mix, Exciter, Dirt, Mod FX Rate, Mod FX Depth, ...) · **Step 1 Depth** (-100 to 100 %) · **Step 1 Rate** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **Step 1 Swing** (0 to 100 %) · **Step 1 Attack** (0 to 100 %) · **Step 1 Hold** (0 to 100 %) · **Step 1 Release** (0 to 100 %) · **Step 1 Length** (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...) · **Step Modulator 2** (Off, On) · **Step 2 Target** (Off, Gate (Mix), Gate (Add), Filter Cutoff, Filter Res, Filter Mix, Exciter, Dirt, Mod FX Rate, Mod FX Depth, ...) · **Step 2 Depth** (-100 to 100 %) · **Step 2 Rate** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **Step 2 Swing** (0 to 100 %) · **Step 2 Attack** (0 to 100 %) · **Step 2 Hold** (0 to 100 %) · **Step 2 Release** (0 to 100 %) · **Step 2 Length** (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...) · **Step Modulator 3** (Off, On) · **Step 3 Target** (Off, Gate (Mix), Gate (Add), Filter Cutoff, Filter Res, Filter Mix, Exciter, Dirt, Mod FX Rate, Mod FX Depth, ...) · **Step 3 Depth** (-100 to 100 %) · **Step 3 Rate** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **Step 3 Swing** (0 to 100 %) · **Step 3 Attack** (0 to 100 %) · **Step 3 Hold** (0 to 100 %) · **Step 3 Release** (0 to 100 %) · **Step 3 Length** (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...)

Each step (32, 1 to 32): Level, Length

Remix FX



A DJ-style multi-effect: two XY pads, a gater and a downsampler, reverse, scratch and tape stop.

> **How to use it:** Perform it live on the master or a bus: gater, filter, tape stop and scratch for transitions.

Controls

Gater (0 to 100 %) · **Gater Rate** (1/1, 1/2, 1/4, 1/8, 1/8 T, 1/16, 1/16 T, 1/32, 1/64) · **Downsampler** (0 to 100 %) · **Downsampler Type** (Downsample, Crush, Lo-Fi) · **Reverse** (Off, On) · **Reverse Length** (1/1, 1/2, 1/4, 1/8, 1/16) · **Scratch** (Off, On) · **Scratch Speed** (1/1, 1/2, 1/4, 1/8, 1/16) · **Tape Stop** (Off, On) · **Tape Stop Length** (1/1, 1/2, 1/4, 1/8, 1/16) · **Pad A Effect** (Filter, Repeater, Wobble, Reverb, Orbit, Delay) · **Pad A X** (0 to 1) · **Pad A Y** (0 to 1) · **Pad A Active** (Off, On) · **Pad A Lock** (Off, On) · **Pad B Effect** (Filter, Repeater, Wobble, Reverb, Orbit, Delay) · **Pad B X** (0 to 1) · **Pad B Y** (0 to 1) · **Pad B Active** (Off, On) · **Pad B Lock** (Off, On) · **Filter Type** (Phat 24 dB, Classic 12 dB) · **Reverb Color** (Dark, Medium, Bright) · **Orbit Type** (Flanger, Phaser)

Beat Breaker



Beat Breaker: the incoming audio, a pattern long, played back in slices - each from its own place in the input, at its own speed (backwards too), repeated, filtered, faded and panned.

> **How to use it:** Chop and rearrange a drum loop live: choose a pattern and edit slices.

Controls

Length (1 Beat, 2 Beats, 4 Beats, 8 Beats, 16 Beats) · **De-click** (0 to 100 %) · **Bypass Below** (Off, On) · **Bypass Below Freq** (20 to 500 Hz) · **Mix** (0 to 100 %) · **Edit Mode** (Time, Repeat, Cutoff, Resonance, Volume, Pan) · **Slice** (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...) · **Slices** (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ...) · **Pattern** (Straight, Stutter, Reverse, Halftime, Double, Tape Stop, Gated, Shuffle, Filter Sweep, Ping Pong) · **Pattern Loaded** (0 to 100) · **Pattern Off** (Off, On) · **Randomize** (Off, On) · **Randomize All** (Off, On)

Each slice (32, 1 to 32): Output Beat, Input Beat, Speed, Repeat, Cutoff, Resonance, Volume, Volume Slope, Pan

Specialized Effects

Effects for special jobs: vocoding, adding brightness, a sub octave, widening.

In this chapter: **Vocoder**, **Exciter**, **Sub Bass**.

Vocoder



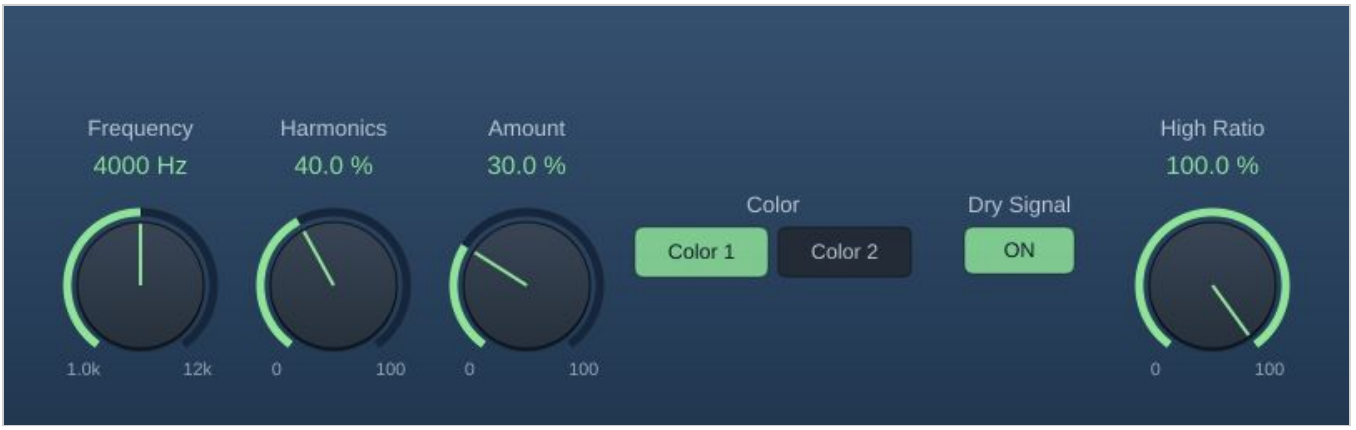
The classic vocoder: the track's spectrum (bands followed) shapes a carrier - the side chain or a built-in synth.

> **How to use it:** Put it on a synth track, set the side chain to your vocal track, and play chords: the synth talks.

Controls

Bands (8, 12, 16, 24, 32) · **Lowest** (50 to 500 Hz) · **Highest** (2000 to 14000 Hz) · **Attack** (0.5 to 50 ms) · **Release** (5 to 500 ms) · **Formant** (-12 to 12 st) · **Carrier** (Side Chain, Saw, Saw + Noise, Noise) · **Carrier Note** (C0, C#0, D0, D#0, E0, F0, F#0, G0, G#0, A0, ...) · **Unvoiced** (0 to 100 %) · **Mix** (0 to 100 %)

Exciter



Adds brightness: new harmonics made from the highs only.

> **How to use it:** A little on a dull vocal or overheads adds air and presence.

Controls

Frequency (1000 to 12000 Hz) · **Harmonics** (0 to 100 %) · **Amount** (0 to 100 %) · **Color** (Color 1, Color 2) · **Dry Signal** (Off, On) · **High Ratio** (0 to 100 %)

Sub Bass



An octave (or two) below the bass, made from its zero crossings and following its level.

> **How to use it:** Adds weight under a thin bass line or kick; keep it subtle and check on headphones.

Controls

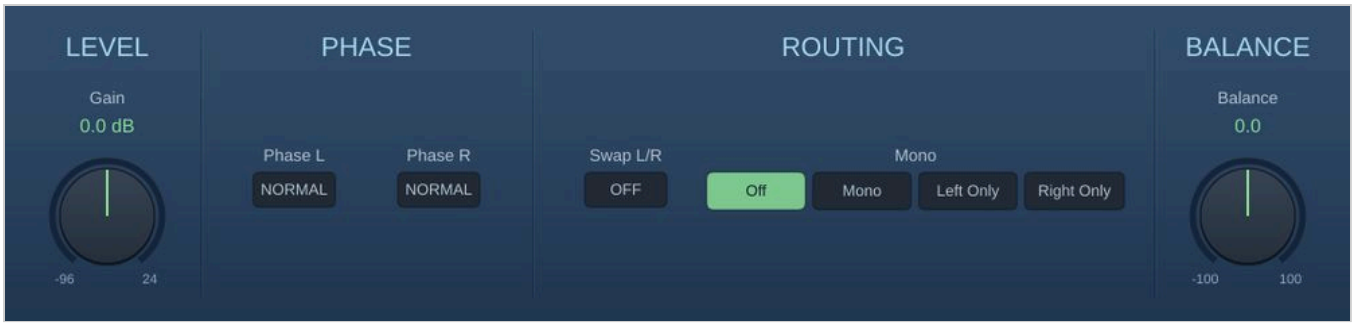
Bass Range (60 to 300 Hz) · **Octave** (-1, -2) · **Sub Level** (-60 to 6 dB) · **Dry Level** (-60 to 6 dB) · **Shape** (Sine, Square)

Utility Effects

Tools for fixing and testing: level, polarity, mono, test tones.

In this chapter: **Gain**, **Test Oscillator**.

Gain



Gain: level up or down, invert the phase of either side, swap left and right, sum to mono (or keep just the left or right), and balance. The utility for fixing polarity and checking mono.

> **How to use it:** Mono on the master checks how the mix sounds on a phone speaker; Phase invert fixes two mics cancelling each other.

Controls

Gain (-96 to 24 dB) · **Phase L** (Normal, Inverted) · **Phase R** (Normal, Inverted) · **Swap L/R** (Off, On) · **Mono** (Off, Mono, Left Only, Right Only) · **Balance** (-100 to 100)

Test Oscillator



A test tone, noise or a sweep, in place of (or added to) the track.

Controls

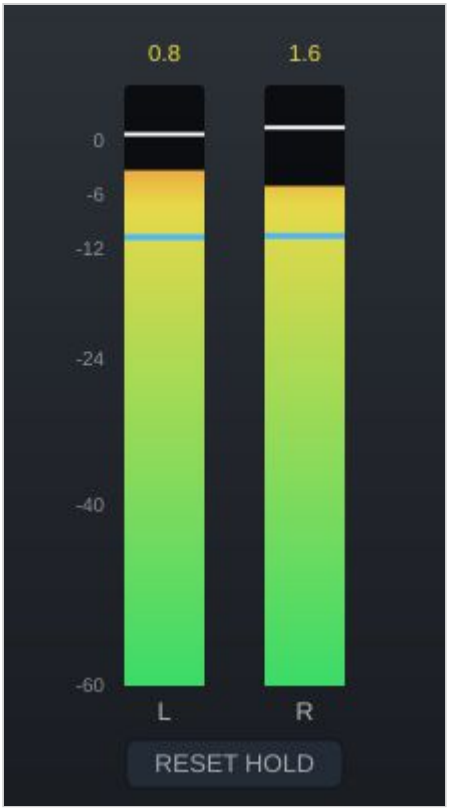
Waveform (Sine, Square, Saw, Triangle, White Noise, Pink Noise, Sweep) · **Frequency** (20 to 20000 Hz) · **Level** (-60 to 0 dB) · **Sweep Time** (0.5 to 30 s) · **Signal** (Replace, Add)

Meters and Mastering

Meters and analyzers that show what you hear: levels, loudness, stereo image, tuning, tempo - and the Mastering Assistant.

In this chapter: **Level Meter**, **Loudness Meter**, **Correlation Meter**, **MultiMeter**, **Tuner**, **BPM Counter**, **Mastering Assistant**.

Level Meter

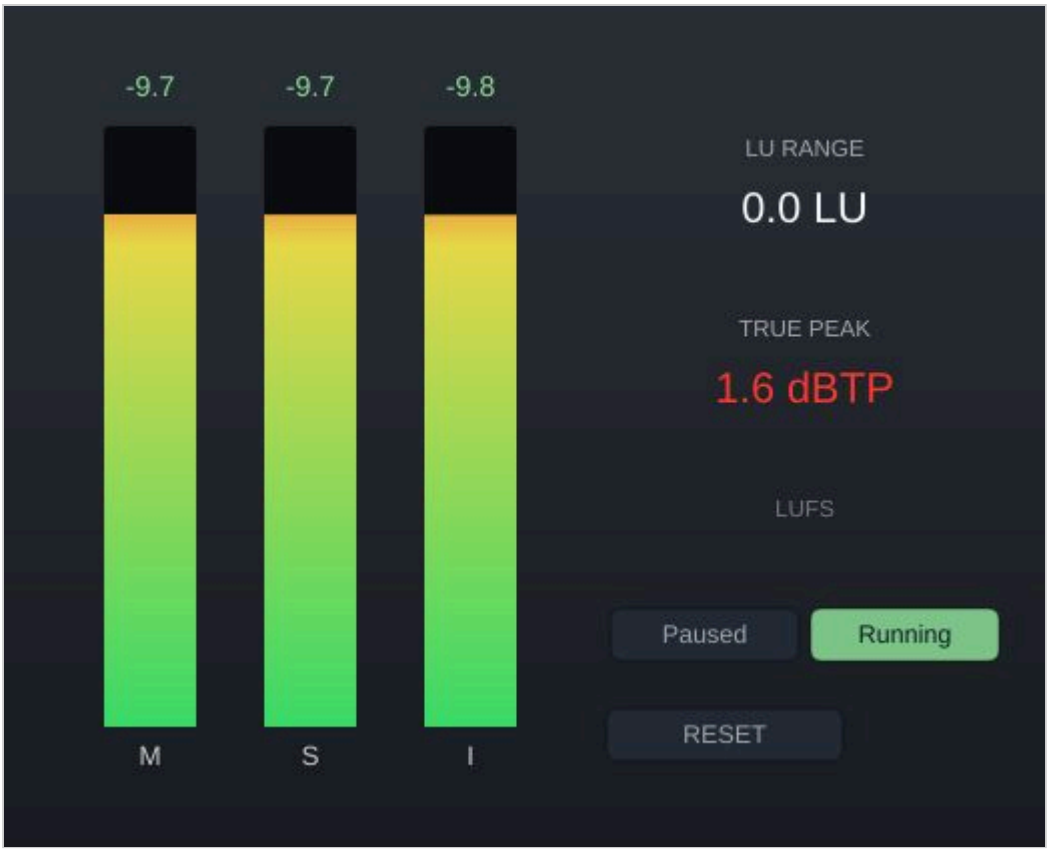


Level Meter: peak and RMS of each side, with the highest peak held (for the window).

Controls

Reset Hold (Off, On)

Loudness Meter



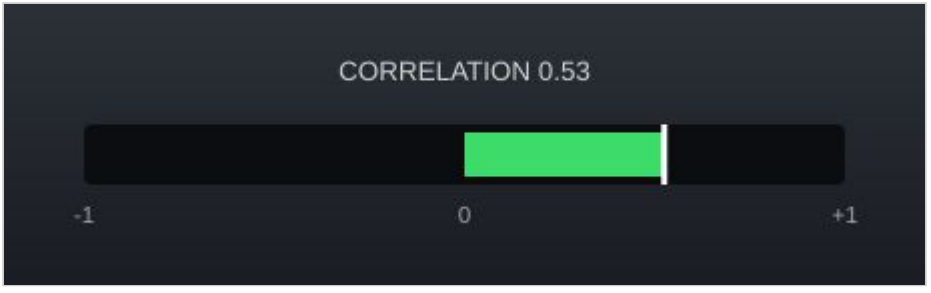
Loudness Meter: M, S and I in LUFS, the loudness range in LU, the true peak; Start/Pause and Reset.

> **How to use it:** Play the whole song and read I (integrated): -14 LUFS for streaming.

Controls

Start/Pause (Paused, Running) · **Reset** (Off, On) · **Scale** (EBU +9, EBU +18)

Correlation Meter



Correlation Meter: +1 in phase (mono), 0 wide, -1 out of phase.

MultiMeter



MultiMeter: a 31-band third-octave analyzer, a goniometer, correlation and peak levels.

> **How to use it:** Check the tonal balance and stereo image of your mix against a reference song.

Controls

Peak Hold (Off, On) · **View** (Analyzer, Goniometer) · **Tilt** (Off, 1.5 dB/oct, 3 dB/oct, 4.5 dB/oct, 6 dB/oct)

Tuner



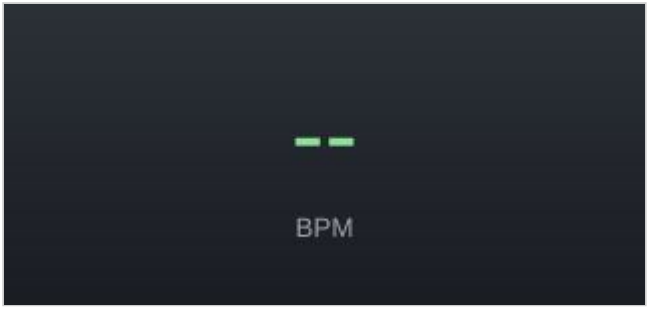
Tuner: the note played and how far off it is, against the reference pitch.

> **How to use it:** Put it on a guitar or bass track and tune before recording.

Controls

Reference (415 to 466 Hz)

BPM Counter



BPM Counter: the tempo found from the onsets of what's playing.

Mastering Assistant



Mastering Assistant: listens to the mix (the first 30 seconds played after Reanalyze), then sets a corrective EQ (Auto EQ, how much of it), the gain to reach the loudness you choose, and runs the mix through a character - Clean, Valve, Punch or Transparent - into a true peak limiter at -1 dBTP; Excite adds upper-mid saturation, Width moves the sides (above 150 Hz), a custom EQ (low, mid, high); Loudness Compensation plays it back at the original loudness for a fair comparison.

> **How to use it:** Put it last on the master. Play the loudest part after Reanalyze; then choose a character and adjust Loudness by ear.

Controls

Character (Clean, Valve, Punch, Transparent) · **Auto EQ** (0 to 100 %) · **Loudness** (-24 to -6 LUFS) · **Loudness Compensation** (Off, On) · **Excite** (Off, On) · **Width** (0 to 200 %) · **Custom EQ** (Off, On) · **Custom Low** (-6 to 6 dB) · **Custom Mid** (-6 to 6 dB) · **Custom High** (-6 to 6 dB) · **Reanalyze** (Off, On) · **Bypass** (Off, On)

Legacy Effects

Older designs kept for their sound and for older projects: simpler reverbs, compressors, gates and amps.

In this chapter: **EQ**, **Denoiser**, **AVerb**, **GoldVerb**, **Silver Compressor**, **Silver Gate**, **DeEsser**, **Speech Enhancer**, **Grooveshifter**, **Guitar Amp Pro**, **Bass Amp**.

EQ



The simple three-band EQ: a low shelf, a mid bell (with Q) and a high shelf, each with its frequency and gain (up to 18 dB).

Controls

Low Freq (20 to 1000 Hz) · **Low** (-18 to 18 dB) · **Mid Freq** (100 to 10000 Hz) · **Mid** (-18 to 18 dB) · **Mid Q** (0.2 to 10) · **High Freq** (1000 to 18000 Hz) · **High** (-18 to 18 dB)

Denoiser



Spectral noise reduction: each frequency below the threshold is turned down, smoothly over time.

Controls

Threshold (-100 to -20 dB) · **Reduction** (0 to 40 dB) · **Smoothing** (0 to 100 %)

AVerb



AVerb: one Density/Time control for both the reflections and the tail.

Controls

Predelay (0 to 200 ms) · **Reflectivity** (0 to 100 %) · **Room Size** (0 to 100 %) · **Density/Time** (0 to 100 %) · **Mix** (0 to 100 %)

GoldVerb



GoldVerb: early reflections from a room of 3 to 7 corners, and a separate tail, balanced.

Controls

Predelay (0 to 200 ms) · **Room Shape** (3 Corners, 4 Corners, 5 Corners, 6 Corners, 7 Corners) · **Room Size** (1 to 50 m) · **Stereo Base** (0 to 200 cm) · **Initial Delay** (0 to 200 ms) · **Spread** (0 to 200 %) · **High Cut** (200 to 20000 Hz) · **Density** (0 to 100 %) · **Reverb Time** (0.3 to 20 s) · **Diffusion** (0 to 100 %) · **Balance ER/Reverb** (-100 to 100 %) · **Mix** (0 to 100 %)

Silver Compressor



Silver Compressor: threshold, ratio, attack, release, a soft knee and makeup gain.

Controls

Threshold (-60 to 0 dB) · **Ratio** (1 to 30 :1) · **Attack** (0.1 to 200 ms) · **Release** (5 to 2000 ms) · **Gain** (0 to 30 dB) · **Auto Gain** (Off, On) · **Soft Knee** (Off, On)

Silver Gate



Silver Gate: below the threshold the sound drops by the reduction, with attack, hold and release, and lookahead.

Controls

Threshold (-100 to 0 dB) · **Reduction** (-100 to 0 dB) · **Attack** (0 to 100 ms) · **Hold** (0 to 2000 ms) · **Release** (2 to 2000 ms) · **Lookahead** (0 to 10 ms)

DeEsser

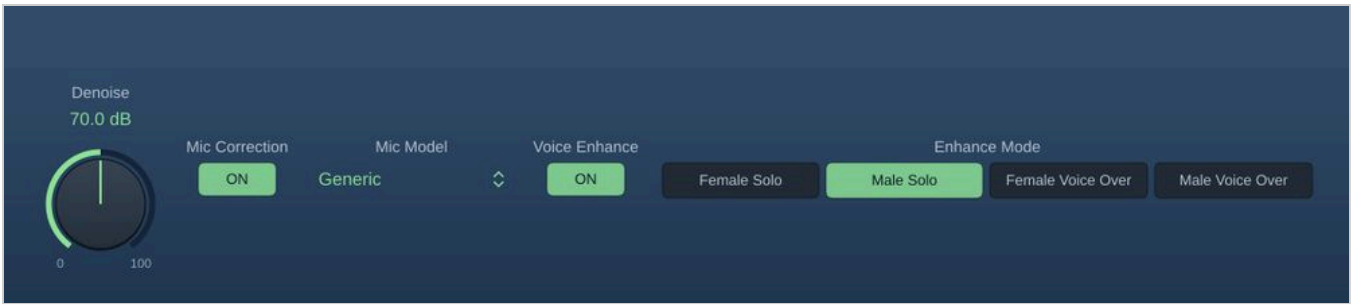


The original DeEsser: a detector band and a separate suppressor band; strength, smoothing, monitors.

Controls

Detector Frequency (500 to 20000 Hz) · **Sensitivity** (0 to 100 %) · **Suppressor Frequency** (500 to 20000 Hz) · **Strength** (0 to 50 dB) · **Smoothing** (1 to 200 ms) · **Monitor** (Off, Detector, Suppressor)

Speech Enhancer



Speech Enhancer: noise taken down, the microphone's colour corrected, the voice brought forward.

Controls

Denoise (0 to 100 dB) · **Mic Correction** (Off, On) · **Mic Model** (Generic, Laptop, Desktop, Webcam, Headset) · **Voice Enhance** (Off, On) · **Enhance Mode** (Female Solo, Male Solo, Female Voice Over, Male Voice Over)

Grooveshifter



Grooveshifter: swing put into straight playing - every other eighth (or sixteenth) moved later, without changing the pitch (grains of the sound played at their speed while time is stretched and squeezed).

Controls

Source Material (Beat, Tonal) · **Grain Size** (0 to 100 ms) · **Grid** (1/8, 1/16) · **Swing** (50 to 85 %) · **Accent** (-12 to 12 dB)

Guitar Amp Pro



Guitar Amp Pro: its amps, speakers and EQs are among Amp Designer's.

Controls

Amp (US Combo, UK Combo, UK Top, US Top Clean, US Top Lead, Small Combo, Brown Top, Modern Top) · **Speaker** (UK 1 x 12, UK 4 x 12, US 1 x 12 Open, US 1 x 12 Closed, US 2 x 12, US 4 x 10, Small 1 x 8, DI Box) · **EQ** (British, American, Modern) · **Gain** (0 to 10) · **Bass** (0 to 10) · **Mids** (0 to 10) · **Treble** (0 to 10) · **Presence** (0 to 10) · **Master** (0 to 10) · **Mic Position** (Centre, Off-Axis) · **Mic** (Dynamic, Condenser) · **Reverb** (Off, On) · **Reverb Type** (Spring 1, Spring 2, Spring 3) · **Reverb Level** (0 to 100 %) · **Effects** (Off, On) · **Trem/Vib** (Tremolo, Vibrato) · **Depth** (0 to 100 %) · **Speed** (0.1 to 12 Hz) · **Sync** (Free, Sync) · **Output** (-24 to 12 dB)

Bass Amp



The original Bass Amp: its amps and controls, played by Bass Amp Designer.

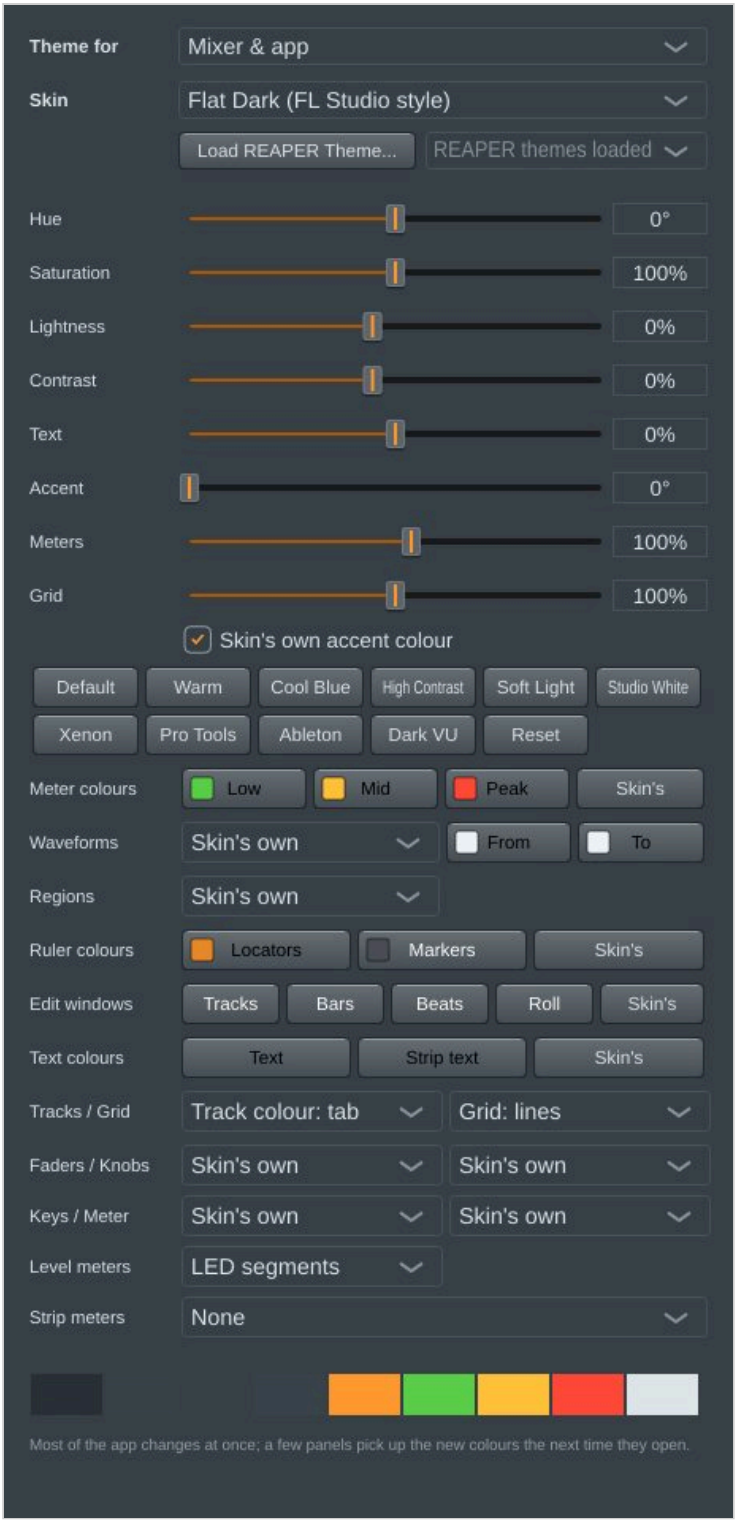
Controls

Amp Model (American Basic, American Deluxe, British Top, Modern) · **Pre Gain** (0 to 10) · **Bass** (0 to 10) · **Mid** (0 to 10) · **Mid Frequency** (100 to 1000 Hz) · **Treble** (0 to 10) · **Output** (-24 to 12 dB)

More

Looks and Themes

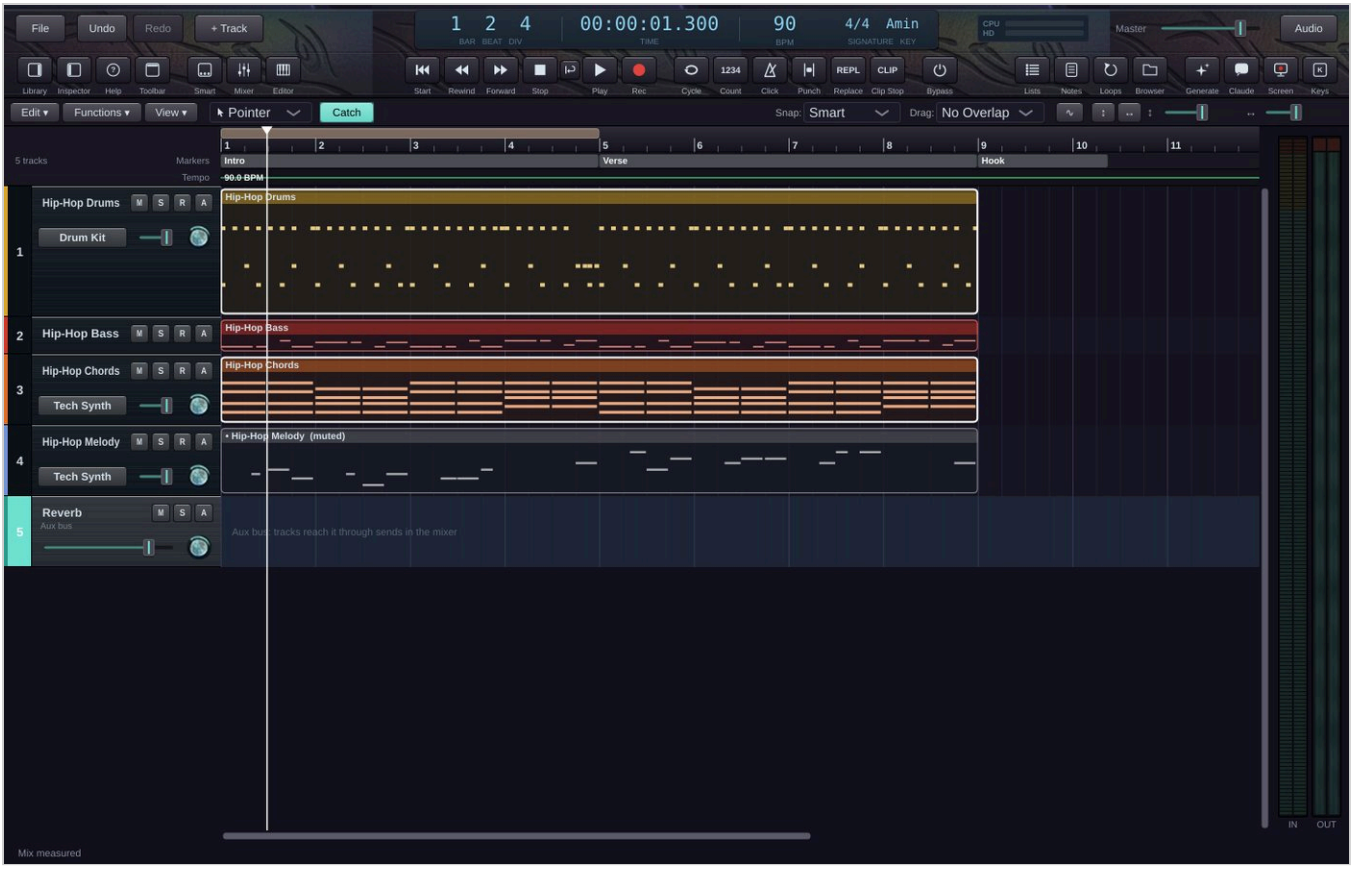
View > Theme... changes how the whole app looks.

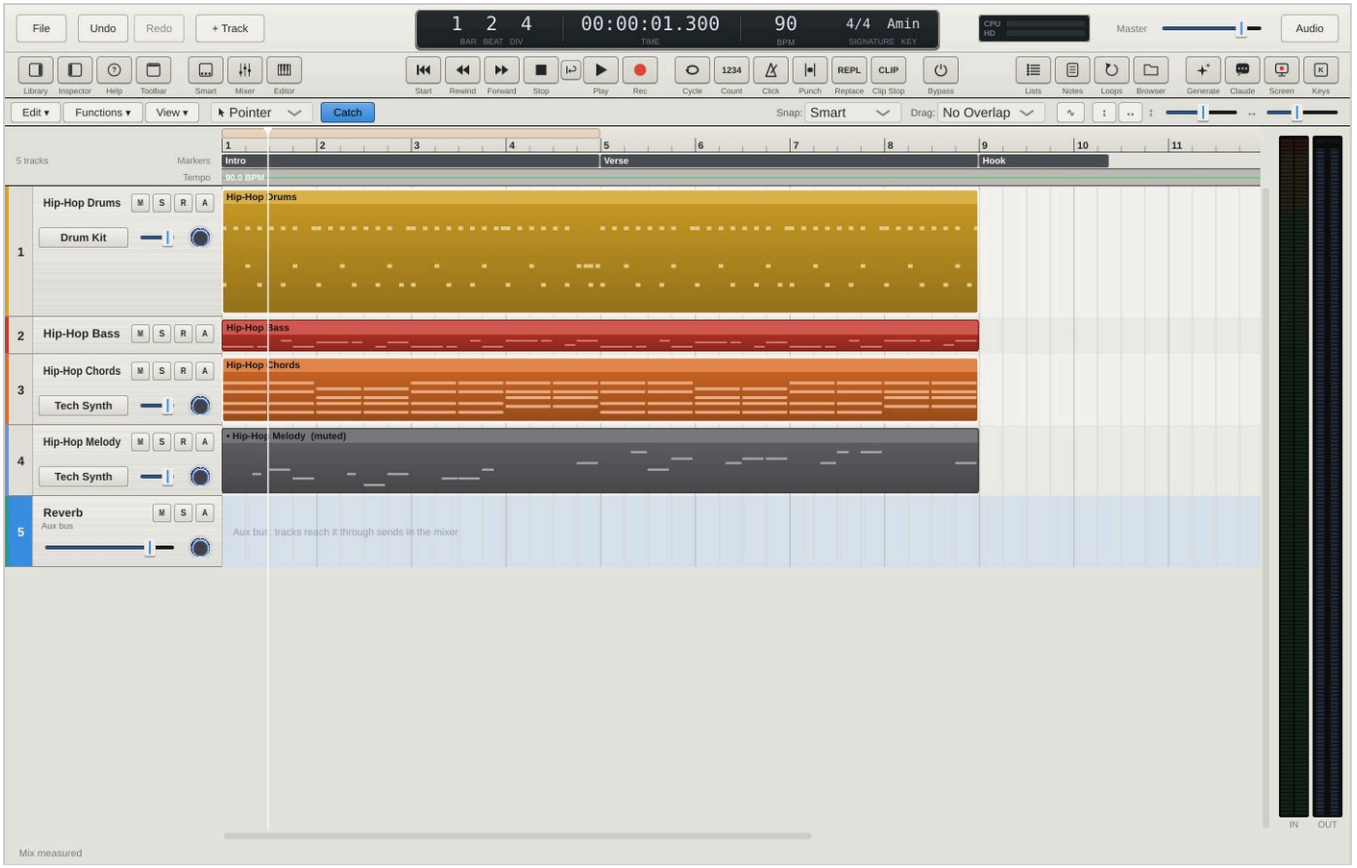


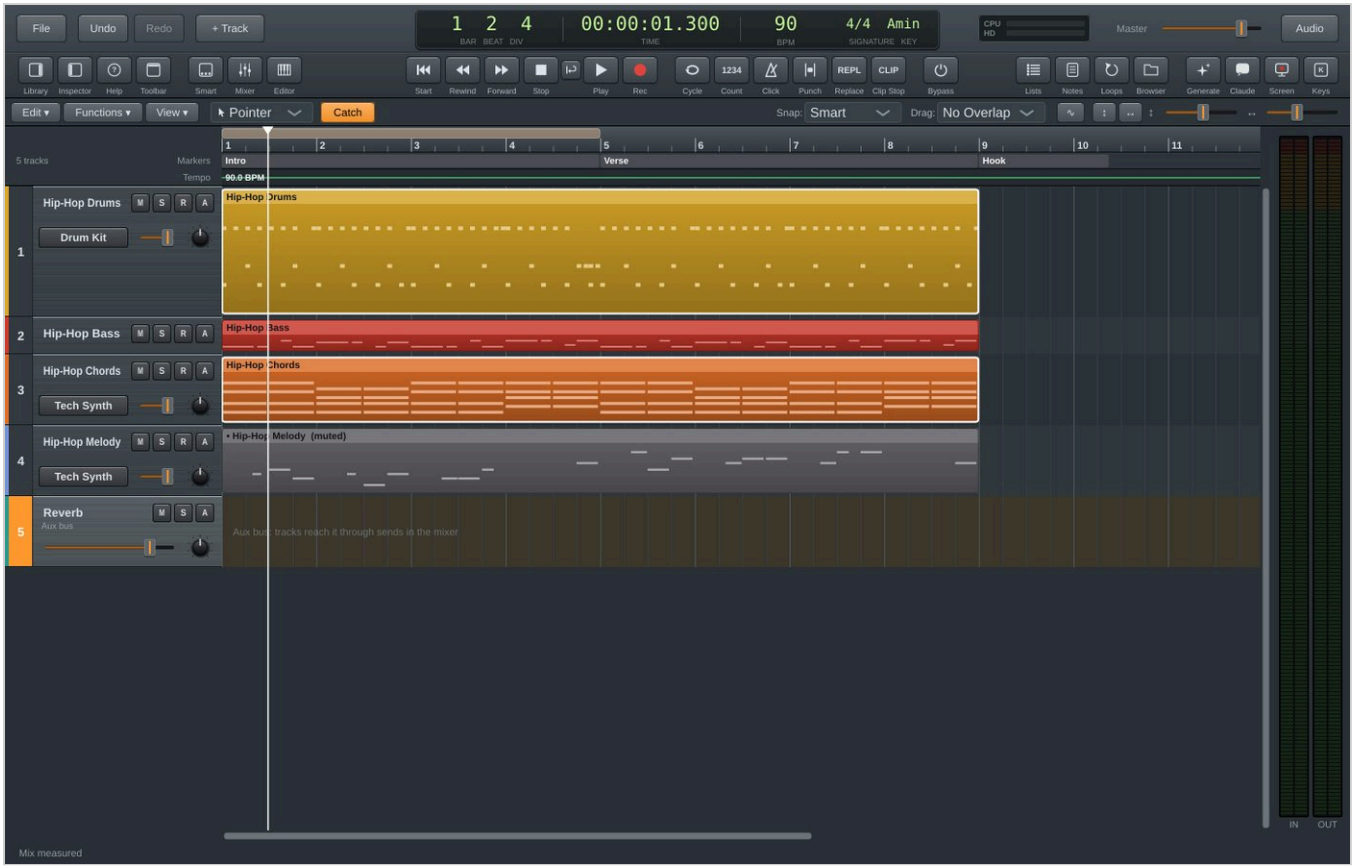
- **Theme for** - edit the mixer's theme (and the rest of the app), or give the editor, the toolbar or the MultiMeter a theme of its own.
- **Skin** - Studio Grey, Flat Dark (FL Studio style), Studio White, Xenon (dark glass console), Pro Tools style, Ableton style, or a **REAPER theme** (load a `.ReaperThemeZip`: its pictures

and colours skin the app).

- **Presets** - Default, Warm, Cool Blue, High Contrast, Soft Light, Studio White, Xenon, Pro Tools, Ableton and Dark VU: click one to try it.
- **Colours** - hue, lightness, contrast, text, accent, meters, grid; the locator and marker colours; how waveforms are coloured (the track's colour, your own colour, or a gradient).
- **Parts** - the **fader caps**, **knobs**, channel **keys** and **meters** whatever the skin (Xenon with black long-throw caps, say), an amber **VU meter** at the top of each mixer strip, and LED or smooth level meters.







Window layout

- **View > Show All Panels / Hide All Panels**.

- **Show Control Bar**, **Show Toolbar**, **Show Status Bar**, **Show Track Headers**, **Show Input and Output Meters**.
- **View > Show Inserts in Edit Window** and **Show I/O in Edit Window** - each track's insert slots, input and output beside its header.
- Open the **mixer** (`Cmd/Ctrl+2`) and the **editor** (`Cmd/Ctrl+4`) in their own windows to use a second screen.
- **Draw with the Graphics Card** - the window is drawn by the graphics card: smoother meters and scrolling with many tracks.

Settings and Performance

Settings

Settings... (`Cmd/Ctrl+Option/Alt+,`):

- **Autosave** - Off, or every 1, 2, 5 or 10 minutes.
- **Backups kept** - 3 to 50.
- **Automatic backups** - only when saving, or every 5, 10 or 15 minutes.
- **Recording format** - 16-bit, 24-bit or 32-bit float WAV.
- **When you stop** - stay where it stopped, or go back to where play started.
- **Eco mode** - Off, On (for slower computers), or When on battery.

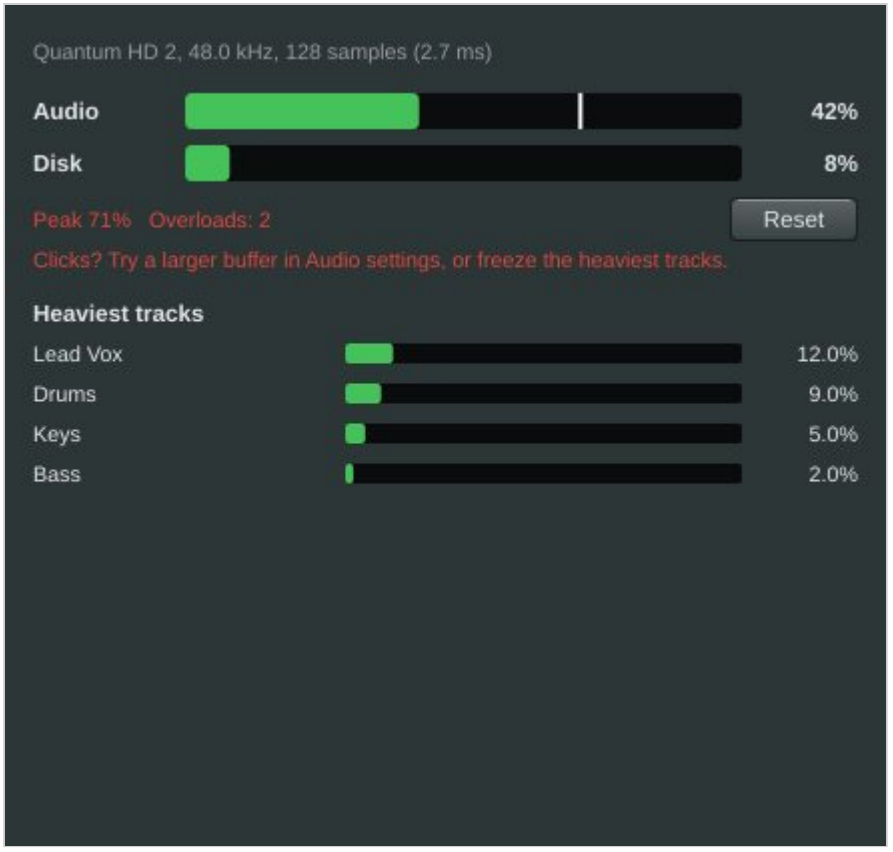
Key commands

Key Commands... (`Option/Alt+K`) shows every command with its keys. Choose a **preset** - Logic Pro, Pro Tools, Ableton Live, FL Studio, Cubase or LUNA - to use the keys you already know. Search by name, menu or key. Click a command and press a key to add it; **Reset to Default** puts them back.



Performance

View > Performance... shows how hard the computer works: the **audio** load and its peak, **overloads** (clicks), the **disk**, and the **heaviest tracks**.



If you hear clicks or dropouts:

1. Raise the **buffer size** in Audio Settings (256, 512 or 1024).
2. **Freeze** the heaviest tracks.
3. Turn on **Mix > Suspend Plug-ins When Silent**.
4. Put reverbs on a **bus** instead of on every track.
5. Close other programs (web browsers especially).

Eco Mode

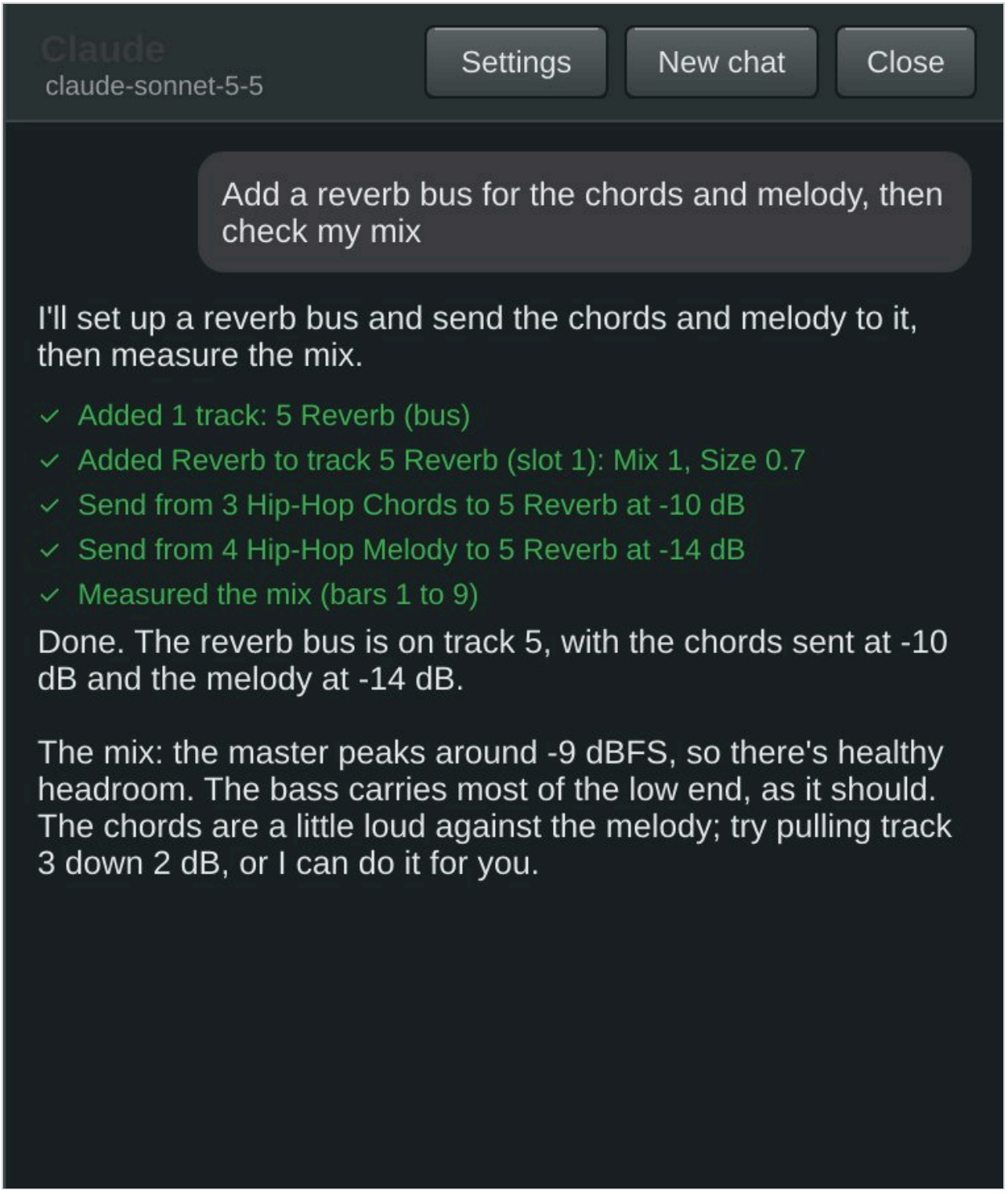
View > Eco Mode is for slower computers or a laptop on battery: meters and the playhead are drawn 15 times a second instead of 30, they stop when the app isn't in front, and other drawing is lighter - so more of the computer goes to the audio. In Settings you can choose **When on battery** so it turns on by itself when you unplug a laptop.

Quick Help

View > Quick Help (the **?** key) - a box that explains whatever is under the mouse. Leave it on while you learn the app.

The Claude Assistant

Click **Claude** in the control bar (**View > Claude Assistant**) for a panel where you can ask questions about your project - or tell it what to do, in your own words.



Setting it up

The assistant uses your own **Anthropic API key**, and its use is billed to your Anthropic account. Click **Settings** in the panel, paste your key (it starts with `sk-ant-`) and choose a model: **Sonnet** is fast and capable (recommended), **Opus** is the most capable but slower and costs more, **Haiku** is the quickest and cheapest.

What it can do

It can read your whole project (tempo, tracks, regions, mixer, effects) and change it:

- Set the tempo, the cycle, the click and count-in; play, stop, go to a bar.
- Add, change and delete tracks: names, faders, pans, mute, solo, record arm.
- Write MIDI, read the notes of a part, or generate drums, bass, chords and a melody in a genre.
- Set the channel strip EQ and dynamics, add and set effects, add sends to buses, set the master bus compressor and the master volume.
- Run any menu command.
- **Analyze the mix** - it renders the song and measures every track (peak, loudness and frequency balance), then tells you what to fix, or fixes it.

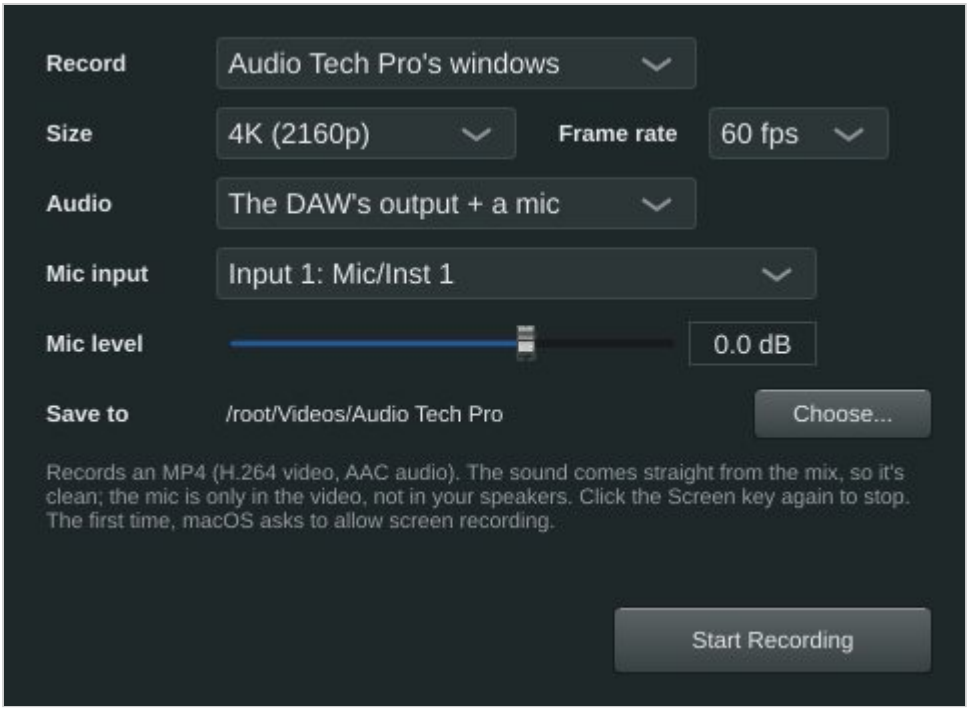
Examples

- "Make a trap beat at 140 BPM in F minor."
- "Add a reverb bus and send the vocals and the keys to it."
- "Why does my mix sound muddy?"
- "Turn the hi-hats down 3 dB and pan them a little left."
- "How do I record several takes and pick the best parts?"

Each change it makes is one step you can undo with `Cmd/Ctrl+Z`. **New chat** starts a new conversation.

Screen Recording

View > Record Screen... (or the **Screen** key) records the app - or the whole screen - to a video with the mix's sound. Use it for tutorials, beat videos and social media.



- **Record** - Audio Tech Pro's windows, or the whole screen.
- **Size** - up to 4K (2160p). **Frame rate** - 30 or 60 fps.
- **Audio** - the DAW's output, or the DAW's output plus a mic (your voice explaining).
- **Mic input** and **Mic level** - which interface input is the mic, and how loud.
- **Save to** - the folder for the videos.

Click **Start Recording**. The sound comes straight from the mix, so it's clean; the mic is only in the video, not in your speakers. Click the **Screen** key again to stop. The video is an MP4 (H.264 video, AAC audio).

The first time on a Mac, macOS asks you to allow screen recording (System Settings > Privacy & Security > Screen Recording). Allow it, then start again.

Troubleshooting

The Mac says the app "can't be opened" or "is damaged"

macOS blocks apps that aren't from the App Store. Open **Terminal** and paste this one line, press Return, and type your password (nothing shows while you type):

```
APP="/Applications/Audio Tech Pro.app"; sudo xattr -cr "$APP"; sudo codesign --force --deep --sign - "$APP"; open "$APP"
```

Do it once after each new version.

No sound

- Click **Audio** (top right) and check your interface is chosen. The status bar says "No audio device open" when none is.
- Check the **master volume** and the track's **mute** and **solo**.
- On an instrument track, check an instrument is chosen and your MIDI keyboard is connected (play and watch the track's meter).
- On Windows, choose the interface's **ASIO** driver.

I can't hear myself while recording

Turn on **I** (input monitoring) on the track, or use your interface's direct monitoring. If you hear yourself late, lower the buffer size.

Clicks, pops and dropouts

Raise the buffer size, freeze heavy tracks, and see **View > Performance...** (see *Settings and Performance*).

Recordings are out of time

Run **Calibrate Recording Latency...** once for your interface.

The project can't find its audio

Use **File > Find Missing Audio Files...** and choose the folder where the files are now. To avoid it, use **File > Consolidate Project...** or **Save a Copy As...** before moving a project.

A plugin doesn't appear

Run **Plug-in Manager...** to scan again. Check the format filter in **Mix > Plugin Formats in Menus**.

Nemix's recorded instruments are silent

The Nemix Library is installed with the app. If it was removed, reinstall the app, or put the **Nemix Library** folder in your Music folder under *Audio Tech Pro*.

Something went wrong

`Cmd/Ctrl+Z` undoes almost anything. **File > Revert to Backup...** opens an earlier save.

Key Commands

Every command, by menu, with its keys. `Cmd/Ctrl` is Cmd on a Mac and Ctrl on Windows; `Option/Alt` is Option on a Mac and Alt on Windows. Change any of them in **Key Commands...** (`Option/Alt+K`), or choose the keys of another program there.

File

- **New** `Cmd/Ctrl+N` - Start a new, empty project
- **Open...** `Cmd/Ctrl+O` - Open a saved project
- **Save** `Cmd/Ctrl+S` - Save the project
- **Save As...** `Cmd/Ctrl+Shift+S` - Save the project under a new name
- **Consolidate Project...** - Copy every audio file the project uses into its folder (Audio Files), so it's all in one place
- **Save as Template...** - Save this project as a template to start new ones from (File > New from Template)
- **New Alternative...** - Save a copy of the project beside it, to try ideas and switch between them (File > Alternatives)
- **Revert to Backup...** - Open one of the backups made each time you saved
- **Import Audio File...** `Cmd/Ctrl+Shift+I`, `Cmd/Ctrl+I` - Put audio files on the selected track at the playhead
- **Bounce Song to WAV...** `Cmd/Ctrl+B`, `Cmd/Ctrl+E` - Export the song, or the cycle range, to a WAV file
- **Export All Tracks as Audio Files...** `Cmd/Ctrl+Shift+E` - Each track (and bus) to its own WAV in a folder, with its plugins, fader and pan, ignoring solo
- **Import Tracks from Project...** - Bring tracks (regions, plugins, sends, automation) in from another project
- **Open Project from Another Program...** - Open an Ableton Live, FL Studio, Studio One, Pro Tools, Logic, Cubase, DAWproject or MIDI file project
- **Save a Copy As...** - A copy of the project with every audio file it uses, in one folder (and a .zip to send if you like): it opens on any computer
- **Close Project** - Closes the project (asking to save changes first)
- **Find Missing Audio Files...** - Looks for audio files the project can't find, in a folder you choose

Edit

- **Undo** `Cmd/Ctrl+Z` - Undo the last edit
- **Redo** `Cmd/Ctrl+Shift+Z`, `Cmd/Ctrl+Y` - Redo the edit you undid
- **Delete** `Backspace`, `Delete` - Delete the selected region
- **Split at Playhead** `Cmd/Ctrl+T` - Split regions at the playhead
- **Quantize** `Q` - Quantize the selected MIDI region to 16th notes

- **Set Locators by Selection** `Cmd/Ctrl+U` - Set the cycle to the selected region and turn Cycle on
- **Cut** `Cmd/Ctrl+X` - Cut the selected regions
- **Copy** `Cmd/Ctrl+C` - Copy the selected regions
- **Paste** `Cmd/Ctrl+V` - Paste at the playhead, on the selected track
- **Select All** `Cmd/Ctrl+A` - Select every region
- **Repeat** `Cmd/Ctrl+R` - Repeat the selected regions once, right after them
- **Join Regions** `Cmd/Ctrl+J` - Join the selected regions on a track into one (audio: consolidated into one new file)
- **Consolidate Clips** `Cmd/Ctrl+Shift+J` - The selected audio regions on each track become one region playing one new file (gain, fades and transpose in it; gaps silent)
- **Align Audio to Guide...** - Lines the timing of the selected regions up with a guide region (the top one), as VocAlign and Cubase's AudioAlign do
- **Detect Song Key...** - Listens to the song (or the selected regions) and finds its key and scale, to set as the project's key
- **Split to Stems...** - Splits the selected audio regions into vocals, drums, bass and the rest (or six stems, with guitar and piano), each on a new track, as Logic's Stem Splitter does
- **Mute Regions** - Mute or unmute the selected regions
- **Nudge Left** `Option/Alt+Left` - Move the selected regions a 16th note earlier
- **Nudge Right** `Option/Alt+Right` - Move the selected regions a 16th note later
- **Trim Region Start to Playhead** `Option/Alt+[` - Move the selected regions' start to the playhead
- **Trim Region End to Playhead** `Option/Alt+]` - Move the selected regions' end to the playhead
- **Loop Regions** `L` - Repeat the selected regions up to the next region (drag the upper right edge to set how far)
- **Region Gain...** - Turn the selected audio regions up or down
- **Remove Fades** - Take the fades off the selected regions
- **Split Regions by Locators** - Split every region at the cycle's start and end
- **Cut Section Between Locators** - Remove the cycle's stretch of the song, and close the gap
- **Insert Silence Between Locators** - Open a gap the length of the cycle at its start
- **Repeat Section Between Locators** - Play the cycle's stretch twice: a copy right after it
- **Tool Menu** `T` - Choose a tool: pointer, pencil, eraser, scissors, glue, mute
- **AudioSuite...** - Process the selected audio regions offline with an effect or plugin (VST3, AU, CLAP), into new files
- **Snap to Grid** `N` - Snap on or off: regions and notes land on the grid, or anywhere (Cmd while dragging does the opposite once)

Track

- **Bounce in Place** - Render the selected track to a new audio track and mute the original
- **New Audio Track** `Cmd/Ctrl+Option/Alt+A` - Add an audio track

- **New Software Instrument Track** `Cmd/Ctrl+Option/Alt+S` - Add an instrument (MIDI) track
- **New Aux Bus** - Add an aux bus for sends
- **Delete Track** `Cmd/Ctrl+Backspace` - Delete the selected track
- **Mute Track** `M` - Mute or unmute the selected track
- **Solo Track** `S` - Solo or unsolo the selected track
- **Record Enable Track** - Arm or disarm the selected track for recording
- **Show Automation** `A` - Show or hide automation on the selected track
- **Select Previous Track** `Up` - Select the track above
- **Select Next Track** `Down` - Select the track below
- **Hide Track** - Hide the selected tracks and select the next one (they still play)
- **Show Hidden Tracks** `H` - Show or leave out the tracks you hid
- **New Track With Duplicate Settings** `Cmd/Ctrl+D` - A new track like the selected one (instrument, fader, console, inserts, sends), without its regions
- **Track Colours...** `Option/Alt+C` - Choose a colour for the selected tracks
- **New Drummer Track** - A drummer who plays the song in a genre: set how busy and how loud, fills and swing
- **New Bass Player Track** - A session bass player, in the song's key
- **New Keyboard Player Track** - A session keyboard player playing chords in the song's key
- **Create Track Stack...** `Cmd/Ctrl+Shift+D` - Put the selected tracks in a stack: a folder stack (one fader on top of theirs) or a summing stack (they play into a bus)
- **Create Folder Stack** - Put the selected tracks in a folder stack: one fader, mute and solo for them all
- **Create Summing Stack** - Put the selected tracks in a summing stack: they play into one aux bus
- **Freeze Track** - Render the selected tracks with their instrument and plugins to audio, to save processing (again to unfreeze)
- **New VCA** - A VCA fader for the selected tracks: it moves them all, keeping their balance
- **Generate MIDI...** - Write drums, bass, chords and a melody in a genre's style

Transport

- **Play or Stop** `Space` - Start or stop playback
- **Record** `R` - Record on armed tracks, or punch out
- **Go to Beginning** `Return` - Move the playhead to the start
- **Rewind One Bar** `,` - Move the playhead back one bar
- **Forward One Bar** `.` - Move the playhead forward one bar
- **Create Marker** `Option/Alt+'` - A marker at the playhead (double-click the marker track for one anywhere)
- **Go to Next Marker** `Option/Alt+.` - Move the playhead to the next marker
- **Go to Previous Marker** `Option/Alt+,` - Move the playhead to the previous marker
- **Delete Marker** - Delete the selected marker
- **Cycle** `C` - Turn Cycle (loop) on or off
- **Metronome** `K` - Turn the click on or off

- **Count-in** `Shift+K` - One bar of click before recording
- **Autopunch** `Cmd/Ctrl+Option/Alt+I` - Record only between the punch locators (the red bar under the cycle; drag it in the ruler)
- **Stop Recording on Clip** - Recording stops by itself if a mic or instrument input clips (hits 0 dB); the take keeps what came before
- **Replace** - A recording replaces the parts of regions it covers
- **Merge MIDI Recordings** - MIDI recorded over a MIDI region is merged into it

Mix

- **Exclusive Solo** - Soloing a track unsolos the others (as in LUNA)
- **MultiMeter** - The master's meters: levels, loudness (LUFS), spectrum, goniometer and phase correlation
- **Automix...** - A balanced starting mix for a genre: each track measured and its fader set (one undo step)
- **Bypass All (Compare)** - Hear the mix with every channel's processing bypassed - preamps, channel strips, inserts, tape - and the master's inserts, bus compressor and tape; again to hear it all back (as one switch, to compare)
- **Reset Channels...** - Every channel (or the selected ones) and the master back to a fresh start: faders, pans, mute and solo, the channel strips, inserts and sends (as Cubase's Reset MixConsole Channels)
- **Suspend Plug-ins When Silent** - Plug-ins on a track that's silent stop working until sound comes again, to save CPU (as Cubase's VST3 suspend and FL Studio's Smart Disable)
- **Group Settings...** - Name the groups and choose what each links: volume, mute, solo, record
- **I/O Labels...** - Your own names for the interface's inputs (they show in the input menus)
- **Signal Flow** - The selected track's path from its input to the main output, with every stage

View

- **Show Inserts in Edit Window** - Each track's insert slots beside its header (as in Pro Tools' Edit window)
- **Show I/O in Edit Window** - Each track's input and output beside its header, to choose them there (as on the mixer strips)
- **Show Comments in Mixer** - A notes field under each mixer channel (as in Pro Tools)
- **Open Mixer Window** `Cmd/Ctrl+2` - The mixer in a window of its own (resize it, put it on another screen)
- **Open Editor Window** `Cmd/Ctrl+4` - The piano roll (and the other region editors) in a window of its own
- **Theme...** - The skin (studio grey or flat dark) and its colours: hue, lightness, contrast, text, accent, meters, grid
- **Performance...** - How hard the computer works: audio load, its peak, overloads, the disk, and the heaviest tracks

- **Notepad** `Cmd/Ctrl+Option/Alt+P` - Project notes: lyrics, ideas, what to fix (saved with the project)
- **Browser** `F` - Find audio files on your computer and put them on the selected track
- **Quick Help** - A box that explains whatever is under the mouse
- **Waveform Zoom** - Draw the waveforms taller, to see quiet parts (click again for more, then back)
- **Zoom to Fit** `Z` - The whole song across the tracks area, and every track fitting the height
- **Show Control Bar** - The bar along the top (display, transport, tools): show or hide it
- **Show Toolbar** - A row of keys for the common edits (split, join, repeat, quantize, bounce, freeze...)
- **Show Status Bar** - The line of hints along the bottom
- **Draw with the Graphics Card** - The window is drawn by the graphics card (smooth meters and scrolling with many tracks) instead of the processor
- **Show Input and Output Meters** - The meters to the right of the tracks
- **Show Track Headers** - The track names, buttons and faders at the left of the tracks
- **Show All Panels** - Inspector, control bar, toolbar, Smart Controls, meters and the status bar
- **Hide All Panels** - Just the tracks: every panel around them hidden
- **Live Loops** - A grid of loops per track: start and stop cells and scenes in time, on the next bar
- **Catch Playhead** - The tracks area follows the playhead while it plays (Logic's Catch); off, the view stays where you put it
- **Eco Mode** - For slower computers or a laptop on battery: meters and the playhead drawn 15 times a second instead of 30, still while the app isn't in front, strips out of view left alone - the sound is the same (Settings has On when on battery, too)
- **Mixer** `X` - Show or hide the mixer
- **Piano Roll** `P`, `E` - Open the selected MIDI region in the piano roll, or close it
- **Zoom In** `Cmd/Ctrl+Right` - Zoom in horizontally
- **Zoom Out** `Cmd/Ctrl+Left` - Zoom out horizontally
- **Inspector** `I` - Show or hide the Inspector: the selected track's channel strip and the master's, on the left
- **Zoom Vertical In** `Cmd/Ctrl+Down` - Make every track taller
- **Zoom Vertical Out** `Cmd/Ctrl+Up` - Make every track shorter
- **Undo History...** `Cmd/Ctrl+Option/Alt+Z` - Every edit you can undo; click one to go back (or forward) to it
- **Settings...** `Cmd/Ctrl+Option/Alt+,` - Autosave, backups, recording format and more
- **Smart Controls** `B` - Eight knobs for the selected track's sound (right-click one to map it)
- **Library** `Y` - Patches: save a track's sound and put it on other tracks
- **Loop Browser** `O` - Browse a folder of loops and samples: click to hear, drag onto a track
- **Controller Assignments...** - The knobs and faders on your MIDI controller that move tracks (right-click a track > Learn MIDI Controller to add one)
- **Event List** `Cmd/Ctrl+Option/Alt+D` - The selected MIDI region's notes and controllers as a list you can edit
- **Key Commands...** `Option/Alt+K` - Show and edit the key commands

- **Audio Settings...** `Cmd/Ctrl+,` - Choose the audio interface, sample rate and buffer size
- **Calibrate Recording Latency...** - Measure the round trip of your interface
- **Plug-in Manager...** - Scan for VST3 and Audio Unit plugins
- **Combine Audio Interfaces...** - Use several audio interfaces at once, as one
- **Record Screen...** - Record the app (or the whole screen) to a video with the mix's sound; again to stop
- **Claude Assistant** - Show or hide the Claude panel: ask questions, or tell it what to do

Piano Roll Keys

The keys of the piano roll (press `?` in the piano roll to see them there too).

Tools

- P Draw (pencil; it remembers the last note's length)
- B Paint (drag to lay notes along the grid)
- D Delete
- C Slice
- T the tool menu (T again: Mute)
- E Select
- Z Zoom (Option-click zooms out)
- Right-click or right-drag: delete notes, with any tool

Editing

- Arrows: Up/Down a semitone (Shift: an octave), Left/Right a grid step
- Q Quantize
- Delete delete the selected notes
- Cmd/Ctrl+A Select all
- Esc deselect
- Cmd/Ctrl+C / Cmd/Ctrl+X / Cmd/Ctrl+V Copy, cut, paste (at the playhead)
- Cmd/Ctrl+D Duplicate
- Option-scroll: taller or shorter rows
- Cmd-scroll: zoom in time

Note Tools (on The Selected Notes, or All)

- Cmd/Ctrl+L Quick Legato
- Option/Alt+U Chop (at the quantize grid)
- Cmd/Ctrl+G Glue
- Option/Alt+S Strum...
- Option/Alt+A Arpeggiate...
- Option/Alt+F Flam
- Option/Alt+R Randomize...
- Option/Alt+X Scale Levels (velocity)...
- Option/Alt+K Limit (fold into a range)...
- Option/Alt+C Claw Machine (thin out)...
- Option/Alt+L Articulate (note lengths)...
- Option/Alt+M Mute / Unmute
- Option/Alt+Y Flip Horizontally (reverse)

- Shift+Option/Alt+Y Flip Vertically (invert)
- Cmd/Ctrl+B Duplicate (also Cmd+D)